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Appendix A – Piping Datasheets

Appendix B – Valve Datasheets

1 SCOPE

This Specification defines the technical and material requirements for pipe, fittings, and valves. It provides specifications for materials, dimensions and pressure ratings as well as jointing methods. It also provides the following:

- A classification index correlating pipe specification with fluid service.
- An explanation of the line, valve and pipe specials numbering system.
- Reference to the relevant piping system datasheets.
- Reference to the relevant valve datasheets.

The following is excluded from the scope of this specification:

- Instrument and control tubing downstream of last piping isolation valve.
- All flexible hoses.

2 DEFINITIONS

2.1 GENERAL

word	DEFINITION
Contractor	The party engaged by the Superintendent to carry out the Works
Superintendent	EPC Group or an approved representative
Supplier	Any Vendor contracted to supply the Project

Table 2-1 General Definitions

2.2 ABBREVIATIONS

word	DEFINITION
ANSI	American National Standards Institute
API	American Petroleum Institute
AS	Australian Standard
ASME	American Society of Mechanical Engineers
ASTM	American Society of Testing and Materials
BB	Bolted Bonnet
BW	Butt Weld
BSP	British Standard Pipe Thread
BSPf	British Standard Pipe Thread – Female
BSPm	British Standard Pipe Thread – Male
DN	Nominal Diameter
CA	Corrosion Allowance
CS	Carbon Steel
ERW	Electrical Resistance Welded
FF	Flat Faced

word	DEFINITION
HDPE	High Density Polyethylene
ID	Internal Diameter
LR	Long Radius
OD	Outer Diameter
OS&Y	Outside Screw & Yoke
NBR	Nitrile
NDE	Non-Destructive Examination
NPS	Nominal Pipe Size
NPT	National Pipe Thread
NPTf	National Pipe Thread - Female
NPTm	National Pipe Thread - Male
P&ID	Piping and Instrumentation Diagram
PE	Plain End
PTFE	Polytetrafluoroethylene (Fluoroplastic)
PN	Pressure Number
RB	Reduced Bore
RF	Raised Face
RTJ	Ring Type Joint
Sch	Schedule (Wall Thickness)
SMLS	Seamless
Std Wt.	Standard Weight
SP	Special Piping Item
SR	Short Radius
SS	Stainless Steel
SW	Socket Weld
TE	Threaded End
UNC	United National Course Thread
UNS	Unified Numbering System
WN	Weld Neck
XS	Extra Strong
XXS	Double Extra Strong
3D	Elbow bend radius nominally 3 x pipe diameter
5D	Elbow bend radius nominally 5 x pipe diameter

Table 2-2 – Abbreviations and Acronym List

3 REFERENCE DOCUMENTS

All work performed and procedures followed shall be in accordance with the relevant standards where applicable unless otherwise noted on the drawings or ordered by the Superintendent.

All work shall comply with the Federal, State, or local governing bodies legislation & regulations as follows:

- Work Health and Safety Act 2020
- Work Health and Safety (General) Regulations 2022
- Work Health and Safety (Mines) Regulations 2022
- Mines Safety and Inspection Act 1994
- Mines Safety and Inspection Levy Regulations 2010

Where no standard exists, or in the case of conflicting information, clarification shall be sought from the Superintendent. However, the following precedence shall apply:

1. Compliance with local statutory regulations
2. Design Drawings
3. Equipment Specifications
4. Technical Specifications
5. Shop Detail Drawings

3.1 CODES, STANDARDS AND REFERENCES

All works shall comply with the requirements of the current revision relevant Australian Standards. Where there is no applicable Australian Standard the appropriate British, German, American or other approved international standard shall apply. The use of other standards must be approved by the Superintendent.

The relevant standards include, but are not limited to, the following:

3.1.1 Australian Standards

standard no.	TITLE
AS 1110	ISO Metric Hexagon Bolts and Screws - Product Grades A and B
AS 1111	ISO Metric Hexagon Bolts and Screws - Product Grade C
AS 1112	ISO Metric Hexagon Nuts
AS 1214	Hot-Dip Galvanised Coatings on Threaded Fasteners
AS 1345	Identification of the Contents of Pipes, Conduits and Ducts
AS 1528	Tubes (stainless steel) and tube fittings for the food industry
AS 1722	Pipe Threads of Whitworth Form, part 1 – Sealing Pipe Threads (based on ISO 7)
AS 2129	Flanges for pipes, valves and fittings
AS 2177	Non-Destructive Testing - Radiography of Welded Butt Joints in Metal
AS 2544	Grey Iron Pressure Fittings
AS 2700	Colour Standards for General Purposes
AS 4041	Pressure Piping
AS 4129	Fittings for polyethylene (PE) pipes for pressure applications
AS 4130	Polyethylene (PE) pipes for pressure applications
AS 4131	Polyethylene (PE) compounds for pressure pipes and fittings

standard no.	TITLE
AS 4343	Pressure equipment – Hazard levels
AS 4680	Hot Dipped Galvanised (Zinc) Coatings on Fabricated Ferrous Articles
AS ISO 7.1	Pipe Threads where Pressure-Tight Joints are made on the Threads - Dimensions, Tolerances and Designation

Table 3-1 – Australian Standards List

3.1.2 American Society of Mechanical Engineers

standard no.	TITLE
ASME B1.1	Unified Inch Screw Threads (UN and UNR Thread Form)
ASME B1.20.1	Pipe Threads, General Purpose (Inch)
ASME B16.5	Pipe Flanges & Flanged Fittings
ASME B16.9	Factory-Made Wrought Buttwelding Fittings
ASME B16.11	Forged Fittings, Socket-Welding and Threaded
ASME B16.20	Metallic Gaskets for Pipe Flanges – Ring Joint, Spiral Wound and Jacketed
ASME B16.21	Non-metallic Flat Gaskets for Pipe Flanges
ASME B16.25	Buttwelding Ends
ASME B16.34	Valves – Flanged, Threaded and Welding End
ASME B16.47	Large Diameter Steel Flanges (Series A)
ASME B16.48	Line Blanks
ASME B31.3	Process Piping
ASME B31.12	Hydrogen Piping and Pipelines
ASME B36.10	Welded and Seamless Wrought Steel Pipe
ASME B36.19M	Stainless Steel Pipe

Table 3-2 – American Society of Mechanical Engineers Standards List

3.1.3 American Petroleum Institute

standard no.	TITLE
API RP 941	Steels for Hydrogen Service at Elevated Temperatures and Pressures in Petroleum Refineries and Petrochemical Plants
API SPEC 5L	Line Pipe
API SPEC 6A	Wellhead and Tree Equipment
API SPEC 6D	Pipeline and Piping Valves
API SPEC 6FA	Fire Test for Valves
API STD 594	Check Valves—Flanged, Lug, Wafer, and Butt-welding
API STD 598	Valve Inspection and Testing
API STD 600	Steel Gate Valves—Flanged or Butt-welding Ends, Bolted Bonnets
API STD 602	Gate, Globe, and Check Valves for Sizes DN100 (NPS 4) and Smaller for the Petroleum and Natural Gas Industries
API STD 607	Fire Test for Quarter-turn Valves and Valves Equipped with Non-metallic Seats
API STD 608	Metal Ball Valves—Flanged, Threaded, and Welding Ends
API STD 623	Steel Globe Valves—Flanged and Butt-welding Ends, Bolted Bonnets

Table 3-3 – American Petroleum Institute Standards List

3.1.4 American Society for Testing and Materials

standard no.	TITLE
ASTM A105	Forgings, Carbon Steel, for Piping Components
ASTM A106	Seamless Carbon Steel Pipe for High Temperature Service
ASTM A123	Zinc (Hot-Dip Galvanised) Coatings on Iron and Steel Products
ASTM A182	Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service
ASTM A193	Alloy-Steel and Stainless Steel Bolting for High Temperature or High Pressure Service
ASTM A194	Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High Pressure or High Temperature Service
ASTM A213	Seamless Ferritic and Austenitic Alloy-Steel Boiler, Superheater, and Heat-Exchanger Tubes
ASTM A234	Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service
ASTM A240	Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM A269	Seamless and Welded Austenitic Stainless Steel Tubing for General Service
ASTM A312	Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes
ASTM A320	Alloy-Steel and Stainless Steel Bolting for Low-Temperature Service
ASTM A333	Seamless and Welded Steel Pipe for Low-Temperature Service and Other Applications with Required Notch Toughness
ASTM A335	Seamless Ferritic Alloy-Steel Pipe for High Temperature Service

standard no.	TITLE
ASTM A350	Carbon and Low-Alloy Steel Fittings, Requiring Notch Toughness Testing for Piping Components
ASTM A351	Castings, Austenitic for Pressure-Containing Parts
ASTM A380	Cleaning, Descaling, and Passivation of Stainless Steel Parts, Equipment and Systems
ASTM A403	Wrought Austenitic Stainless Steel Piping Fittings
ASTM A494	Castings, Nickel and Nickel Alloy
ASTM A790	Seamless and Welded Ferritic/Austenitic Stainless Steel Pipe
ASTM A815	Wrought Ferritic, Ferritic/Austenitic, and Martensitic Stainless Steel Piping Fittings
ASTM A967	Chemical Passivation Treatments for Stainless Steel Parts
ASTM B167	Nickel-Chromium-Iron Alloys and Nickel-Chromium-Cobalt-Molybdenum Alloy Seamless Piping and Tube (abbreviated name)
ASTM B366	Factory-Made Wrought Nickel and Nickel Alloy Fittings
ASTM B564	Nickel Alloy Forgings

Table 3-4 – American Society for Testing and Materials Standards List

3.1.5 Manufacturers Standardization Society

standard no.	TITLE
MSS SP-55	Quality Standard for Steel Castings for Valves, Flanges, Fittings and Other Piping Components – Visual Method for Evaluation of Surface Irregularities.
MSS SP-97	Integrally Reinforced Forged Branch Outlet Fittings - Socket Welding, Threaded and Buttwelding Ends

Table 3-5 – Manufacturers Standardization Society Standards List

3.1.6 National Associate of Corrosion Engineers

standard no.	TITLE
NACE MR0175	Metals for Sulphide Stress Cracking and Stress Corrosion Cracking Resistance in Sour Oilfield Environments

Table 3-6 – National Associate of Corrosion Engineers Standards List

3.1.7 European Standards

standard no.	TITLE
BS 6374-5	Lining of Equipment with Polymeric Materials for the Process Industries – Specification for Lining with Rubbers
BS EN 10241	Steel Threaded Pipe Fittings
BS EN 10312	Welded Stainless Steel Tubes for the Conveyance of Water and other Aqueous Liquids - Technical Delivery Conditions
BS EN 10088-1	Stainless Steels – Part 1: List of Stainless Steels
BS EN 10204	Metallic Materials – Type of Inspection Documents

Table 3-7 – European Standards List

3.2 EPC DOCUMENTS

All works shall comply with the latest revisions of the following documentation which form the project piping deliverables:

3.2.1 Technical Specifications

All works shall comply with the latest revisions of the following specifications:

specification no.	TITLE
SPEC-00-G-05	Owner Protective Coating Specification
2000-F00-STG-GE-10002	Site Specific Information
2000-F00-STG-GE-10003	Plant Numbering Specification
2000-F00-STG-ME-10003	Rubber Lining Specification
2000-F00-STG-ME-10005	Insulation of Tanks and Piping Specification
2000-F00-STG-PP-10002	Piping Fabrication, Installation and Testing Specification

Table 3-8 – Technical Specifications

3.2.2 Standard Drawings

The following piping standard drawings are applicable to the project:

Drawing no.	TITLE
2000-F00-STD-PP-10001	General Notes and Standard Drawing Index

Table 3-9 – Standard Drawings

3.2.3 Project Documents

Document no.	TITLE
2000-F00-LST-PP-10001	Piping Line List
2000-F00-LST-PP-10002	Manual Valve List
2000-F00-LST-PP-10003	Control Valve List
2000-F00-LST-PP-10005	Special Piping Item List

Table 3-10 – Project Documents

3.2.4 Piping System Datasheets

Datasheet no.	TITLE
2000-F00-DSH-PP-CL1	Fully flanged spun ceramic lined steel piping system with flat face flanges to AS2129 Table D for low pressure slurry applications.
2000-F00-DSH-PP-ECTFEDC1	Cascade butt weld / induction weld ECTFE SDR 21 piping system and secondary contained PE100 SDR17 outer piping with AS2129 Table E flanges for low pressure and high temperature reagent service applications.
2000-F00-DSH-PP-MS1	Welded carbon steel piping system with flanges to AS2129 Table D for low pressure liquid or gas applications.

Datasheet no.	TITLE
2000-F00-DSH-PP-MS2	Welded carbon steel piping system with flanges to AS2129 Table E for medium pressure liquid or gas applications.
2000-F00-DSH-PP-MSG1	Welded galvanised carbon steel piping system with flanges to AS2129 Table D for low pressure liquid or gas applications.
2000-F00-DSH-PP-MSG2	Welded galvanised carbon steel piping system with flanges to AS2129 Table E for medium pressure liquid or gas applications.
2000-F00-DSH-PP-MSG3	Viega Megapress (cold press) galvanised carbon steel piping system with flanges to AS2129 Table E for medium pressure liquid or gas applications.
2000-F00-DSH-PP-PE8	Butt welded HDPE PN10 (SDR17) piping system, short radius bends, with flanges to AS2129 Table D for low pressure liquid applications.
2000-F00-DSH-PP-PE10	Butt Welded HDPE PN10 (SDR17) piping system, sweep bends, with flanges to AS 2129 Table D for low pressure slurry applications.
2000-F00-DSH-PP-PE12.5	Butt Welded HDPE PN12.5 (SDR13.6) piping system with flanges to AS 2129 Table E for medium pressure slurry tails pipeline applications or liquid applications.
2000-F00-DSH-PP-PE16	Butt Welded HDPE PN16 (SDR11) piping system with flanges to AS 2129 Table E for medium pressure slurry or liquid applications.
2000-F00-DSH-PP-PE20	Butt Welded HDPE PN20 (SDR 9) piping system with flanges to AS 2129 Table E for high pressure slurry or liquid applications.
2000-F00-DSH-PP-PEDC1	IR fusion PE100 SDR11 piping system and secondary contained PE100 SDR17 outer piping with AS2129 Table E flanges for low pressure and ambient reagent service applications
2000-F00-DSH-PP-RL1	Fully flanged 6mm thick natural rubber lined steel piping system with flat face flanges to AS2129 Table D for low pressure slurry applications.
2000-F00-DSH-PP-RL2	Fully flanged 6mm thick natural rubber lined steel piping system with flat face flanges to AS2129 Table E for medium pressure slurry applications.
2000-F00-DSH-PP-RL3	Fully flanged 12mm thick natural rubber lined steel piping system with flat face flanges to AS2129 Table D for low pressure slurry applications.
2000-F00-DSH-PP-RL4	Not Used
2000-F00-DSH-PP-RL5	Fully flanged 6mm thick butyl rubber lined steel piping system with flat face flanges to AS2129 Table D for low pressure acidic slurry applications.
2000-F00-DSH-PP-SS1	Welded Gr316 stainless steel piping system, Schedule 10S, with flanges to AS 2129 Table D for low pressure liquid or gas applications.
2000-F00-DSH-PP-SS2	Welded Gr316 stainless steel piping system, Schedule 10S, with flanges to AS 2129 Table E for medium pressure liquid or gas applications.
2000-F00-DSH-PP-SS3	Viega Megapress (cold press) Gr316 stainless steel piping system with flanges to AS 2129 Table E for medium pressure liquid or gas applications.
2000-F00-DSH-PP-SS4	Vic-Press Grade 316L stainless steel piping system with flanges to AS 2129 Table D for low pressure potable water applications.
2000-F00-DSH-PP-SS5	Butt welded Gr316 stainless steel piping system, Schedule 10S, with flanges to AS 2129 Table E for hydrogen peroxide duty.
2000-F00-DSH-PP-SS6	Butt welded Gr316 stainless steel piping system, Schedule 10S, with flanges to AS 2129 Table E for oxygen duty.
2000-F00-DSH-PP-SS7	Butt welded Duplex 2205 stainless steel piping system, Schedule 10S, with flanges to AS 2129 Table E

Datasheet no.	TITLE
2000-F00-DSH-PP-SS8	Swagelok fitting Gr316 stainless steel tube system, with SAE flanges for standard pressure hydraulic/lubrication applications.
2000-F00-DSH-PP-SS9	Swagelok fitting Gr316 stainless steel tube system, with SAE flanges for standard pressure hydraulic/lubrication applications.
2000-F00-DSH-PP-SS10	Welded Gr316 stainless steel piping system, Schedule 40S, with flanges to AS 2129 Table H for high pressure liquid or gas applications.
2000-F00-DSH-PP-XS	Welded carbon steel piping system XS schedule, with flanges to AS2129 Table D for low pressure, low abrasion slurry applications.

Table 3-11 – Piping System Data Sheets

3.2.5 Manual Valve Standard Datasheets

Document no.	title
2000-F00-DSH-PP-VB03N1	Ball Valve – Full bore, three piece body, Hastelloy, BSP threaded for isolation duty in elution and corrosive duties at elevated temperatures.
2000-F00-DSH-PP-VB03S1	Ball Valve – Full bore, three piece body, stainless steel, BSP threaded for general service and non-hazardous reagent applications
2000-F00-DSH-PP-VB03S2	Not used
2000-F00-DSH-PP-VB03S3	Ball Valve – Full bore, three piece body, stainless steel, BSP threaded, fire-safe/anti-static for hydrocarbon applications.
2000-F00-DSH-PP-VB03S4	Ball Valve – Full bore, three piece body, stainless steel, BSP threaded, fire-safe/anti-static for oxygen service.
2000-F00-DSH-PP-VB03S5	Ball Valve – Full bore, three piece body, stainless steel, BSP threaded, WaterMark Certified for potable water service.
2000-F00-DSH-PP-VB03S6	Double Block and Bleed Valve- Two-valve single outlet instrumentation gauge manifold, stainless steel, BSP threaded for reagent applications.
2000-F00-DSH-PP-VB03S7	Ball Valve – Full bore, two piece body, stainless steel, BSP threaded, vented for Hydrogen Peroxide service.
2000-F00-DSH-PP-VB09S1	Ball Valve – Full bore, block body, double block and bleed, stainless steel, BSP threaded for high pressure scheme water
2000-F00-DSH-PP-VBD1C1	Ball Valve – Full bore, two piece body, carbon steel, flanged to AS 2129 Table D for caustic and cyanide applications.
2000-F00-DSH-PP-VBD1N1	Ball Valve – Full bore, two piece body, Hastelloy, flanged to AS 2129 Table D for isolation duty in elution and corrosive duties at elevated temperatures.
2000-F00-DSH-PP-VBD1S1	Ball Valve – Full bore, two piece body, stainless steel, flanged to AS 2129 Table D for general service and non-hazardous reagent applications.
2000-F00-DSH-PP-VBD1S2	Ball Valve – Full bore, two piece body, stainless steel, flanged to AS 2129 Table D, fire-safe/anti-static for hydrocarbon applications.
2000-F00-DSH-PP-VBD1S3	Ball Valve – Full bore, two piece body, Stainless Steel, flanged to AS 2129 Table D, vented, for Reagents service, with extended stem.
2000-F00-DSH-PP-VBD1S4	Ball Valve – V-notch ball rotary globe valve, full-bore, one-piece body, Stainless Steel, flanged to ANSI CL150 RF, for flow control duty for general service.
2000-F00-DSH-PP-VBD1T1	Ball Valve – Full bore, three piece body, polypropylene body, flanged to AS 2129 Table D for Hydrochloric acid service.

Document no.	title
2000-F00-DSH-PP-VBE1C1	Ball Valve – Full bore, two piece body, carbon steel, flanged to AS 2129 Table E for cyanide applications.
2000-F00-DSH-PP-VBE1N1	Ball Valve – Full bore, two piece body, Hastelloy, flanged to AS 2129 Table E for isolation duty in elution and corrosive duties at elevated temperatures.
2000-F00-DSH-PP-VBE1S1	Ball Valve – Full bore, two piece body, stainless steel, flanged to AS 2129 Table E for general service and non-hazardous reagent applications.
2000-F00-DSH-PP-VBE1S2	Not used
2000-F00-DSH-PP-VBE1S3	Ball Valve – Full bore, two piece body, stainless steel, flanged to AS 2129 Table E, Fire-safe/Anti-static for Oxygen Service.
2000-F00-DSH-PP-VBE1S4	Ball Valve – Full bore, two piece body, stainless steel, flanged to AS 2129 Table E, vented for Hydrogen Peroxide service.
2000-F00-DSH-PP-VBE1S5	Ball Valve – Full bore, two piece body, Stainless Steel, flanged to AS 2129 Table D, vented, for Reagents service, with extended stem.
2000-F00-DSH-PP-VBFD1L1	Butterfly Valve - Ductile iron, EPDM liner & seat, wafer to AS2129 Table D for isolation and flow control duty for general service and non-hazardous reagent applications.
2000-F00-DSH-PP-VBFD1L2	Butterfly Valve - Ductile iron, EPDM liner & seat, lugged to AS2129 Table D for tank nozzles or end of line isolation duty for general service and non-hazardous reagent applications.
2000-F00-DSH-PP-VBFD1L3	Butterfly Valve - Ductile iron, FKM liner & seat, wafer to AS2129 Table D for isolation and flow control duty for low pressure air applications.
2000-F00-DSH-PP-VBFD1L4	Butterfly Valve - Ductile iron, FKM liner & seat, lugged to AS2129 Table D for end of line isolation duty for low pressure air applications.
2000-F00-DSH-PP-VBFD1L5	Butterfly Valve - Ductile iron, PTFE lined seat and disc, lugged to AS2129 Table D for Hydrochloric acid service.
2000-F00-DSH-PP-VBFD1L6	Butterfly Valve – Ductile Iron, EPDM liner & seat, flanged to AS2129 Table D for tank nozzles or end of line isolation duty for general service and non-hazardous reagent applications.
2000-F00-DSH-PP-VBFD1S1	Butterfly Valve – Stainless steel, EPDM liner & seat, wafer to AS2129 Table D for isolation and flow control duty for general service and non-hazardous reagent applications.
2000-F00-DSH-PP-VBFD1S2	Butterfly Valve – Stainless steel, EPDM liner & seat, lugged to AS2129 Table D for tank nozzles or end of line isolation duty for general service and non-hazardous reagent applications.
2000-F00-DSH-PP-VBFD1S3	Butterfly Valve – Stainless steel, FKM liner & seat, wafer to AS2129 Table D for isolation and flow control duty for general service and non-hazardous reagent applications.
2000-F00-DSH-PP-VBFD1S4	Butterfly Valve – Stainless steel, FKM liner & seat, lugged to AS2129 Table D for tank nozzles or end of line isolation duty for general service and non-hazardous reagent applications.
2000-F00-DSH-PP-VBFE1L1	Butterfly Valve - Ductile iron, EPDM liner & seat, wafer to AS2129 Table E for isolation and flow control duty for general service.
2000-F00-DSH-PP-VBFE1L2	Butterfly Valve - Ductile iron, EPDM liner & seat, lugged to AS2129 Table E for tank nozzles or end of line isolation duty for general service.
2000-F00-DSH-PP-VBFE1L3	Butterfly Valve - Ductile iron, FKM liner & seat, wafer to AS2129 Table E for isolation and flow control duty for compressed air service applications.

Document no.	title
2000-F00-DSH-PP-VBFE1L4	Butterfly Valve - Ductile iron, FKM liner & seat, lugged to AS2129 Table E for end of line isolation duty for compressed air service applications.
2000-F00-DSH-PP-VBFE1S1	Butterfly Valve – Stainless steel, EPDM liner & seat, wafer to AS2129 Table E for isolation and flow control duty for general service.
2000-F00-DSH-PP-VBFE1S2	Butterfly Valve – Stainless steel, EPDM liner & seat, lugged to AS2129 Table E for tank nozzles or end of line isolation duty for general service.
2000-F00-DSH-PP-VBFE1S3	Not used
2000-F00-DSH-PP-VBFE1S4	Butterfly Valve – Stainless steel, FKM liner & seat, lugged to AS2129 Table E for tank nozzles or end of line isolation duty for general service and non-hazardous reagent applications.
2000-F00-DSH-PP-VC01P1	Swing Check Valve – HDPE body, 316 SS components, ANSI flanged for saline water applications.
2000-F00-DSH-PP-VC01S1	Swing Check Valve – Stainless steel, BSP threaded for general service and non-hazardous reagent applications.
2000-F00-DSH-PP-VC01S2	Swing Check Valve – Stainless steel, BSP threaded, WaterMark certified for potable water service.
2000-F00-DSH-PP-VCD1C1	Dual Disc Check Valve – Carbon steel, wafer to AS2129 Table D for general service applications.
2000-F00-DSH-PP-VCD1S1	Dual Disc Check Valve – Stainless steel, wafer to AS2129 Table D for general service and hazardous reagent applications.
2000-F00-DSH-PP-VCD1S2	Swing Check Valve – Stainless steel, flanged to AS2129 Table D, fire-safe/anti-static for hydrocarbon applications.
2000-F00-DSH-PP-VCD1S3	Dual Disc Check Valve – Stainless steel, wafer to AS2129 Table D for potable water applications.
2000-F00-DSH-PP-VCD1S4	Swing Check Valve – Stainless steel, flanged to AS2129 Table D, for hazardous reagent applications.
2000-F00-DSH-PP-VCD1S5	Dual Disc Check Valve – Stainless steel, wafer to AS2129 Table D for general service and non-hazardous reagent applications.
2000-F00-DSH-PP-VCD1T1	Swing Check Valve – Polypropylene, flanged to AS2129 Table D for Hydrochloric acid service.
2000-F00-DSH-PP-VCE1C1	Dual Disc Check Valve – Carbon steel, wafer to AS2129 Table E for general service applications.
2000-F00-DSH-PP-VCE1S1	Dual Disc Check Valve – Stainless steel, wafer to AS2129 Table E for general service and hazardous reagent applications.
2000-F00-DSH-PP-VCE1S2	Not Used
2000-F00-DSH-PP-VCE1S3	Swing Check Valve – Stainless steel, flanged to AS2129 Table E, Fire-safe/Anti-static for Oxygen Service.
2000-F00-DSH-PP-VCE1S4	Inline Single Disc Check Valve – Stainless steel, wafer to AS2129 Table E for Hydrogen Peroxide
2000-F00-DSH-PP-VCE1T1	Swing Check Valve – Polypropylene, flanged to AS2129 Table E for Hydrochloric acid service.
2000-F00-DSH-PP-VDD1L1	Diaphragm Valve - Straight through type, cast iron, rubber lined, flanged to AS 2129 Table D for slurry service applications.
2000-F00-DSH-PP-VDD1L2	Diaphragm Valve - Weir type, cast iron, butyl rubber diaphragm and lining, flanged to AS 2129 Table D for Acid/Reagent service applications.

Document no.	title
2000-F00-DSH-PP-VDD1S1	Diaphragm Valve - Weir type, stainless steel, unlined, rubber diaphragm, flanged to AS 2129 Table D for general service and non-hazardous reagent applications.
2000-F00-DSH-PP-VDD1S2	Diaphragm Valve - Weir type, stainless steel, unlined, butyl rubber diaphragm, flanged to AS 2129 Table D for pregnant eluate and electrowinning service.
2000-F00-DSH-PP-VDE1L1	Not used
2000-F00-DSH-PP-VDE1L2	Diaphragm Valve - Weir type, cast iron, butyl rubber diaphragm and lining, flanged to AS 2129 Table E for Acid/Reagent service applications.
2000-F00-DSH-PP-VDE1S1	Diaphragm Valve - Weir type, stainless steel, unlined, rubber diaphragm, flanged to AS 2129 Table E for general service.
2000-F00-DSH-PP-VDE1S2	Diaphragm Valve - Weir type, stainless steel, unlined, FKM rubber diaphragm, flanged to AS 2129 Table E for high temperature eluate service.
2000-F00-DSH-PP-VDTD1L1	Dart Valve - yDart type, mild steel, rubber lined, flanged to AS 2129 Table D for slurry service applications.
2000-F00-DSH-PP-VF03S1	Float Valve - Stainless Steel, Viton Seal, BSP threaded for scheme water applications.
2000-F00-DSH-PP-VGD1C1	Gate Valve - Bonnet, carbon steel, flanged to AS2129 Table D for Reagents services.
2000-F00-DSH-PP-VGD1S1	Gate Valve - Bonnet, stainless steel, flanged to AS2129 Table D for general services and low hazard reagents.
2000-F00-DSH-PP-VGE1C1	Gate Valve - Bonnet, carbon steel, flanged to AS2129 Table D for Reagents services.
2000-F00-DSH-PP-VGE1F1	Gate Valve - OS&Y, Ductile Iron, flanged to AS2129 Table E for isolation duty of fire water service.
2000-F00-DSH-PP-VGLE1S1	Globe Valve - Stainless Steel, PTFE Seat, flanged to AS 2129 Table E for oxygen applications.
2000-F00-DSH-PP-VKGD1C1	Knife Gate Valve - Delta, Bidirectional shear, Ni resist, Metal Seat, lugged to AS2129 Table D for severe slurry service applications.
2000-F00-DSH-PP-VKGD1L1	Knife Gate Valve - Bidirectional, ductile iron, natural rubber replaceable sleeve, narrow body, lugged to AS2129 Table D for slurry service applications.
2000-F00-DSH-PP-VKGD1L2	Knife Gate Valve - Bidirectional, ductile iron, natural rubber replaceable sleeve, wide body, flanged to AS2129 Table D for slurry service applications.
2000-F00-DSH-PP-VKGD1L3	Knife Gate Valve - Bidirectional, ductile iron body, 2205 duplex gate, butyl rubber replaceable sleeve, wide body, flanged to AS2129 Table D for acidic slurry service applications.
2000-F00-DSH-PP-VKGE1C1	Knife Gate Valve - Delta, Bidirectional shear, Ni resist, Metal Seat, lugged to AS2129 Table E for slurry service applications.
2000-F00-DSH-PP-VKGE1L1	Knife Gate Valve - Bidirectional, ductile iron, natural rubber replaceable sleeve, wide body, flanged to AS2129 Table E for high pressure slurry service applications.
2000-F00-DSH-PP-VPID1L1	Pinch Valve – Closed body type, cast iron body, replaceable natural rubber sleeve, flanged to AS2129 Table D for isolation and flow control duty for slurry service applications.
2000-F00-DSH-PP-VTWD1C1	Three Way Temperature Control Valve, carbon steel, flanged to AS2129 Table D for cooling water services.

Table 3-12 – Valve Data Sheets

4 FLUID SERVICES SUMMARY

The fluid services applicable for this project, their relevant piping system datasheets and pipe label descriptions are outlined in the Fluid Services Table below.

Fluid Code	Fluid Service	Fluid Design Conditions kPag @ °C	Piping System	Piping System Design Rating kPag @ °C	External Surface Treatment	Gasket	Notes	Permissible Valves
AA	Aeration Air	N/A	N/A	N/A	N/A	N/A	Not used	N/A
AE	Acid Effluent	700 @ 45 °C	PE8	700 @ 45 °C	N/A	TC2003	Threaded connections for drains only	VB03N1 ² , VBD1N1 ³ , VBD1T1, VCD1T1, VDD1L2
	Acid Rinse (Fine Carbon)	700 @ 45 °C	PE10	700 @ 45 °C	N/A	TC2003	Suitable for service combining Acid Rinse Wash and Fine Reject Carbon Slurry	VKGD1L3
			RL5	700 @ 70 °C	CS3	C4430	Suitable for service combining Acid Rinse Wash and Fine Reject Carbon Slurry	VB03N1 ² , VBD1N1 ⁴ , VKGD1L3
AS	Antiscalant	700 @ 45 °C	PE8	700 @ 45 °C	N/A	TC2003		VBD1S1, VCD1S1, VCD1S4
			SS3	1,400 @ 140 °C	N/A	C4430	Preferred for DN≤50 Small Bore	VB03S1, VBE1S1, VC01S1
BA	Blower Air	150 @ 180 °C	MSG1	700 @ 200 °C	N/A	C4430		VBFD1L3, VBFD1L4, VCD1C1
BE	Barren Eluate	690 @ 45 °C	MS1	700 @ 200 °C	CS3	C4430	Allowed for tank drains and overflows only	VBFD1L4
			PE8	700 @ 45 °C	N/A	TC2003		VB03S1, VBD1S1, VBFD1L3, VBFD1L4, VCD1S1, VCD1S4, VDD1L2
			PEDC1	690 @ 45 °C	N/A	TC2003	To be used for piping with no secondary containment. Valving not applicable	N/A
			SS1	700 @ 100 °C	N/A	C4430		VB03S1, VBD1S1, VBFD1S3, VBFD1S4, VCD1S1, VCD1S4, VDD1S1
BW	Bore Water (Raw Water)	700 @ 45 °C	PE16	1,200 @ 45 °C	N/A	C6327	To be used as much as practicable	VBE1S1, VBFE1L1, VBFE1L2, VCE1S1
			SS2	1,400 @ 100 °C	N/A	C4430	Optional. Engineer Selection	VB03S1, VBFE1S1, VBFE1S2, VCE1S1, VDE1S1
			SS3	1,400 @ 100 °C	N/A	C4430	Preferred for DN≤50 Small Bore	VB03S1, VBE1S1, VC01S1, VDE1S1
	Bore Water (Saline Water)	500 @ 45 °C	PE8	700 @ 45 °C	N/A	C6327	To be used as much as practicable	VBD1S1, VBFD1L1, VBFD1L2, VC01P1, VCD1S1, VCD1S4
			SS1	700 @ 100 °C	N/A	C4430	Optional. Engineer Selection	VB03S1, VBFD1S1, VBFD1S2, VCD1S1, VCD1S4, VDD1S1
			SS3	1,400 @ 140 °C	N/A	C4430	Preferred for DN≤50 Small Bore	VB03S1, VBE1S1, VC01S1
CA	Compressed Air	1,000 @ 45 °C	N/A	N/A	N/A	N/A	Not used	N/A

Fluid Code	Fluid Service	Fluid Design Conditions kPag @ °C	Piping System	Piping System Design Rating kPag @ °C	External Surface Treatment	Gasket	Notes	Permissible Valves
CN	Cyanide Solution	690 @ 45 °C	MS2	1,400 @ 80 °C	CS3	C4430	Limited to DN150 maximum. Threaded connections not allowed except for drain points. Drains to be plugged with threaded nipple. Galvanised fittings not permitted.	VB03S1 ¹ , VB03S6 ⁴ , VBE1C1, VBE1S1, VCE1C1, VCE1S1, VDE1L2, VGD1C1
			SS2	1,400 @ 150 °C	N/A	C4430	Limited to DN150 maximum. Threaded connections not allowed except for drain points. Drains to be plugged with threaded nipple. Galvanised fittings not permitted.	VB03S1 ¹ , VB03S6 ⁴ , VBE1S1, VCE1S1, VDE1S2
			PE16	1,200 @ 45 °C	N/A	TC2003		VBE1C1, VBE1S1, VCE1C1, VCE1S1, VDE1L2, VGD1C1
			PEDC1	690 @ 45 °C	N/A	TC2003	To be used for piping with no secondary containment. Valving not applicable	N/A
		400 @ 45 °C	MS1	700 @ 80 °C	CS3	C4430	Limited to DN150 maximum. Threaded connections not allowed except for drain points. Drains to be plugged with threaded nipple. Galvanised fittings not permitted.	VB03S1 ¹ , VB03S6 ⁴ , VBD1C1, VBD1S1, VCD1C1, VCD1S1, VCD1S4, VDD1L2, VGD1C1
			SS1	700 @ 120 °C	N/A	C4430	Limited to DN150 maximum. Threaded connections not allowed except for drain points. Drains to be plugged with threaded nipple. Galvanised fittings not permitted.	VB03S1 ¹ , VB03S6 ⁴ , VBD1S1, VCD1S1, VCD1S4, VDD1S2
			PE8	700 @ 45 °C	N/A	TC2003		VBD1C1, VBD1S1, VCD1C1, VCD1S1, VCD1S4, VGD1C1
			PEDC1	690 @ 45 °C	N/A	TC2003	To be used for piping with no secondary containment. Valving not applicable	N/A
CO	Copper Sulphate	700 @ 45 °C	PE16	1,200 @ 45 °C	N/A	TC2003	Only used for small bore	VB03S1 ¹ , VB03S6 ⁴ , VBE1S1
			PE8	700 @ 45 °C	N/A	TC2003		VB03S1 ¹ , VB03S6 ⁴ , VBD1S1
			SS1	700 @ 200 °C	N/A	C4430		VB03S1 ¹ , VB03S6 ⁴ , VBD1S1, VBD1S3, VC01S1, VCD1S1, VCD1S4
CON	Concentrate Slurry	700 @ 45 °C	MS1	700 @ 80 °C	CS3	C4430	Allowed for tank drains and overflows only	VKGD1L1
			PE10	700 @ 45 °C	N/A	C6327		VB03S1 ² , VDD1L1, VDTD1L1, VKGD1L1, VKGD1L2
			RL1	700 @ 65 °C	CS3	C4430		VB03S1 ² , VBD1S1 ⁴ , VDD1L1, VDTD1L1, VKGD1L1, VKGD1L2, VPID1L1
			SS3	1,400 @ 140 °C	N/A	C4430	Allowed for sample pump drains only	N/A

Fluid Code	Fluid Service	Fluid Design Conditions kPag @ °C	Piping System	Piping System Design Rating kPag @ °C	External Surface Treatment	Gasket	Notes	Permissible Valves
COW	Cooling Water	700 @ 45 °C	MSG1	700 @ 120 °C	N/A	C4430		VB03S1, VBD1S1, VBFD1L1, VBFD1L2, VCD1C1, VTWD1C1
			PE8	700 @ 45 °C	N/A	C6327		VBD1S1, VBFD1L1, VBFD1L2, VCD1C1
			SS1	700 @ 200 °C	N/A	C4430	Option for clean in place acid washing of heat exchangers	VB03S1
			SS3	1,400 @ 140 °C	N/A	C4430	Preferred for DN≤50 Small Bore	VB03S1, VBE1S1
CR	Crusher Oil	N/A	N/A	N/A	N/A	N/A	Not used	N/A
CS	Cyanide Slurry	700 @ 45 °C	PE10	700 @ 45 °C	N/A	C6327		VB03S1 ² , VDD1L1, VDTD1L1, VKGD1L1, VKGD1L2
			RL1	700 @ 65 °C	CS3	C4430		VB03S1 ² , VDD1L1, VDTD1L1, VKGD1L1, VKGD1L2, VPID1L1
CT	Calcine Tailings	N/A	N/A	N/A	N/A	N/A	Not used	N/A
CW	Cyanide Water	700 @ 45 °C	PE8	700 @ 45 °C	N/A	C6327		VB03S1, VBD1S1, VBFD1L1, VBFD1L2, VCD1S1, VCD1S4
			SS1	700 @ 120 °C	N/A	C4430		VB03S1, VBD1S1, VBFD1S1, VBFD1S2, VCD1S1, VCD1S4
			SS3	700 @ 140 °C	N/A	C4430		VB03S1, VBD1S1, VCD1S1, VCD1S4
DCW	Decant Cyanide Water	N/A	N/A	N/A	N/A	N/A	Not used	N/A
DL	Diesel Fuel	N/A	N/A	N/A	N/A	N/A	Not used	N/A
FA	Flotation Air	N/A	N/A	N/A	N/A	N/A	Not used	N/A
FD	Floor Drainage	700 @ 45 °C	MS1	700 @ 80 °C	CS3	C4430	Optional. Sumps discharge only.	VB03S1 ² , VKGD1L1
			MSG1	700 @ 120 °C	N/A	C4430	Optional. Lube Oil Drains Only.	VBFD1L1, VBFD1L2
			MSG3	1,400 @ 140 °C	N/A	C4430	Preferred for DN≤50 Small Bore	VB03S1 ²
			PE10	700 @ 45 °C	N/A	C6327	For slurry services	VKGD1L1
			PE8	700 @ 45 °C	N/A	C6327	For liquor services	VBFD1L2
			RL1	700 @ 70 °C	CS3	C4430		VBD1S1 ⁴ , VKGD1L1
			SS1	700 @ 120 °C	N/A	C4430	Optional. Sumps discharge only. To be used only for Pregnant/Barren Eluate	VB03S1 ² , VBFD1S2

Fluid Code	Fluid Service	Fluid Design Conditions kPag @ °C	Piping System	Piping System Design Rating kPag @ °C	External Surface Treatment	Gasket	Notes	Permissible Valves
FL	Flocculant	700 @ 45 °C	MS1	700 @ 80 °C	CS3	C4430	Option when handling concentrated flocculant	VB03S1, VBD1C1, VCD1C1, VDD1L2, VGD1C1
			PE8	700 @ 45 °C	N/A	TC2003		VBD1C1, VCD1C1, VDD1L2, VGD1C1
			SS1	700 @ 200 °C	N/A	C4430	Option when handling diluted flocculant with saline water	VB03S1, VBD1S1
FR	Frother	700 @ 45 °C	PE8	700 @ 45 °C	N/A	TC2003		VBD1S1, VBFD1L1, VBFD1L2, VCD1S1, VCD1S4
			SS1	700 @ 100 °C	N/A	C4430		VB03S1, VBD1S1, VBD1S3, VBFD1S1, VBFD1S2, VCD1S1, VCD1S4, VDD1S1
			SS3	1,400 @ 140 °C	N/A	C4430	Preferred for DN≤50 Small Bore	VB03S1, VBE1S1, VC01S1
FS	Flotation Slurry	700 @ 45 °C	PE10	700 @ 45 °C	N/A	C6327		VBD1S1, VDD1L1, VKGD1L1, VKGD1L2
			RL1	700 @ 65 °C	CS3	C4430		VB03S1 ² , VBD1S1 ⁴ , VDD1L1, VKGD1L1, VKGD1L2, VPID1L1
			RL3	700 @ 65 °C	CS3	C4430	To be used for pipe launders	VB03S1 ² , VDD1L1, VKGD1L1, VKGD1L2, VPID1L1
			SS3	1,400 @ 140 °C	N/A	C4430	Allowed for sample pump drains	N/A
FT	Final Tailings	1,100 @ 45 °C	PE10	700 @ 45 °C	N/A	C6327	Use where pressures are suitable	VKGD1L1, VKGD1L2
			PE12.5	970 @ 45 °C	N/A	C6327	Use where pressures are suitable	VKGE1L1
			PE16	1,200 @ 45 °C	N/A	C6327		VKGE1L1
			PE20	1,400 @ 45 °C	N/A	C6327		VKGE1L1
			RL1	700 @ 70 °C	CS3	C4430	Use where pressures are suitable	VB03S1 ² , VKGD1L1, VKGD1L2
			RL2	1,400 @ 70 °C	CS3	C4430		VB03S1 ² , VBE1S1 ⁴ , VKGE1L1, VKGE1L1
FW	Fire Water	1,200 @ 45 °C	MSG2	1,400 @ 80 °C	N/A	C4430	Red fluid ID pipe banding required	VB03S1, VGE1F1
			MSG3	1,400 @ 140 °C	N/A	C4430	Red fluid ID pipe banding required	VB03S1
			PE16	1,200 @ 45 °C	N/A	C6327	Shall only be used for buried piping	VGE1F1

Fluid Code	Fluid Service	Fluid Design Conditions kPag @ °C	Piping System	Piping System Design Rating kPag @ °C	External Surface Treatment	Gasket	Notes	Permissible Valves
GW	Gland Water (High Pressure)	1,200 @ 45 °C	PE20	1,400 @ 45 °C	N/A	C6327		VBE1S1, VBFE1L1, VBFE1L2, VCE1S1
			SS2	1,400 @ 100 °C	N/A	C4430	Optional	VB03S1, VBFE1S1, VBFE1S2, VCE1S1, VDE1S1
			SS3	1,400 @ 140 °C	N/A	C4430	Preferred for DN≤50 Small Bore	VB03S1, VBE1S1, VC01S1
	Gland Water (Low Pressure)	700 @ 45 °C	PE8	700 @ 45 °C	N/A	C6327		VBD1S1, VBFD1L1, VBFD1L2, VCD1S1, VCD1S4
			SS1	700 @ 100 °C	N/A	C4430	Optional	VB03S1, VBFD1S1, VBFD1S2, VCD1S1, VCD1S4, VDD1S1
			SS3	1,400 @ 140 °C	N/A	C4430	Preferred for DN≤50 Small Bore	VB03S1, VBE1S1, VC01S1
HCL	Hydrochloric Acid	410 @ 45 °C	PE8	700 @ 45 °C	N/A	TC2003	Threaded connections for drains only.	VB03N1 ² , VBD1N1 ³ , VBD1T1, VBFD1L5, VCD1T1, VDD1L2
			PEDC1	575 @ 45 °C	N/A	TC2003	To be used for piping with no secondary containment. Valving not applicable	N/A
			SS7	700 @ 50 °C	N/A	TC2003	Threaded connections for drains only.	VB03N1 ² , VBE1N1 ³ , VCE1T1
HO	Hydraulic Oil	15,000 @ 45 °C	SS8	30,000 @ 110 °C	N/A	High Pressure Gasket	For <25OD Small Bore. Valve supplied by others	N/A
			SS9	15,838 @ 93 °C	N/A	High Pressure Gasket	For ≤50OD Small Bore. Valve supplied by others	N/A
	Hydraulic Oil (Return)	700 @ 45 °C	SS1	700 @ 100 °C	N/A	C4430	To be used only for return lines if pressure is suitable	VB03S3, VBD1S2
			SS8	700 @ 110 °C	N/A	High Pressure Gasket	For <25OD Small Bore. Valve supplied by others	VBD1S2
			SS9	700 @ 93 °C	N/A	High Pressure Gasket	For ≤50OD Small Bore. Valve supplied by others	VBD1S2
			SS10	3,500 @ 100 °C	N/A	C4430	Drains only. Other valves supplied by others	VB03S3 ²
HP	Hydrogen Peroxide	700 @ 45 °C	SS5	700 @ 180 °C	N/A	TC2003	Butt welded or flanged system only. Threaded connections only for drains. Vented Ball Valves specific for Hydrogen Peroxide service only.	VB03S7 ¹ , VBE1S4, VBE1S5, VCE1S4
HPE	Hydrochloric Acid / Pregnant Eluate	700 @ 130 °C	SS7	700 @ 200 °C	N/A	TC2003	This is specific to the combined Acid Wash and Elution Column handling a combination of PE & HCL. Threaded connections for drains only.	VB03N1 ² , VBE1N1

Fluid Code	Fluid Service	Fluid Design Conditions kPag @ °C	Piping System	Piping System Design Rating kPag @ °C	External Surface Treatment	Gasket	Notes	Permissible Valves
IA	Instrument Air	1,000 @ 45 °C	MSG2	1,400 @ 80 °C	N/A	C4430		VB03S1, VBE1C1, VGE1C1, VCE1C1
			MSG3	1,400 @ 140 °C	N/A	C4430	Preferred for DN≤50 Small Bore	VB03S1, VBE1S1, VC01S1
			PE16	1,200 @ 45 °C	N/A	C6327	Shall only be used for buried piping	VB03S1, VBE1C1, VGE1C1, VCE1C1
LN	Lead Nitrate	700 @ 45 °C	PE8	700 @ 45 °C	N/A	TC2003		VBD1S1, VCD1S1, VCD1S4
			SS1	700 @ 80 °C	N/A	C4430		VB03S1 ¹ , VBD1S1, VCD1S1, VCD1S4, VGD1S1
LO	Lubricating Oil (High Pressure)	15,838 @ 45 °C	SS8	30,000 @ 110 °C	N/A	High Pressure Gasket	For <25OD Small Bore. Valve supplied by others	N/A
			SS9	15,838 @ 93 °C	N/A	High Pressure Gasket	For ≤50OD Small Bore. Valve supplied by others	N/A
	Lubricating Oil (Medium Pressure)	1,800 @ 45 °C	SS8	30,000 @ 110 °C	N/A	High Pressure Gasket	For <25OD Small Bore. Valve supplied by others	N/A
			SS9	15,838 @ 93 °C	N/A	High Pressure Gasket	For ≤50OD Small Bore. Valve supplied by others	N/A
			SS10	3,500 @ 100 °C	N/A	C4430	Primary Crusher Lube Oil. Drains only. Other valves supplied by others	VB03S3 ²
	Lubricating Oil (Low Pressure/Return)	700 @ 45 °C	SS1	700 @ 100 °C	N/A	C4430	To be used only for return lines if pressure is suitable	VB03S3, VBD1S2
			SS2	1,400 @ 200 °C	N/A	C4430	To be used only for return lines if pressure is suitable	VBE1S2
			SS3	1,400 @ 140 °C	N/A	C4430	To be used only for small bore return lines if pressure is suitable	N/A
			SS8	30,000 @ 110 °C	N/A	High Pressure Gasket	For <25OD Small Bore. Valve supplied by others	N/A
			SS9	15,838 @ 93 °C	N/A	High Pressure Gasket	For ≤50OD Small Bore. Valve supplied by others	N/A
LPG	Fuel Gas	250 @ 45 °C	MS1	700 @ 200 °C	CS3	C4430		VBD1S2, VCD1S2
			MSG3	1,400 @ 140 °C	N/A	C4430	Preferred for DN≤50 Small Bore	VB03S1
LS	Lime Slurry	700 @ 45 °C	MS1	700 @ 80 °C	CS3	C4430	Optional. Sumps discharge only	VB03S1 ² , VDD1L1, VKGD1L1, VKGD1L2
			PE10	700 @ 45 °C	N/A	TC2003		VDD1L1, VKGD1L1, VKGD1L2
			RL1	700 @ 70 °C	CS3	C4430		VBD1S1 ⁴ , VKGD1L1

Fluid Code	Fluid Service	Fluid Design Conditions kPag @ °C	Piping System	Piping System Design Rating kPag @ °C	External Surface Treatment	Gasket	Notes	Permissible Valves
MS	Mill Slurry	700 @ 45 °C	CL1	700 @ 70 °C	CS3	C4430	Suitable upstream primary cyclone	VB03S1 ² , VBD1S1 ⁴ , VKGD1C1
			PE10	700 @ 45 °C	N/A	C6327		VKGD1L1, VKGD1L2
			RL1	700 @ 70 °C	CS3	C4430	Only suitable post primary cyclone	VB03S1 ² , VBD1S1 ⁴ , VKGD1C1, VKGD1L1, VKGD1L2, VKGE1C1
			RL3	700 @ 70 °C	CS3	C4430	Typically used for gravity lines	VB03S1 ² , VKGD1C1, VKGD1L1, VKGD1L2, VKGE1C1
NS	Caustic Soda	700 @ 45 °C	MS1	700 @ 80 °C	CS3	C4430	Limited to DN150 maximum. Threaded connections not allowed except for drain points. Drains to be plugged with threaded nipple.	VB03S1 ¹ , VB03S6 ⁴ , VBD1C1, VBD1S1, VCD1C1, VCD1S1, VCD1S4, VDD1L2, VGD1C1
			PE8	700 @ 45 °C	N/A	TC2003		VBD1C1, VBD1S1, VCD1C1, VCD1S1, VCD1S4, VGD1C1
OG	Off Gas	N/A	N/A	N/A	N/A	N/A	Not used	N/A
OXY	Oxygen	1,000 @ 45 °C	SS6	1,400 @ 80 °C	N/A	TC2003	Limit use of threaded fittings	VB03S4, VB03S6 ⁴ , VBE1S3, VCE1S3, VGLE1S1
PE	Pregnant Eluate (Elution)	1,000 @ 130 °C	SS2	1,400 @ 150 °C	N/A	C4430		VB03N1 ³ , VB03S1, VBFE1S3, VBFE1S4, VCE1S1, VDE1S2
	Pregnant Eluate (Electrowinning)	700 @ 70 °C	MS1	700 @ 200 °C	CS3	C4430	Electrowinning tank drains and overflows only	VBD1C1
			ECTFEDC1	700 @ 70 °C	N/A	TC2003	To be used for piping with no secondary containment. Valving not applicable	N/A
			SS1	700 @ 120 °C	N/A	C4430		VB03S1, VBFD1S3, VBFD1S4, VCD1S1, VCD1S4, VDD1S2
			SS3	1,400 @ 140 °C	N/A	C4430	Electrowinning sampling lines only	VB03S1
	Pregnant Eluate (ILR)	690 @ 45 °C	MS1	700 @ 200 °C	CS3	C4430	ILR tank drains and overflows only	VBFD1L4
			PEDC1	690 @ 45 °C	N/A	TC2003	To be used for piping with no secondary containment. Valving not applicable	N/A
			SS1	700 @ 120 °C	N/A	C4430		VB03S1, VBFD1S3, VBFD1S4, VCD1S1, VCD1S4, VDD1S2
POT	Potable Water	700 @ 45 °C	PE8	700 @ 45 °C	N/A	C6327		VB03S5, VBFD1L1, VBFD1L2
			SS1	700 @ 120 °C	N/A	C4430		VB03S5, VBFD1S1, VBFD1S2, VCD1S3
			SS4	700 @ 85 °C	N/A	C4430	Preferred for DN≤50 Small Bore	VB03S5, VC01S2

Fluid Code	Fluid Service	Fluid Design Conditions kPag @ °C	Piping System	Piping System Design Rating kPag @ °C	External Surface Treatment	Gasket	Notes	Permissible Valves
PSW	Process Scheme Water (Normal)	700 @ 45 °C	MS1	700 @ 120 °C	CS3	C4430	Allowed for tank drains and overflows only	VBFD1L2
			MS2	1,400 @ 120 °C	CS3	C4430		VB03S1, VBFE1L1, VBFE1L2, VCE1C1
			MSG1	700 @ 120 °C	N/A	C4430	Allowed for cooling/chiller water circuit	VB03S1, VBFD1L1, VBFD1L2, VCD1C1
			MSG3	1,400 @ 140 °C	N/A	C4430	Preferred for DN≤50 Small Bore	VB03S1, VBE1S1
			PE8	700 @ 45 °C	N/A	C6327	Allowed for tank drains and overflows only	VBFD1L2
			PE16	1,200 @ 45 °C	N/A	C6327		VBFE1L1, VBFE1L2
			SS2	1,400 @ 120 °C	N/A	C4430	Optional. Engineer Selection	VB03S1, VBE1S1, VBFE1S1, VBFE1S2, VCE1S1
			SS3	1,400 @ 140 °C	N/A	C4430	Optional. Engineer Selection	VB03S1, VBE1S1, VC01S1
	Process Scheme Water (High Pressure)	15,000 @ 45 °C	SS8	15,000 @ 100 °C	N/A	High Pressure Gasket	Used for downstream of high pressure washers	VB09S1
PW	Process Water (High)	800 @ 45 °C	PE16	1,200 @ 45 °C	N/A	C6327		VB03S1, VBE1S1, VBFE1L1, VBFE1L2, VCE1S1
			SS2	1,400 @ 100 °C	N/A	C4430	Optional	VB03S1 ² , VBE1S1, VBFE1S1, VBFE1S2, VCE1S1, VDE1S1
			SS3	1,400 @ 100 °C	N/A	C4430	Preferred for DN≤50 Small Bore	VB03S1, VBE1S1, VC01S1, VDE1S1
	Process Water (Low)	700 @ 45 °C	PE8	700 @ 45 °C	N/A	C6327		VB03S1, VBD1S1, VBFD1L1, VBFD1L2, VCD1S1, VCD1S4, VDD1S1
			SS1	700 @ 100 °C	N/A	C4430	Optional	VB03S1 ² , VBD1S1, VBFD1S1, VBFD1S2, VCD1S1, VCD1S4, VDD1S1
			SS3	1,400 @ 100 °C	N/A	C4430	Preferred for DN≤50 Small Bore	VB03S1, VBE1S1, VC01S1, VDE1S1

Fluid Code	Fluid Service	Fluid Design Conditions kPag @ °C	Piping System	Piping System Design Rating kPag @ °C	External Surface Treatment	Gasket	Notes	Permissible Valves
SC	Carbon Slurry	700 @ 45 °C	MS1	700 @ 80 °C	CS3	C4430	Allowed for tank drains and overflows only containing Scheme Water	VB03S1 ² , VKGD1L1
			MS2	1,400 @ 80 °C	CS3	C4430	Only suitable post Elution containing Scheme Water	VB03S1 ² , VKGE1L1
			PE10	700 @ 45 °C	N/A	C6327	Allowed for tank drains, overflows and gravity launders only	VKGD1L1
			PE16	1,200 @ 45 °C	N/A	C6327		VKGE1L1
			RL1	700 @ 70 °C	CS3	C4430		VB03S1 ² , VBD1S1 ⁴ , VKGD1L1, VKGD1L2
			SS1	700 @ 80 °C	N/A	C4430	Only suitable post Kiln containing Raw Water	VKGD1L1
	Carbon Slurry (Acid Rinse)	700 @ 45 °C	PE10	700 @ 45 °C	N/A	TC2003	Suitable for service combining Acid Rinse Wash and Fine Reject Carbon Slurry	VKGD1L3
			RL5	700 @ 70 °C	CS3	C4430	Suitable for service combining Acid Rinse Wash and Fine Reject Carbon Slurry	VB03N1 ² , VBD1N1 ⁴ , VKGD1L3
SW	Shower Water	N/A	N/A	N/A	N/A	N/A	Not used	N/A
TE	Teric	N/A	N/A	N/A	N/A	N/A	Not used	N/A
TH	Thermal Oil	N/A	N/A	N/A	N/A	N/A	Not used	N/A
TW	Transfer Water	700 @ 45 °C	N/A	N/A	N/A	N/A	Not used	N/A
V	Vent	700 @ 200 °C	MS1	700 @ 232 °C	CS3	C4430		N/A
		700 @ 45 °C	PE8	700 @ 45 °C	N/A	C6327		N/A
WM	UFG Media and Process Water	700 @ 45 °C	PE10	700 @ 45 °C	N/A	C6327		VKGD1L1, VKGD1L2
			RL1	700 @ 70 °C	CS3	C4430		VB03S1 ² , VKGD1L1, VKGD1L2
			XS	700 @ 80 °C	CS3	C4430		VB03S1 ² , VKGD1L1, VKGD1L2
WW	Washdown Water	700 @ 45 °C	PE8	700 @ 45 °C	N/A	C6327		VBFD1L1, VBFD1L2
			SS3	1,400 @ 140 °C	N/A	C4430	Preferred for DN≤50 Small Bore	VB03S1, VBE1S1, VC01S1
XA	Xanthate	700 @ 45 °C	PE8	700 @ 45 °C	N/A	TC2003		VBD1S1, VCD1S1, VCD1S4, VGD1S1
			PE16	1,200 @ 45 °C	N/A	TC2003		VBE1S1, VCE1S1, VGE1S1
			SS1	700 @ 80 °C	N/A	C4430		VB03S1 ¹ , VBD1S1, VBD1S3, VCD1S1, VCD1S4, VGD1S1

Table 4-1 – Fluid Services

Valve Notes:

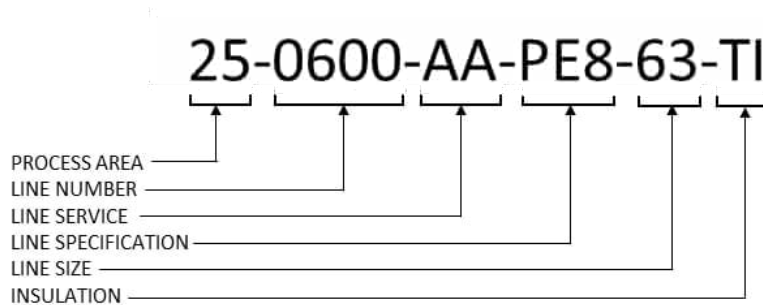
1. To only be used for drain/minsup connections only. Drains to be plugged with threaded nipple
2. To only be used for instrumentation or drain/minsup service points only
3. To be used for piping that is sharing both HCL and PE service
4. To be used for instrumentation only

5 LINE, VALVE AND PIPE SPECIALS DESIGNATION

5.1 LINE DESIGNATION

5.1.1 Line Numbering Methodology

The following alpha-numeric pipeline identification system will be used, comprising of six elements:



Example:

25-0600-AA-PE8-63-TI 25 Cleaner Flotation, 0600 Line Number, AA Aeration Air, PE8 HDPE PE100 PN10 Short Radius Bends, 63 Line Size OD63, TI Thermal Insulation

5.1.1.1 Process Area

The process area codes are a two-digit (numeric) identifier as per Plant Numbering Specification 2000-F00-STG-GE-10003, Area Codes, and shall be allocated based on the area number that the line starts in, with the exception of branches from piperacks, which shall be allocated the area that the line terminates.

5.1.1.2 Line Number

The line number is a four-digit (numeric) identifier will be applied to indicate the sequential line numbers. The combination of both Process Area and Line Number forms the unique line number.

The sequential identifier is changed for each service change or line specification change.

For a given service, the sequential identifier of the line number is changed at a branch of a piping run between equipment connections or between equipment and end of service.

The sequential identifier shall not be changed for a pipe diameter change within the piping run. Line numbers are preferably assigned in flow order.

New lines located within existing Process Areas shall commence with the number '3XXX' to avoid potential duplication of existing line numbers.

In special instances, some unique numbers will have an alpha prefix. These are segmented sections of the same line and will only be done for producing piping isometrics and fabrication. These will not be shown on P&ID's and only shown on the line list. Lines with the prefix or without is considered as the same line number. Refer to Table 5-1 and Figure 5-1 for prefix definitions.

PREFIX	DESCRIPTION
A	Brownfields Piperack SP2
B	Brownfields Piperack
C	Primary Crushing Discharge Conveyor
D	Stockpile Feed Conveyor
E	Coarse Ore Stockpile Reclaim Tunnel
F	Recycle Crushing
G	Milling Conveyors
H	Milling Piperack
I	Not Used
J	Milling, Classification and Trash Screening
K	Process Water
L	Reagents Piperack
M	SP1 Reagents
N	Cooling Water
O	Not Used
P	Greenfields Modular Piperack
Q	Tailings Thickener
R	Rougher and Scavenger Flotation
S	Cleaner and Cleaner-Scavenger Flotation
T	Air Compressors
U	Raw Water
V	Final Tails
W	SP2 Reagents
X	GF/BF Interchange Rack
Y	CIL4
Z	Elution

Table 5-1: Prefix Definition Codes

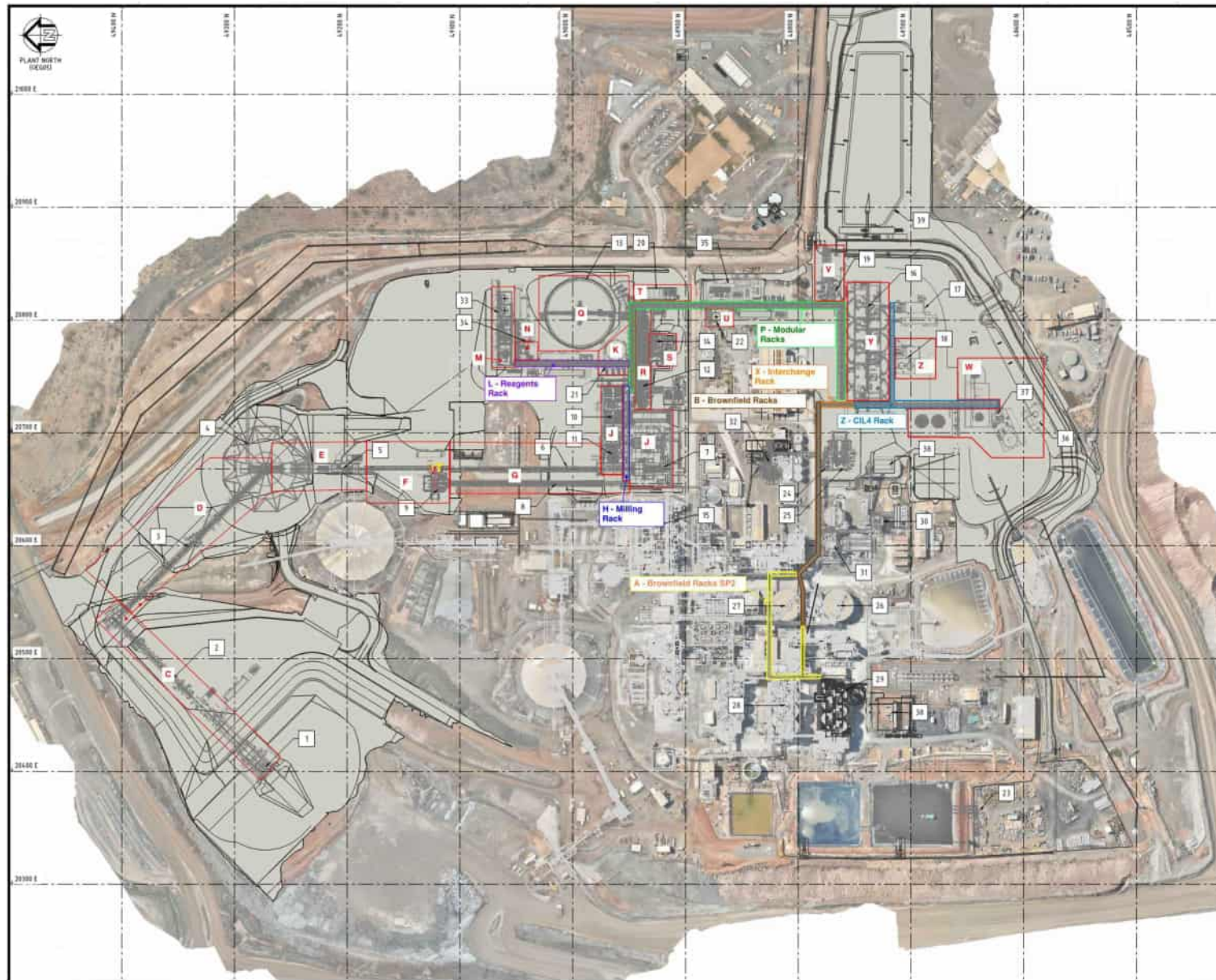


Figure 5-1: Prefix Definition Codes Key Plan

5.1.1.3 Line Service

The line service is a two or three capital letter identifier to indicate the pipeline service code (fluid conveyed in the pipe). Refer to Section 4 for list of Fluid Service Codes.

5.1.1.4 Line Specification

The line specification is a two or three capital letter and one or two-digit (alphanumeric) identifier to indicate the pipeline material specification.

The digit identifier is to differentiate the flange class, pipe thickness and fitting types for the given the piping materials. Refer to Piping System Datasheets for details.

Identifier Code	MATERIAL SPECIFICATION
CL	Ceramic Lined
MH	Mining Hose (Not used, mining hose identified as SPI)
MS(1-2)	Mild Steel, Std. Weight
MSG(1-3)	Mild Steel, Galvanised
PE(8-20)	High Density Polyethylene
PI	Polypropylene Instrument Tubing (Not used)
PP	Polypropylene
PVC	Polyvinyl Chloride
RL(1-6)	Rubber Lined Mild Steel
SS(1-10)	Stainless Steel
XS	Mild Steel, Extra Strong

Table 5-2: Material Specification Identifier Codes

5.1.1.5 Line Size

The line size is a one to four-digit (numeric) identifier to indicate the nominal pipe size. Nominal pipe diameter (DN) for steel pipes, and pipe outer diameter (OD) for polyethylene pipes.

5.1.1.6 Insulation (Alphabetic)

The insulation is a two capital letter identifier to indicate the insulation purpose, definitions as follows:

insulation type	DEFINITION
ET	Electrical Heat Trace
PP	Personnel Protection
ST	Steam Heat Trace
TI	Thermal Insulation

Table 5-3 - Insulation Type Identifiers

Note: Insulation is only nominated where required. The absence of an insulation code indicates no insulation.

5.1.2 Insulation Suffix

Suffix TI – Thermal Insulation

The requirement for insulation on individual lines is nominated on the line number and the Piping Line List 2000-F00-LST-PP-10001, except for safety shower water pipelines, which will be insulated as directed by the Superintendent on site. Refer to Insulation of Tanks and Piping Specification 2000-F00-STS-ME-10005

Generally, safety shower pipelines will only require insulation where the flow through the line is not continuous, and the line is exposed to direct sunlight.

All insulation will be installed after successful completion of service testing and acceptance by the Superintendent.

Suffix PP – Insulation for Personnel Protection

Insulation or guarding of pipes for personnel protection shall be designed such that the surface temperature of the insulation or guarding shall not exceed 60 °C.

Suffix ST – Steam Heat Tracing

The requirement for heat tracing on individual pipelines is nominated on the line number and the Piping Line List. Refer standard piping drawings for details.

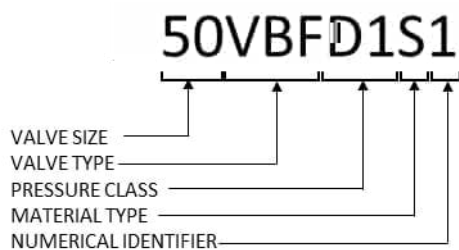
Suffix ET – Electrical Heat Tracing

The requirement for heat tracing on individual pipelines is nominated on the line number and the Piping Line List. Refer standard piping drawings for details.

5.2 VALVE DESIGNATION

5.3 MANUAL VALVES

Manual Valves will be designated by their size and valve type as follows:



Examples:

150VBF D1S1 DN100 Size, VBF Butterfly Valve, D1 AS 2129 Table D, S Stainless Steel, 1 Valve Type Identifier

5.3.1 Valve Size

The valve size is a numeric identifier indicating the valve connection size reflective of the nominal diameter (DN).

5.3.2 Valve Type

A valve identifier comprising of two or three letters as follows:

Valve Type	DEFINITION
RV	Relief Valve
TV	Tech Taylor Valve
VA	Air Release / Vacuum Breaker Valve
VB	Ball Valve
VBF	Butterfly Valve
VC	Check Valve
VD	Diaphragm Valve
VDT	Dart Valve
VG	Gate Valve
VGL	Globe Valve
VF	Float Valve
VKG	Knife Gate Valve
VN	Needle Valve
VP	Valve Pressure / Sustaining
VPI	Pinch Valve
VPL	Plug Valve
VR	Rotary Valve
VTW	3 Way Valve
FLV	Flushing Water Valve
GLV	Gland Water Valve
SCV	Scour/Drain Valve

Table 5-4 - Valve Type Identifiers

5.3.3 Pressure Class

Numeric identifier comprising of two digits representing the valve class is standard and defined as follows:

Pressure Code	VALVE CLASS
01	Socket or Threaded or ASME 150#
03	Socket or Threaded or ASME 300#
09	Threaded or ASME 900#
D1	AS2129 Table D / AS4087 PN14
E1	AS2129 Table E / AS4087 PN18
F1	AS2129 Table F / AS4087 PN21
H1	AS2129 Table H / AS4087 PN35

Table 5-5 - Pressure Rating Identifiers

5.3.4 Material Type

An alpha identifier comprising of one letter representing the valve material is defined as follows:

Material Code	DEFINITION
A	ABS
C	Carbon steel
D	Duplex stainless steel
F	Iron (Ductile & Grey)
G	Galvanised steel
H	Polypropylene
K	Low alloy steel
L	Lined steel/iron
N	Nickel alloy steel
P	HDPE
R	FRP
S	Stainless steel
T	Plastic - Other (PP, PVDF, Nylon)
V	PVC

Table 5-6 - Material Code Identifier

5.3.5 Numerical Identifier

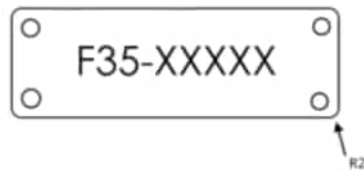
A one-digit (numeric) identifier will be applied for each valve with a certain valve type, pressure class and material type, corresponding to the valve's datasheet.

5.3.6 Unique Valve Identifier

For manual valves, a unique five-digit identifier (alpha-numeric MV-XXXXX) starting from 10,001 - 19,999 will be applied for each unique manual valve and will be used for the valve tag. Unique ID's will be captured on the respective P&IDs. Wherever possible, consecutive sequence numbers in the direction of flow. Suffixes shall not be used.

A permanently attached stainless steel tag(s), minimum 0.9mm thick, sharp edges removed, and stamped with the following information in lettering as per Figure 5-2, shall be provided on each manual valve:

Tagging structure
WBS-5 DIGIT UNIQUE No



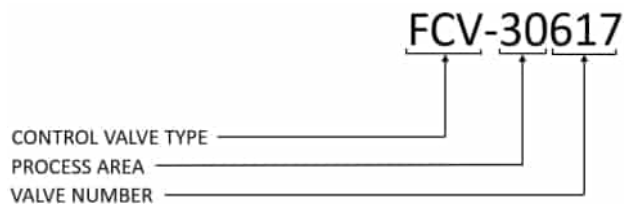
0.9mm Thick 316 Gr Stainless Steel
4 off 5mm holes, suitable for 4.6mm cable ties
5mm high lettering, German Gothic Text
Engraved, Black enamel filled lettering
Rounded corners
4.6mm 316 Stainless cables ties, or 316 braided wire
with swaged joint used to affix to valve or pipework

Figure 5-2: Manual Valve Tag

5.4 ACTUATED (CONTROL) VALVES

Actuated Valves types will be identified by means of pictorial representations on the drawings and are identified by Actuated Valve Type Identifier and a five-digit Unique Identifier as indicated on the P&IDs.

Example of control valve tag number as follows:



Example:

FCV-30617 FCV Flow Control Valve, 30 Concentrate Pre-Leach Thickening, CIL2/3 & Post-Leach Thickening, 617 Valve Number

5.4.1 Control Valve Type

Identifier codes for different types of control valves are indicated in the following table:

Identifier Code	DEFINITION	COMMENTS
XV	Isolation Valve (Open / Close)	Open/Close Actuation for isolation
SOV	Solenoid Valve (Open / Close)	Open/Close Actuation for isolation
DCV	Density Control Valve	Modulating actuation for density control
FCV	Flow Control Valve	Modulating actuation for flow control
LCV	Level Control Valve	Modulating actuation for level control
PCV	Pressure Control Valve	Modulating actuation for pressure control
PRV	Pressure Relief Valve	Automatically opens and relieves pressure at a set point
TCV	Temperature Control Valve	Modulating actuation for temperature control

Table 5-7 - Control Valve Identifier

5.4.2 Process Area

The process area codes are a two-digit (numeric) identifier as per Plant Numbering Specification 2000-F00-STS-GE-10003

5.4.3 Valve Number

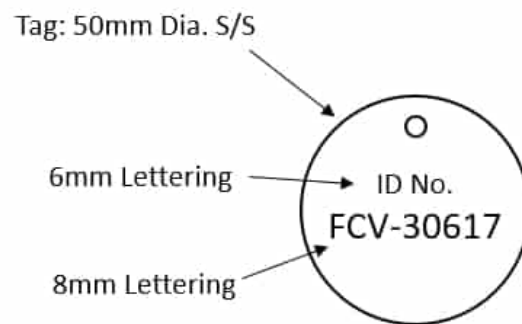
A three-digit (numeric) identifier will be applied, sequential for the in respect to Process Area. Wherever possible, consecutive sequence numbers in the direction of flow. Suffixes shall not be used.

New valves located within existing Process Areas shall commence with the number 401 to avoid potential duplication of existing valve numbers.

5.4.4 Unique Valve Identifier

For control valves, the unique identifier as described in Section 5.4 will be used for the valve tag.

A permanently attached stainless steel tag(s), 50mm diameter, minimum 1.5mm thick, sharp edges removed, and stamped with the following information in lettering as sized below, shall be provided on each actuated valve:



Note: Tag shall be permanently fixed to each valve via 1.6mm – 2.0mm stainless steel wire rope with swaged joint or similar as approved by the Principal.

Figure 5-3: Control Valve Tag

5.4.5 Pipe Special Designation

The following identification system will be used for piping specialty items. Special Piping Item technical details shall be captured on the project specific Specials Piping Items (SPI) List:

SP-WW-XXX SP prefix, WW Pipe Special Code, XXX Unique Sequential Number (common for the same Special Piping Items)

Special Piping Items shall be identified in common groups such that a total quantity can be applied in the Pipe Specials List. To be in a common group, the Pipe Special should have the same diameter, pressure rating, end connections, design requirements and materials of construction.

5.4.6 Pipe Special Code

The pipe special code provides reference to the specific type of pipe special defined as follows:

Pipe Special Type	DEFINITION
EB	Expansion Bellows
FR	Filter / Regulator
FS	Miscellaneous Fire Services Items
HS	Hose
MC	Minsup Coupling
MS	Miscellaneous Items
MV	Maric Valve
OP	Orifice Plate
SC	Straub Coupling
SP	Spray Nozzle
SS	Special Spool
ST	Strainer
VC	Victaulic Coupling

Table 5-8 - Pipe Special Type Identifiers

For example, all project DN25 Y-strainers for a low-pressure raw water application, BSP threaded with the same mesh size would be SP-ST-001 while a size change to DN50 would be SP-ST-002. The description of the item is then covered in the Pipe Specials List.

Examples:

SP-ST-001 Special Pipe Item - Strainer - 001

SP-OP-002 Special Pipe Item – Orifice Plate - 002

6 DESIGN AND MATERIAL REQUIREMENTS

6.1 GENERAL

The materials selection for all piping components are based on a design life as specified on the Piping System Datasheet in Appendix A.

For components/items where the material quality is not specified, the Supplier shall select materials with equal or better mechanical properties and corrosion resistance than the component they are connected to and shall be subject to approval by the Superintendent.

Pipe, valves and fittings in sizes DN32, DN65, DN90, DN125, DN550, DN650, DN700, DN800 and DN850 are not preferred. Where equipment is normally provided by a supplier with connections in any of the above sizes, connections will be made with suitable swages or reducers to pipe of approved size. Refer to Table 6-1 for equivalent HDPE piping and non-preferred sizing.

All piping component wall thicknesses specified in the individual Piping System Datasheets are calculated in accordance with the Design Standard and are based on the internal design pressure, design temperature, specified corrosion allowance, the manufacturing under tolerance and the threading allowance as applicable. All piping subject to any additional loading such as external pressure, thermal loading, live loads, etc. shall be individually assessed to ensure full compliance with the Design Standard.

Steel pipe (dn)	EQUIVALENT HDPE (OD)
15*	20*
20*	25*
25	32
32*	40*
40	50
50	63
65*	75*
80	90
90*	No Equivalent
100	110
125*	125*
	140*
150	160
	180*
200	200*
	225
250	250*
	280
300	315
350	355
400	400
450	450
500	500
550*	560*
600	630
650*	No Equivalent
700*	710
750	No Equivalent
800*	800
850*	No Equivalent
900	900
1050	1000
1200	1200

Table 6-1 – Standard Pipe Sizes and Equivalent HDPE

*Denotes pipe sizes that are non-preferred and shall be avoided.

6.2 SPECIFIC MATERIAL REQUIREMENTS

Material substitutions may only be made with prior approval in writing from the Superintendent.

Harmful and/or toxic materials illegal for import into Australia, such as asbestos, shall not be used.

No cast iron, ductile iron, malleable iron, aluminium, plastic or copper bearing alloy shall be used in hydrocarbon or hydrogen services.

Copper, copper-based alloys, galvanised materials, cadmium plated materials, etc. shall not be exposed to hydrogen sulphide, amine, caustic or similar materials.

6.3 MANUAL VALVES

All manual valves shall comply with the requirements of their associated Valve Datasheet in Appendix B.

6.4 CONTROL VALVES

All control valves shall comply with the requirements of the Control Valve List 2000-F00-LST-PP-10003. The valve list will stipulate control philosophy for valves and allocate a manual valve code for the valve to conform to materials, pressure rating and temperature rating. Refer to Section 6.3 for the requirements.

6.5 SPECIAL PIPING ITEMS

Special piping items will be separately listed and detailed on the Special Piping Items List 2000-F00-LST-PP-10005. All special piping items shall comply with the requirements specified.

6.6 PIPE AND FITTINGS

All pipe and fittings shall comply with the requirements of the associated Piping System Datasheet.

Longitudinal welded pipe where specified on the Piping System Datasheet may be substituted by seamless pipe of the same material grade as specified on the Piping System Datasheet, but not vice versa. Seamless pipe may only be substituted by longitudinal welded pipe with prior written approval from the Superintendent. All pipe or fittings manufactured by welding, shall be subject to 100% radiography to achieve a joint efficiency factor of 1.0.

6.7 FLANGES, SPECTACLE BLINDS, SPADES AND SPACERS

All standard flanges shall be in accordance with AS 2129 and as specified on the Piping System Datasheet.

Details of flanges that mate to a higher rated specification, (i.e. Specification breaks at valves, control valves, etc.) shall be as per the Piping System Datasheet except that the rating and facing shall be specified to match the higher rated specification.

Flat face flanges are preferred. Raised Face to be used only as required e.g. Valves, Instrumentation, Mechanical Packages. Raised Face connecting to Flat Face is not allowed. Flat face flanges shall be used when mating to cast bronze, cast iron and HDPE flanges. Lap joint flanges and raised face flanges shall not be used for these materials. Additionally, when mating to grey cast iron flanges, full face gaskets shall be used and be made of ASME B16.5 Annex C, Group No. 1a materials.

Spectacle blinds, spades and spacers shall be supplied in accordance with AS 2129.

6.8 STUDS, NUTS AND GASKETS

Stud bolts and nuts shall be the type specified on the relevant Piping System Datasheet.

M32 stud bolts and above shall have their length increased by 1.5 additional bolt diameter to accommodate bolt tensioning equipment.

Gaskets shall be the type specified in Table 4-1 and for dimensions refer to Piping System Datasheet.

Where specified, gasket thickness refers to uncompressed thickness.

6.9 PIPE THREADS

The pipe thread form shall be in accordance with the Piping System Datasheet.

All threaded fittings shall be installed with PTFE (Poly tetra fluoro ethylene) tape and Loctite 577 thread sealant (or equivalent approved by the Superintendent) excluding Potable Water.

Threaded fittings in Potable Water service shall use a Watermark approved thread sealant.

6.10 BRANCH CONNECTIONS

Branch connections shall comply with the branch schedule specified on the Piping System Datasheet.

6.11 WELDED ATTACHMENTS

All direct welded attachments to piping (i.e. for support services) shall be of compatible material and shall be identified and traced as per the piping material.

6.12 PIPING FABRICATION AND INSTALLATION

For details of fabrication, installation and NDE, refer to Piping Fabrication Specification 2000-F00-STS-PP-10002.

6.13 EXTERNAL SURFACE PROTECTION

For details of external surface protection requirements, refer to Technical Specification for Protective Coating SPEC-00-G-05.

6.14 INSULATION

For details of piping insulation requirements refer to Insulation of Tanks and Piping Specification 2000-F00-STS-ME-10005.

6.15 SEMI-RIGID TUBING

Semi-rigid tubing (i.e., Nylon) shall be used for instrument air supply to pneumatically actuated equipment and supply of gland water to pumps. Semi-rigid tubing shall be continuously supported where possible (i.e., cable ladder or inside conduit) for longer runs. Short runs less than 1.5m, such as instrument air connections between control valve and air header, can be point to point and unsupported. Fittings for semi-rigid tubing shall be SMC, KQ2 series or approved equivalent.

6.16 SERVICE IDENTIFICATION

Service identification and flow directional arrows are required on all pipes as per AS 1345.

Refer to Technical Specification for Protective Coating SPEC-00-G-05 Appendix B-3 for colour and identification requirements.

7 PIPE INTERNAL LININGS

7.1 RUBBER LINING

Rubber lining installation requirements are as per Rubber Lining Specification 2000-F00-STS-ME-10003.

The rubber lining material, thickness and hardness are specified on the Piping System Datasheets for lined pipe systems and shall be in accordance with the respective datasheet and standard drawings unless otherwise specified.

Thickness of rubber facing on flanges shall be 3.0 mm unless otherwise specified.

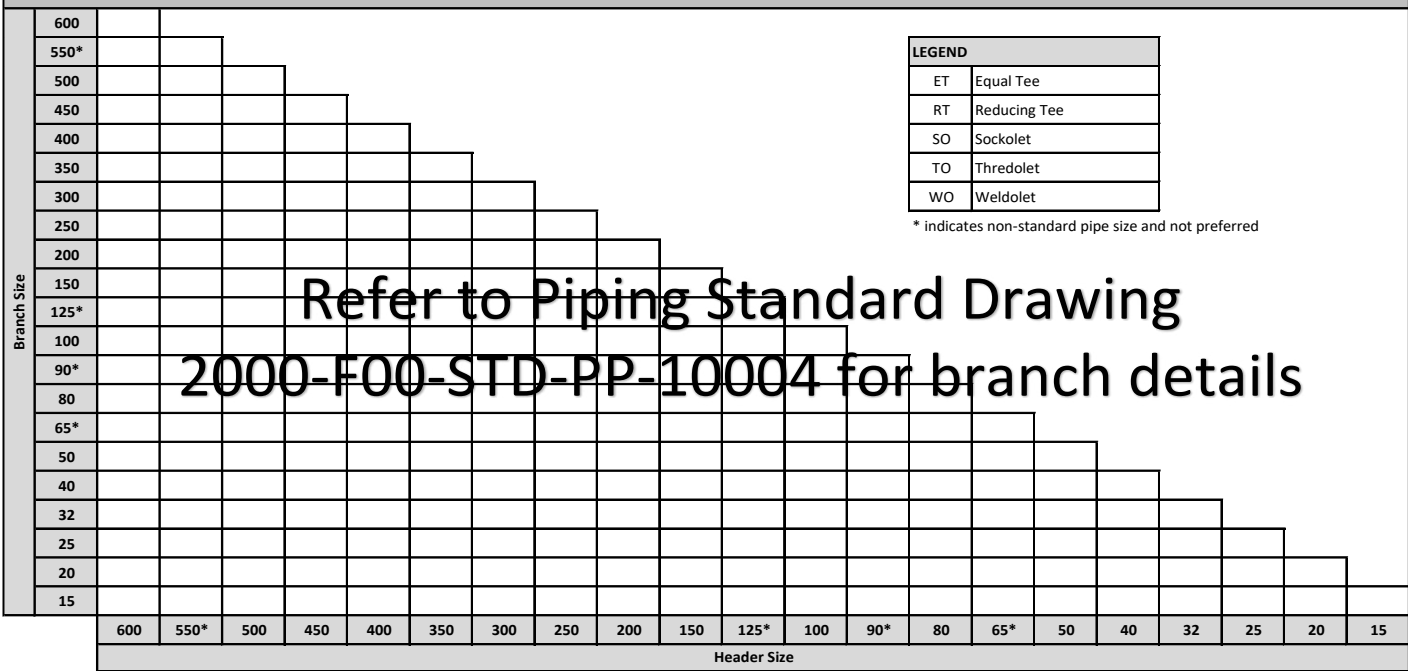
Rubber lined spools shall be maximum 6 m long for greater than DN80 in size, and maximum 3 m long for DN80.

APPENDIX A – PIPING DATASHEETS

CL1	Fully flanged spun ceramic lined steel piping system with flat face flanges to AS2129 Table D for low pressure slurry applications.				
Material :		Carbon Steel - Spun Ceramic Lined – 12mm Thick Lining ⁴			
Design Codes :		AS 4041, Class 3 (Note 11)			
Design Temperature (°C) ¹ :	0	70			
Design Pressure (kPa(g)) ¹ :	700	700			
Corrosion Allowance (mm) :	0.0				
Pressure Testing :	Initial service leak testing as per AS 4041. Hydrostatic pressure testing not required unless directed by the Engineer.				
Non-Destructive Testing :	Visual inspection only				
Description	Size Range	Dimensional Specification	Type	End Preparations	Material Specification
Pipe	DN50 ³	ASME B36.10M	STD WT, SMLS	Plain ends	ASTM A53 Gr B / API 5L Grade B
	DN65 ³ – 600		STD WT, ERW		
Induction Bend, 3D or 5D ⁸	DN80 – 600	As per pipe ²	As per pipe	Plain ends	As per pipe
Elbow, 1.5D ⁹	DN80 – 600	ASME B16.9 ²	Butt Weld	Plain ends	ASTM A234 WPB
Lateral, 45°	DN80 – 600	As per pipe ²	As per pipe	Plain ends	As per pipe
Lateral, 60°					
Wye, 90°	DN80 – 600	As per pipe ²	As per pipe	Plain ends	As per pipe
Wye, 60°					
Tees ⁵	DN65 ³ – 600	As per pipe ²	As per pipe	Plain ends	As per pipe
Reducer, eccentric	DN50 ³ – 600	ASME B16.9 ²	As per large end piping	Plain ends	ASTM A234 WPB
Reducer, concentric	DN50 ³ – 600				
Flange, Slip-On	DN50 ³ – 600	AS 2129 ²	Table D, FF	Plate	ASTM A105
Flange, Spinning	DN50 ³ – 600			Backing Ring c/w Shoulder Plate	AS/NZS 3678 Gr 250
Flange, Blind	DN50 ³ – 600			Blank	ASTM A105
Pipe Nipple ³	DN15 – 50	ASTM A733	XS WT, SMLS, 100mm	Threaded, BSP	ASTM A53 Gr B
Plugs ³	DN15 – 50	ASME B16.11	Class 3000	Threaded, BSP	ASTM A105
Stud Bolts, Type A	DN50 – 600	AS/NZS 1252.1	Galvanised, Gr. 8.8 bolts		AS 4291.1
Machine Bolts, Metric ⁶	DN50 – 600		Galvanised, Hex head, Gr. 8.8 bolts		
Nuts	DN50 – 600		Galvanised, Gr. 8 hex nuts		AS 4291.2
Gaskets	DN50 – 600	AS 2129	Table D, Klinger C4430 or approved equivalent, 1.5mm thick, flat face		
Ceramic Lining	DN50 – 600	12mm spun ceramic. Refer to Note 4 for lining thickness of fixed flanges.			

1	The design pressure and temperature conditions are based on plate flange design, STD WT fittings with no branch reinforcement and the maximum temperature limitation of the specified rubber lining. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	Refer to design drawings and the Standard Piping Drawings for dimensional details for fittings and plate flanges.
3	Pipe diameters less than DN80 shall not be used for main pipe runs and shall only be permitted for branch connections for drains, flushing points or instrumentation connections unless approved by the engineer.
4	No lining on flange faces. For details refer to the Standard Piping Drawings.
5	For preferred branch details refer to the Standard Piping Drawings. Use only laterals or Y-piece fittings for branches except for flushing connections, vents or instrument connections.
6	Not Used
7	Length of flanged spools shall be 3m maximum for DN80 and 6m maximum for DN100 and above.
8	Induction 3D bends to be used as priority. Induction 5D bends shall only be used with the approval by the Engineer.
9	1.5D elbows shall only to be used with approval by the Engineer.
10	For external surface treatment requirements refer to the Project Pipe Materials and Valve Specification.
11	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343.

BRANCH CHART



ECTFEDC1	Cascade butt weld / induction weld ECTFE SDR 21 piping system and secondary contained PE100 SDR17 outer piping with AS2129 Table E flanges for low pressure and high temperature reagent service applications										
Material :		Ethylene chlorotrifluroethylene PN16 (SDR21) - Extruded Pipe									
Design Codes :		AS 4041, Class 3									
Design Life (Yr) :		50	50	50	25	25	25	25	15		
Design Temperature (°C) ¹ :		0	10	30	45	60	70	80	90		
Design Pressure (kPa(g)) ¹ :		1,000	1,000	1,000	800	600	490	390	300		
Corrosion Allowance (mm) :		0 mm									
Pressure Testing :		930 kPa(g) @ 30 °C Hydrostatic Pressure Test as per manufacturers recommendation									
Non-Destructive Testing :		100% Visual Inspection									
Description		Size Range		Dimensional Specification		Type		End Preparations		Material Specification	
Pipe ⁵		OD90 ³		Manufacturer Std ²		Manufacturer Std ²		Plain end		Manufacturer Std ²	
Elbow, 90°, Short Radius		OD90 ³		Manufacturer Std ²		Manufacturer Std ²		Plain end		Manufacturer Std ²	
Elbow, 45°		OD90 ³		Manufacturer Std ²		Manufacturer Std ²		Plain end		Manufacturer Std ²	
Tee, equal ⁴		OD90 ³		Manufacturer Std ²		Manufacturer Std ²		Plain end		Manufacturer Std ²	
Cap		OD90 ³		Manufacturer Std ²		Manufacturer Std ²		Plain end		Manufacturer Std ²	
Reducer, concentric		OD90 ³		Manufacturer Std ²		Manufacturer Std ²		Plain end		Manufacturer Std ²	
Dog Bone, Restraint, with cutout		OD90 ³		Manufacturer Std ²		Manufacturer Std ²		Plain end		Manufacturer Std ²	
Dog Bone, Restraint, without cutout		OD90 ³		Manufacturer Std ²		Manufacturer Std ²		Plain end		Manufacturer Std ²	
Dog Bone, Transition to Single Tube ⁸		OD90 ³		Manufacturer Std ²		Manufacturer Std ²		Plain end		Manufacturer Std ²	
Measuring Fitting		OD90 ³		Manufacturer Std ²		Manufacturer Std ²		Plain end		Manufacturer Std ²	
Induction E-Coupler		OD90 ³		Manufacturer Std ²		Manufacturer Std ²		Plain end		Manufacturer Std ²	
Stub Flange c/w groove for O-ring & Stainless Steel backing ring		OD90 ³		Manufacturer Std ² AS 2129 ⁶		SDR21 Stub, Table E backing ring		Plain end		Manufacturer Std ² , ASTM A240 GR 316L	
Stub Flange c/w Stainless Steel backing ring ⁸		OD90 ³		Manufacturer Std ² AS 2129 ⁶		SDR21 Stub, Table E backing ring		Plain end, short spigot		Manufacturer Std ² , ASTM A240 GR 316L	
Flange, blind		DN100		AS 2129 ⁶		Table E, flat face (blind), Stainless Steel			ASTM A105		
Machine Bolts, Metric		DN100		ISO 965-1		304SS, Hex Head, Gr. A2-70 bolts			ISO 3506-1		
Stud Bolts		DN100				304SS, Gr. A2-70 bolts					
Nuts		DN100				304SS, Gr. A2-70 hex nuts			ISO 3506-2		
O-ring/Seal		OD90 ³		Manufacturer Std ²		FPM O-rings					
Gasket ⁸		DN100		AS 2129		Table E, Klinger TC2003 or approved equivalent, 1.5mm thick, flat face					
Thread Sealant		DN15 – 50		-		Thread tape or liquid sealant approved to AS4020, WaterMark Certified for Potable Water service.				PTFE	

NOTES													
1	The design pressures stated are permissible for liquids only and are based on temperature deratings for the specified ECTFE pipe material in accordance with ISO 10931 for the specified design life. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.												
2	Preferred manufacturer is Agru Double Containment Piping												
3	Refer to inner pipe and corresponding outer pipe (containment pipe) below <table> <tr> <th>Inner Pipe (OD mm)</th><th>Outer Pipe (OD mm)</th></tr> <tr> <td>32</td><td>90</td></tr> <tr> <td>63</td><td>125</td></tr> <tr> <td>90</td><td>160</td></tr> <tr> <td>110</td><td>200</td></tr> <tr> <td>160</td><td>280</td></tr> </table>	Inner Pipe (OD mm)	Outer Pipe (OD mm)	32	90	63	125	90	160	110	200	160	280
Inner Pipe (OD mm)	Outer Pipe (OD mm)												
32	90												
63	125												
90	160												
110	200												
160	280												
4	Moulded, machined or formed fittings shall be used as a priority for this piping system. Special fittings requiring fabrication such as reducing tees and laterals may not achieve the design pressure ratings of this pipe system and shall only be permitted with approval by the Engineer.												
5	Pipe is only available in maximum 5m lengths												
6	Thickness to manufacturer standards, drilling to AS 2129 Table E.												
7	Not used												
8	To transition from Dual Contained to Single Wall piping systems, use Dog Bone, Transition to Single Tube.												
9	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343.												

MS1	Welded carbon steel piping system with flanges to AS2129 Table D for low pressure liquid or gas applications.					
Material :		Carbon Steel				
Design Codes :		AS 4041, Class 3 (Note 8)				
Design Temperature (°C) ¹ :		- 50	232	250	275	300
Design Pressure (kPa(g)) ¹ :		700	700	650	600	570
Corrosion Allowance (mm) :		1.5				
Pressure Testing :		Initial service leak testing as per AS 4041. Hydrostatic pressure testing not required unless directed by the Engineer.				
Non-Destructive Testing :		Visual inspection only				

Description	Size Range	Dimensional Specification	Type	End Preparations	Material Specification
Pipe	DN25 – 50	ASME B36.10M	STD WT, SMLS	Plain Ends	ASTM A53 Gr B / API 5L Gr B
	DN65 – 600		STD WT, ERW	Bevelled Ends	
	DN650 – 900		STD WT, LSAW		
Elbow, 90°	DN25 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A105
	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB
Elbow, 45°	DN25 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A105
	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB
Tee, equal	DN25 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A105
	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB
Tee, reducing	DN32 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A105
	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB
Cap	DN25 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A105
	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB
Coupling, reducing	DN32 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A105
Reducer, eccentric	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB
Reducer, concentric	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB
Flange, Plate, Slip-On	DN25 – 900	AS 2129	Table D, FF	Plate	AS 3678 GR250
Flange, Blind	DN25 – 900			Blank	
Flange, Slip-On	DN25 – 900	AS 2129	Table D, RF ⁷	Plate	AS 3678 GR250
Flange, Blind	DN25 – 900			Blank	
Thredolet ⁵	DN15 – 20 ⁴	MSS-SP-97	Class 3000	Threaded, BSP	ASTM A105
Sockolet	DN25 – 50 ⁴	MSS-SP-97	Class 3000	Socket Weld	ASTM A105
Weldolet	DN65 – 350 ⁴	MSS-SP-97	STD WT	Bevelled	ASTM A105
Pipe Nipple ⁵	DN15 – 50	ASTM A733	XS WT, SMLS, 100mm	Threaded, BSP	ASTM A53 Gr B
Hex Nipple ⁵	DN15 – 50	ASME B16.11	Class 3000	Threaded, BSP	ASTM A105
Hex Head Bushing ⁵	DN15 – 50	ASME B16.11	Class 3000	Threaded, BSP	ASTM A105
Hex Head Plug ⁵	DN15 – 50	ASME B16.11	Class 3000	Threaded, BSP	ASTM A105

MS2	Welded carbon steel piping system with flanges to AS2129 Table E for medium pressure liquid or gas applications.					
Material :		Carbon Steel				
Design Codes :		AS 4041, Class 3 (Note 7)				
Design Temperature (°C) ¹ :		- 50	232	250	275	300
Design Pressure (kPa(g)) ¹ :		1400	1400	1300	1200	1100
Corrosion Allowance (mm) :		1.5				
Pressure Testing :		Initial service leak testing as per AS 4041. Hydrostatic pressure testing not required unless directed by the Engineer.				
Non-Destructive Testing :		Visual inspection only				

Description	Size Range	Dimensional Specification	Type	End Preparations	Material Specification
Pipe	DN25 – 50	ASME B36.10	STD WT, SMLS	Plain Ends	ASTM A53 Gr B / API 5L Gr B
	DN65 – 600		STD WT, ERW	Bevelled Ends	
	DN650 – 900		STD WT, LSAW		
Elbow, 90°	DN25 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A105
	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB
Elbow, 45°	DN25 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A105
	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB
Tee, equal	DN25 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A105
	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB
Tee, reducing	DN32 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A105
	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB
Cap	DN25 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A105
	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB
Coupling, reducing	DN32 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A105
Reducer, eccentric	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB
Reducer, concentric	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB
Flange, Plate, Slip-On	DN25 – 900	AS 2129	Table E, FF	Plate	AS 3678 GR250
Flange, Blind	DN25 – 900			Blank	
Flange, Slip-On	DN25 – 900	AS 2129	Table E, RF ⁷	Plate	AS 3678 GR250
Flange, Blind	DN25 – 900			Blank	
Threadolet ⁵	DN15 – 20 ⁴	MSS-SP-97	Class 3000	Threaded, BSP	ASTM A105
Sockolet	DN25 – 50 ⁴	MSS-SP-97	Class 3000	Socket Weld	ASTM A105
Weldolet	DN65 – 350 ⁴	MSS-SP-97	STD WT	Bevelled	ASTM A105
Pipe Nipple ⁵	DN15 – 50	ASTM A733	XS WT, SMLS, 100mm	Threaded, BSP	ASTM A53 Gr B
Hex Nipple ⁵	DN15 – 50	ASME B16.11	Class 3000	Threaded, BSP	ASTM A105
Hex Head Bushing ⁵	DN15 – 50	ASME B16.11	Class 3000	Threaded, BSP	ASTM A105
Hex Head Plug ⁵	DN15 – 50	ASME B16.11	Class 3000	Threaded, BSP	ASTM A105

MSG1	Welded galvanised carbon steel piping system with flanges to AS2129 Table D for low pressure liquid or gas applications.					
Material :		Carbon Steel, Hot Dipped Galvanised				
Design Codes :		AS 4041, Class 3 (Note 8)				
Design Temperature (°C) ¹ :		- 50	200			
Design Pressure (kPa(g)) ¹ :		700	700			
Corrosion Allowance (mm) :		1.0				
Pressure Testing :		Initial service leak testing as per AS 4041. Hydrostatic pressure testing not required unless directed by the Engineer.				
Non-Destructive Testing :		Visual inspection only				
Description		Size Range	Dimensional Specification	Type	End Preparations	Material Specification
Pipe	DN25 – 50		ASME B36.10	STD WT, SMLS	Plain Ends	ASTM A53 Gr B ² / API 5L Gr B ²
	DN65 – 600			STD WT, ERW	Bevelled Ends	
	DN650 – 900			STD WT, LSAW		
Elbow, 90°	DN25 – 50		ASME B16.11	Class 3000	Socket Weld	ASTM A105 ²
	DN65 – 900		ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB ²
Elbow, 45°	DN25 – 50		ASME B16.11	Class 3000	Socket Weld	ASTM A105 ²
	DN65 – 900		ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB ²
Tee, equal	DN25 – 50		ASME B16.11	Class 3000	Socket Weld	ASTM A105 ²
	DN65 – 900		ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB ²
Tee, reducing	DN32 – 50		ASME B16.11	Class 3000	Socket Weld	ASTM A105 ²
	DN65 – 900		ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB ²
Cap	DN25 – 50		ASME B16.11	Class 3000	Socket Weld	ASTM A105 ²
	DN65 – 900		ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB ²
Coupling, reducing	DN32 – 50		ASME B16.11	Class 3000	Socket Weld	ASTM A105 ²
Reducer, eccentric	DN65 – 900		ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB ²
Reducer, concentric	DN65 – 900		ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB ²
Flange, Slip-On	DN25 – 900		AS 2129	Table D, FF	Plate	ASTM A105 ²
Flange, Blind	DN25 – 900				Blank	
Flange, Slip-On	DN25 – 900		AS 2129	Table D, RF ⁷	Plate	ASTM A105 ²
Flange, Blind	DN25 – 900				Blank	
Thredolet ⁵	DN15 – 20 ⁴		MSS-SP-97	Class 3000	Threaded, BSP	ASTM A105 ²
Sockolet	DN25 – 50 ⁴		MSS-SP-97	Class 3000	Socket Weld	ASTM A105
Weldolet	DN65 – 350 ⁴		MSS-SP-97	STD WT	Bevelled	ASTM A105
Pipe Nipple ⁵	DN15 – 50		ASTM A733	XS WT, SMLS, 100mm	Threaded, BSP	ASTM A53 Gr B ⁹
Hex Nipple ⁵	DN15 – 50		ASME B16.11	Class 3000	Threaded, BSP	ASTM A105 ⁹
Hex Head Bushing ⁵	DN15 – 50		ASME B16.11	Class 3000	Threaded, BSP	ASTM A105 ⁹
Hex Head Plug ⁵	DN15 – 50		ASME B16.11	Class 3000	Threaded, BSP	ASTM A105 ⁹

Description	Size Range	Specification	Type	End Preparations	Specification
Stud Bolts, Type A	DN25 – 900	AS/NZS 1252.1	Galvanised, Gr. 8.8 bolts		AS 4291.1
Machine Bolts, Metric ⁶	DN25 – 900		Galvanised, Hex head, Gr. 8.8 bolts		
Nuts	DN25 – 900		Galvanised, Gr. 8 hex nuts		AS 4291.2
Gaskets ¹⁰	DN25 – 900	AS 2129	Table D, Klinger C4430 or approved equivalent, 1.5mm thick, flat face		
			Table D, Klinger TC2003 or approved equivalent, 1.5mm thick, flat face		

NOTES

1	The design pressure and temperature conditions are based on AS2129 flange ratings and material type. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	Welded spools and headers to be hot dip galvanised after fabrication. All hot dip galvanising to be in accordance with AS/NZS 4680.
3	The maximum length of flanged spools shall be 6 m for DN80 and 12 m maximum for DN100 and above.
4	Sizing refers to maximum branching size allowed. Refer to branching chart for more detail
5	This specification is strictly to be welded fittings only. Threaded fittings only to be used for instrumentation or tie-in for mechanical packages where required. PTFE Thread tape or liquid sealant to be used.
6	Not Used
7	Flat Face is preferred. Raised Face to be used only as required e.g. Valves, Instrumentation, Mechanical Packages. Raised Face connecting to Flat Face is not permitted.
8	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343. Pipe system not suitable for Potable Water.
9	Threaded fittings are to be hot-dip galvanised as per the requirements of ASTM A123.
10	TC2003 to be used on all reagent services. Refer to Project Pipe Materials and Valve Specification for gasket selection

BRANCH CHART

[illegible]

MSG2	Welded galvanised carbon steel piping system with flanges to AS2129 Table E for medium pressure liquid or gas applications.				
Material :		Carbon Steel, Hot Dipped Galvanised			
Design Codes :		AS 4041, Class 3 (Note 8)			
Design Temperature (°C) ¹ :		- 50	200		
Design Pressure (kPa(g)) ¹ :		1400	1400		
Corrosion Allowance (mm) :		1.0			
Pressure Testing :		Initial service leak testing as per AS 4041. Hydrostatic pressure testing not required unless directed by the Engineer.			
Non-Destructive Testing :		Visual inspection only			
Description	Size Range	Dimensional Specification	Type	End Preparations	Material Specification
Pipe	DN25 – 50	ASME B36.10	STD WT, SMLS	Plain Ends	ASTM A53 Gr B ² / API 5L Gr B ²
	DN65 – 600		STD WT, ERW	Bevelled Ends	
	DN650 – 900		STD WT, LSAW		
Elbow, 90°	DN25 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A105 ²
	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB ²
Elbow, 45°	DN25 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A105 ²
	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB ²
Tee, equal	DN25 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A105 ²
	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB ²
Tee, reducing	DN32 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A105 ²
	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB ²
Cap	DN25 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A105 ²
	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB ²
Coupling, reducing	DN32 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A105 ²
Reducer, eccentric	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB ²
Reducer, concentric	DN65 – 900	ASME B16.9	STD WT	Bevelled Ends	ASTM A234 Gr WPB ²
Flange, Slip-On	DN25 – 900	AS 2129	Table E, FF	Plate	ASTM A105 ²
Flange, Blind	DN25 – 900			Blank	
Flange, Slip-On	DN25 – 900	AS 2129	Table E, RF ⁷	Plate	ASTM A105 ²
Flange, Blind	DN25 – 900			Blank	
Thredolet ⁵	DN15 – 20 ⁴	MSS-SP-97	Class 3000	Threaded, BSP	ASTM A105 ²
Sockolet	DN25 – 50 ⁴	MSS-SP-97	Class 3000	Socket Weld	ASTM A105
Weldolet	DN65 – 350 ⁴	MSS-SP-97	STD WT	Bevelled	ASTM A105
Pipe Nipple ⁵	DN15 – 50	ASTM A733	XS WT, SMLS, 100mm	Threaded, BSP	ASTM A53 Gr B ⁹
Hex Nipple ⁵	DN15 – 50	ASME B16.11	Class 3000	Threaded, BSP	ASTM A105 ⁹
Hex Head Bushing ⁵	DN15 – 50	ASME B16.11	Class 3000	Threaded, BSP	ASTM A105 ⁹
Hex Head Plug ⁵	DN15 – 50	ASME B16.11	Class 3000	Threaded, BSP	ASTM A105 ⁹

MSG3	Viega Megapress (cold press) galvanised carbon steel piping system with flanges to AS2129 Table E for medium pressure liquid or gas applications.				
Material :		Carbon Steel, Hot Dipped Galvanised Pipe, Zinc -Nickle Coated Fittings			
Design Codes :		AS 4041, Class 3 (Note 8)			
Design Temperature (°C) ¹ :		-5	140		
Design Pressure (kPa(g)) ¹ :		1400	1400		
Corrosion Allowance (mm) :		1.0			
Pressure Testing :		Initial service leak testing as per AS 4041 ⁷ . Hydrostatic pressure testing not required unless directed by the Engineer.			
Non-Destructive Testing :		Visual inspection only			
Description	Size Range	Dimensional Specification	Type	End Preparations	Material Specification
Pipe	DN15 – 50	ASME B36.10	Sch 40, SMLS	Plain Ends ³	ASTM A53 Gr B ² / API 5L Gr B ²
Elbow, 90°	DN15 – 50	Manufacturers Standard	Model 4316	Press Connector	Manufacturer Std
			Model 4316.1	Plain End x Press Connector	
Elbow, 45°	DN15 – 50	Manufacturers Standard	Model 4326	Press Connector	Manufacturer Std
			Model 4326.1	Plain End x Press Connector	
Tee, equal ⁴	DN15 – 50	Manufacturers Standard	Model 4318	Press Connector	Manufacturer Std
			Model 4317.2	Plain End x Threaded, BSP	
Tee, reducing ⁴	DN15 – 50	Manufacturers Standard	Model 4318	Press Connector	Manufacturer Std
			Model 4317.2	Plain End x Threaded, BSP	
Cap	DN15 – 50	Manufacturers Standard	Model 4356	Press Connector	Manufacturer Std
Union, Screw Fitting ⁴	DN15 – 50	Manufacturers Standard	Model 4365	Press Connector x Threaded, BSP Male	Manufacturer Std
			Model 4363	Press Connector x Threaded, BSP Female	
Coupling	DN15 – 50	Manufacturers Standard	Model 4315	Press Connector	Manufacturer Std
Coupling, Sliding	DN15 – 50	Manufacturers Standard	Model 4315.5	Press Connector	Manufacturer Std
Reducer	DN20 - 50	Manufacturers Standard	Model 4315.1	Plain End x Press Connector	Manufacturer Std
Adapter, Threaded ⁴	DN15 – 50	Manufacturers Standard	Model 4311	Press Connector x Threaded, BSP Male	Manufacturer Std
			Model 4312	Press Connector x Threaded, BSP Female	
Flange	DN15 - 25	Manufacturers Standard & AS 2129	4359, Table E FF	Press Connector	Manufacturer Std
			4359, Table E RF ⁶		
	DN32 - 50		4359.1, Table E FF		
			4359.1, Table E RF ⁶		
Press-in Connection ⁴	DN20	Manufacturers Standard	Model 4312.2	Threaded, BSP	Manufacturer Std

PE8	Butt welded HDPE PN10 (SDR17) piping system, short radius bends, with flanges to AS2129 Table D for low pressure liquid applications.							
Material :		High Density Polyethylene PE100 PN10 (SDR 17) - Coextruded White Outer/Black Inner Pipe						
Design Codes :		AS 4041, Class 3 (Note 12)						
Design Life (Yr) :		100	100	50	35	22	15	7
Design Temperature (°C) ¹ :		20	30	35 - 40	45	50	55	60
Design Pressure (kPa(g)) ¹ :		700	700	700	700	690	520	450
Corrosion Allowance (mm) :		0.0						
Pressure Testing :		Initial service leak testing as per AS 4041. Hydrostatic pressure testing of below ground services only, unless directed by the Engineer. Maximum test pressure 1050 kPa(g) at 20°C.						
Non-Destructive Testing :		Visual inspection only						
Description	Size Range	Dimensional Specification		Type	End Preparations		Material Specification	
Pipe ²	OD25 - 1200	AS/NZS 4130		SDR17	Plain end		AS/NZS 4131	
Elbow, 90°, Short Radius	OD25 – 90	Manufacturer Std		SDR17	Electrofusion Sockets ¹⁰		AS/NZS 4129 AS/NZS 4131	
	Plain end, long spigot							
	Plain end							
Elbow, 90°, Sweep (fabricated)	OD560 - 900				Plain end			
Elbow, 45°	OD25 – 90	Manufacturer Std		SDR17	Electrofusion Sockets ¹⁰		AS/NZS 4129 AS/NZS 4131	
	Plain end, long spigot							
Elbow, 45°, Sweep (fabricated)	OD355 - 900				Plain end			
Tee, equal ⁵	OD25 – 500	Manufacturer Std		SDR17	Plain end, long spigot		AS/NZS 4129 AS/NZS 4131	
Tee, equal (fabricated) ⁵	OD560 - 900				Plain end			
Tee, reducing ⁵	OD25 – 500				Plain end, long spigot			
Tee, reducing (fabricated) ⁵	OD560 - 900				Plain end			
Y Piece (fabricated) ⁵	OD90 – 900	Manufacturer Std		SDR17	Plain end		AS/NZS 4129 AS/NZS 4131	
Junction (fabricated), 45° ⁵	OD90 – 900							
Junction (fabricated), 60° ⁵	OD90 – 900							
Cap	OD25 – 90	Manufacturer Std		SDR17	Electrofusion Sockets ¹⁰		AS/NZS 4129 AS/NZS 4131	
	OD25 – 315				Plain end, long spigot			
	OD355 - 900				Plain end			
Coupling	OD25 – 90	Manufacturer Std		SDR17	Electrofusion Sockets ¹⁰		AS/NZS 4129 AS/NZS 4131	
Reducer, eccentric	OD50 – 250	Manufacturer Std		SDR17	Plain end, long spigot		AS/NZS 4129 AS/NZS 4131	
Reducer, concentric	OD25 – 400				Plain end, long spigot			
	OD450 – 900				Plain end, short spigot ⁸			

Adaptor, Male BSP Thread	OD20 - OD63	Manufacturer Std	SDR17	Plain end, long spigot	AS/NZS 4129 AS/NZS 4131
Stub Flange c/w galvanised backing ring	OD25 – 400	AS/NZS 4129 AS 2129 ⁹	SDR17 Stub, Table D backing ring	Plain end, long spigot	AS/NZS 4129 AS/NZS 4131
	OD450 – 1200			Plain end, short spigot ⁸	AS 3678 Gr 250
Flange, blind	DN25 – 1200	AS 2129 ⁹	Table D, flat face (blind), galvanised		ASTM A105
Tapping Saddle ⁷	DN15 – 315	AS/NZS 4129	SDR17	BSP Female Tapping	AS/NZS 4129
Stud Bolts, Type A	DN15 – 1200	AS 2528	Galvanised, Gr. 4.6 bolts		AS 1111.1
Machine Bolts, Metric	DN15 – 1200		Galvanised, Hex head, Gr. 4.6 bolts		
Nuts	DN15 – 1200		Galvanised, Gr. 5 hex nuts		AS 1112.3
Gaskets ¹⁵	DN15 – 1200	AS 2129	Klinger C6327 or approved equivalent, Table D, 2mm thick full face.		
			Table D, Klinger TC2003 or approved equivalent, 1.5mm thick, flat face		
Thread Sealant	DN15 – 50	-	Thread tape or liquid sealant approved to AS4020, WaterMark Certified for Potable Water service.		PTFE
Butterfly Valve Spacer ⁴	OD110 - 630	Manufacturer Std	SDR17	Spacer Ring	AS/NZS 4129 PE100

1	The design pressures stated are permissible for liquids only and are based on temperature deratings for the specified polyethylene pipe material in accordance with ISO 9080 and AS 4130 for the specified design life. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	PE100 Series 1 polyethylene is the minimum material grade for supply of pipe and fittings. All HPDE material shall be 100% virgin. Black polyethylene pipe is permissible for underground service and short length process plant spools where temperature derating is not an issue. White co-extruded is preferred for long above ground service pipeline runs. Continuous coils only to be used underground, use of coil pipe for above ground overland pipelines shall only be used with the approval of the Engineer.
3	Diameters nominated are pipe outer diameter (OD).
4	Polyethylene spacer ring may be required for butterfly or swing check valves to allow disc operation, dependent on pipe diameter.
5	Moulded, machined or formed fittings shall be used as a priority for this piping system. Special fittings requiring fabrication such as reducing tees, equal tees, Y-pieces and laterals may not achieve the design pressure ratings of this pipe system and shall only be permitted with approval by the Engineer.
6	Non-preferred sizes can be used as required to suit slurry settling velocities.
7	Where nominated, the minimum branch size is OD63 to avoid potential mechanical damage. Branch connection to be flanged with reduction added downstream. The use of tapping saddles/threaded metallic transition spigots requires Engineering approval.
8	Long spigot fittings to be used in preference to normal spigot fittings.
9	Thickness to manufacturer standards, drilling to AS 2129 Table D.
10	Butt welded fittings shall be used as a priority, however electrofusion welded fittings may be used with engineering approval.
11	An equal tee and reducer tee with reducer(s) may be substituted with a moulded reducing tee if available from the Supplier.
12	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343.
13	For use of this piping system for potable water service, all pipe, fittings, valves and instrumentation shall be WaterMark Certified and approved to AS 4020.
14	Use of this Piping System for compressed air service duty is permitted for buried pipelines only where pressure rating has been de-rated accordingly. Above ground service will require further pressure derating and shall only be used where approved by the Engineer.
15	TC2003 to be used on all reagent services. Refer to Project Pipe Materials and Valve Specification for gasket selection

BRANCH CHART																										
Branch Size	900	ET																								
	800	ET+RED	ET																							
	710	ET+RED	ET+RED	ET																						
	630	RT	ET+RED	ET+RED	ET																					
	560	RT	ET+RED	ET+RED	ET+RED	ET																				
	500	RT	RT	ET+RED	ET+RED	ET+RED	ET																			
	450	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET																		
	400	RT	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET																	
	355	RT	RT	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET																
	315	RT	RT	RT	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET															
	280	RT	RT	RT	RT	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET+RED	ET													
	250*	RT	RT	RT	RT	RT	RT	RT	RT	ET+RED	RT	RT	ET													
	225	RT+RED	RT+RED	RT	RT	RT	RT	RT	RT	RT	RT	RT	RT	ET												
	200*	RT+RED	RT+RED	RT	RT	RT	RT	RT	RT	RT	RT	RT	RT	ET+RED	ET											
	180*	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	ET+RED	ET+RED	ET										
	160	RT+RED	RT+RED	RT+RED	RT	RT	RT	RT	RT	RT	RT	RT	RT	ET+RED	ET+RED	RT	ET									
	140*	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	ET+RED	RT	RT	RT	RT	ET							
	125*	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT	RT	RT	ET								
	110	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT	RT	RT	RT	RT	RT	RT	RT	RT	RT	ET							
	90	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT	RT	RT	RT	RT	RT	ET						
	75*	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT+RED	RT	RT+RED	RT+RED	RT	RT	ET					
	63	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT+RED	RT	RT+RED	RT+RED	RT	RT	ET				
	50	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	RT+RED	RT+RED	RT+RED	RT	ET				
	40*	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	RT+RED	RT+RED	RT+RED	RT+RED	ET+RED	ET		
	32	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	RT+RED	RT+RED	RT+RED	RT+RED	ET+RED	ET+RED	ET	
	25	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	RT+RED	RT+RED	RT+RED	RT+RED	ET+RED	ET+RED	ET+RED	
	900	800	710	630	560*	500	450	400	355	315	280	250*	225	200*	180*	160	140*	125*	110	90	75*	63	50	40*	32	25
	Header Size																									

LEGEND

ET

Equal Tee - moulded

RT

Reducing Tee - moulded

ET+RED

Equal Tee + Reducer (+ Reducer) ¹¹

RT+RED

Reducing Tee + Reducer (+ Reducer) ¹¹

* indicates non-standard pipe size and not preferred where practical ⁶

LEGEND	
ET	Equal Tee - moulded
RT	Reducing Tee - moulded
ET+RED	Equal Tee + Reducer (+ Reducer) ¹¹
RT+RED	Reducing Tee + Reducer (+ Reducer) ¹¹

* indicates non-standard pipe size and not preferred where practical ⁶

PE10	Butt Welded HDPE PN10 (SDR17) piping system, sweep bends, with flanges to AS 2129 Table D for low pressure slurry applications.								
Material :		High Density Polyethylene PE100 PN10 (SDR 17) - Coextruded White Outer/Black Inner Pipe							
Design Codes :		AS 4041, Class 3 (Note 11)							
Design Life (Yr) :		100	100	50	35	22	15	7	
Design Temperature (°C) ¹ :		20	30	35 - 40	45	50	55	60	
Design Pressure (kPa(g)) ¹ :		700	700	700	700	690	520	450	
Corrosion Allowance (mm) :		0.0							
Pressure Testing :		Initial service leak testing as per AS 4041. Hydrostatic pressure testing of below ground services only, unless directed by the Engineer. Maximum test pressure 1050 kPa(g) at 20°C.							
Non-Destructive Testing :		Visual inspection only							

Description	Size Range	Dimensional Specification	Type	End Preparations	Material Specification
Pipe ²	OD25 - 1200	AS/NZS 4130	SDR17	Plain end	AS/NZS 4131
Sweep Bend, 30°	OD25 - 1200	Manufacturer Std	SDR17	Plain end	AS/NZS 4129 AS/NZS 4131
Sweep Bend, 45°					
Sweep Bend, 60°					
Sweep Bend, 90°					
Tee, equal ⁵	OD25 – 500	Manufacturer Std	SDR17	Plain end, long spigot	AS/NZS 4129 AS/NZS 4131
Tee, equal (fabricated) ⁵	OD560 - 1200			Plain end	
Tee, reducing ⁵	OD25 – 500			Plain end, long spigot	
Tee, reducing (fabricated) ⁵	OD560 - 1200			Plain end	
Y Piece (fabricated) ⁵	OD90 – 1200	Manufacturer Std	SDR17	Plain end	AS/NZS 4129 AS/NZS 4131
Junction (fabricated), 45° ⁵	OD90 – 1200				
Junction (fabricated), 60° ⁵					
Cap	OD25 – 315	Manufacturer Std	SDR17	Plain end, long spigot	AS/NZS 4129 AS/NZS 4131
	OD355 - 1200			Plain end	
Reducer, eccentric	OD50 – 250	Manufacturer Std	SDR17	Plain end, long spigot	AS/NZS 4129 AS/NZS 4131
Reducer, concentric	OD25 – 400			Plain end, long spigot	
	OD450 – 1200			Plain end, short spigot ⁸	
Stub Flange c/w galvanised backing ring	OD25 – 400	AS/NZS 4129 AS 2129 ⁹	SDR17 Stub, Table D backing ring	Plain end, long spigot	AS/NZS 4129 AS/NZS 4131 AS 3678 Gr 250
	OD450 – 1200			Plain end, short spigot ⁸	
Flange, blind	DN20 – 1200	AS 2129 ⁹	Table D, flat face (blind), galvanised		ASTM A105
Tapping Saddle ⁷	OD25 – 315	AS/NZS 4129	SDR17	BSP Female Tapping	AS/NZS 4129
Stud Bolts, Type A	DN15 – 1200	AS 2528	Galvanised, Gr. 4.6 bolts		AS 1111.1
Machine Bolts, Metric ⁴	DN15 – 1200		Galvanised, Hex head, Gr. 4.6 bolts		
Nuts	DN15 – 1200		Galvanised, Gr. 5 hex nuts		AS 1112.3
Gaskets ¹²	DN15 – 1200	AS 2129	Klinger C6327 or approved equivalent, Table D, 2mm thick full face.		
			Table D, Klinger TC2003 or approved equivalent, 1.5mm thick, flat face		
Thread Sealant	DN15 – 50	-	Thread tape or liquid sealant		PTFE

1	The design pressures stated are permissible for liquids only and are based on temperature deratings for the specified polyethylene pipe material in accordance with ISO 9080 and AS 4130 for the specified design life. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	PE100 Series 1 polyethylene is the minimum material grade for supply of pipe and fittings. All HPDE material shall be 100% virgin. Black polyethylene pipe is permissible for underground service and short length process plant spools where temperature derating is not an issue. White co-extruded is preferred for long above ground service pipeline runs. Continuous coils only to be used underground, use of coil pipe for above ground overland pipelines shall only be used with the approval of the Engineer.
3	Diameters nominated are pipe outer diameter (OD).
4	Not Used
5	Moulded, machined or formed fittings shall be used as a priority for this piping system. Special fittings requiring fabrication such as reducing tees, equal tees, Y–pieces and laterals may not achieve the design pressure ratings of this pipe system and shall only be permitted with approval by the
6	Non-preferred sizes can be used as required to suit slurry settling velocities.
7	Where nominated, the minimum branch size is OD63 to avoid potential mechanical damage. Branch connection to be flanged with reduction added downstream. The use of tapping saddles/threaded metallic transition spigots requires Engineering approval.
8	Long spigot fittings to be used in preference to normal spigot fittings.
9	Thickness to manufacturer standards, drilling to AS 2129 Table D.
10	An equal tee and reducer tee with reducer(s) may be substituted with a moulded reducing tee if available from the Supplier.
11	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343.
12	TC2003 to be used on all reagent services. Refer to Project Pipe Materials and Valve Specification for gasket selection

BRANCH CHART																													
Branch Size	1200	ET																											
	1000	ET+RED	ET																										
	900	ET+RED	ET+RED	ET																									
	800	ET+RED	ET+RED	ET+RED	ET																								
	710	RT	ET+RED	ET+RED	ET+RED	ET																							
	630	RT	RT	RT	ET+RED	ET+RED	ET																						
	560	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET																					
	500	RT	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET																				
	450	RT	RT	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET																			
	400	RT	RT	RT	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET																		
	355	RT	RT	RT	RT	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET																	
	315	RT	RT	RT	RT	RT	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET																
	280	RT	RT	RT	RT	RT	RT	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET															
	250*	RT+RED	RT	RT	RT	RT	RT	RT	RT	RT	RT	ET+RED	RT	RT	ET														
	225	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT	RT	RT	RT	RT	RT	RT	RT	RT	ET													
	200*	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT	RT	RT	RT	RT	RT	RT	RT	RT	RT	ET+RED	ET											
	180*	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	ET+RED	ET+RED	ET										
	160	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT	RT	RT	RT	RT	RT	RT	RT	RT	ET+RED	ET+RED	RT	ET									
	140*	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	ET+RED	RT	RT	ET									
	125*	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT	RT	RT	ET								
	110	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT	RT	RT	RT	RT	RT	RT	RT	RT	ET								
	90	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT	RT	RT	RT	RT	RT	ET								
	75*	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT+RED	RT	RT+RED	RT+RED	RT	ET							
	63	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT+RED	RT	RT+RED	RT+RED	RT	RT	ET							
	50	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	RT+RED	RT+RED	RT+RED	RT	ET				
	40*	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	RT+RED	RT+RED	RT+RED	RT+RED	ET+RED	ET		
	32	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	RT+RED	RT+RED	RT+RED	RT+RED	ET+RED	ET+RED	ET	
	25	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	RT+RED	RT+RED	RT+RED	RT+RED	ET+RED	ET+RED	ET+RED	ET
		1200	1000	900	800	710	630	560*	500	450	400	355	315	280	250*	225	200*	180*	160	140*	125*	110	90	75*	63	50	40*	32	25
		Header Size																											

LEGEND	
ET	Equal Tee - moulded
RT	Reducing Tee - moulded
ET+RED	Equal Tee + Reducer (+ Reducer) ¹⁰
RT+RED	Reducing Tee + Reducer (+ Reducer) ¹⁰

* indicates non-standard pipe size and not preferred where practical⁶

PE12.5	Butt Welded HDPE PN12.5 (SDR13.6) piping system with flanges to AS 2129 Table E for medium pressure slurry tails pipeline applications.							
Material :		High Density Polyethylene PE100 PN 12.5 (SDR 13.6) - Coextruded White Outer/Black Inner Pipe						
Design Codes :		AS 4041, Class 3 (Note 10)						
Design Life (Yr) :		100	100	50	35	22	15	7
Design Temperature (°C) ¹ :		20	30	35 - 40	45	50	55	60
Design Pressure (kPa(g)) ¹ :		1240	1120	1040	970	890	660	570
Corrosion Allowance (mm) :		0.0						
Pressure Testing :		Initial service leak testing as per AS 4041. Hydrostatic pressure testing of below ground services only, unless directed by the Engineer. Maximum test pressure 1550 kPa(g) at 20°C.						
Non-Destructive Testing :		Visual inspection only						

Description	Size Range	Dimensional Specification	Type	End Preparations	Material Specification
Pipe ²	OD90 - 900 ⁷	AS/NZS 4130	SDR13.6	Plain end	AS/NZS 4131
Sweep Bend, 30°	OD90 - 900	AS/NZS 4130	SDR13.6	Plain end	AS/NZS 4129 AS/NZS 4131
Sweep Bend, 45°					
Sweep Bend, 60°					
Sweep Bend, 90°					
Reducer, eccentric	OD90 – 250	Manufacturer Std	SDR13.6	Plain end, long spigot	AS/NZS 4129 AS/NZS 4131
Reducer, concentric	OD90 – 400			Plain end, long spigot	
	OD450 – 900			Plain end, short spigot ⁸	
Stub Flange c/w galvanised backing ring	OD90 – 400	AS/NZS 4129 AS 2129 ⁹	SDR13.6 Stub, Table E backing ring	Plain end, long spigot	AS/NZS 4129 AS/NZS 4131 AS 3678 Gr 250
	OD90 – 900			Plain end, short spigot ⁸	
Flange, blind	DN80 – 900	AS 2129 ⁹	Table E, flat face (blind), galvanised		ASTM A105
Tapping Saddle ⁵	OD25 – 315	AS/NZS 4129	SDR13.6	BSP Female Tapping	AS/NZS 4129
Stud Bolts, Type A	DN80 – 900	AS 2528	Galvanised, Gr. 4.6 bolts		AS 1111.1
Machine Bolts, Metric ⁴	DN80 – 900		Galvanised, Hex head, Gr. 4.6 bolts		
Nuts	DN80 – 900		Galvanised, Gr. 5 hex nuts		AS 1112.3
Gaskets ¹¹	DN80 – 900	AS 2129	Klinger C6327 or approved equivalent, Table E, 2mm thick full face.		
			Table E, Klinger TC2003 or approved equivalent, 1.5mm thick, flat face		
Thread Sealant	DN15 – 50	-	Thread tape or liquid sealant		PTFE

1	The design pressures stated are permissible for liquids only and are based on temperature deratings for the specified polyethylene pipe material in accordance with ISO 9080 and AS 4130 for the specified design life. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	PE100 Series 1 polyethylene is the minimum material grade for supply of pipe and fittings. All HPDE material shall be 100% virgin. Black polyethylene pipe is permissible for underground service and short length process plant spools where temperature derating is not an issue. White co-extruded is preferred for long above ground service pipeline runs. Continuous coils only to be used underground, use of coil pipe for above ground overland pipelines shall only be used with the approval of the Engineer.
3	Diameters nominated are pipe outer diameter (OD).
4	Not Used
5	This piping system is designed for straight run applications (i.e. tailings pipelines) and branches will require either fabricated tees or a specification break to another piping system. Special fittings requiring fabrication such as reducing tees, equal tees, Y-pieces and laterals may not achieve the design pressure ratings of this pipe system and shall only be permitted with approval by the Engineer.
6	Non-preferred sizes can be used as required to suit slurry settling velocities.
7	Smallest pipe size is limited to 90 OD due to lack of availability of PN12.5 stub flanges for smaller pipe sizes.
8	Long spigot fittings to be used in preference to short spigot fittings.
9	Thickness to manufacturer standards, drilling to AS 2129 Table E.
10	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343.
11	TC2003 to be used on all reagent services. Refer to Project Pipe Materials and Valve Specification for gasket selection

PE16	Butt Welded HDPE PN16 (SDR11) piping system with flanges to AS 2129 Table E for medium pressure slurry or liquid applications.								
Material :		High Density Polyethylene PE100 PN16 (SDR 11) - Coextruded White Outer/Black Inner Pipe							
Design Codes :		AS 4041, Class 3 (Note 13)							
Design Life (Yr) :		100	100	50	35	22	15	7	
Design Temperature (°C) ¹ :		20	30	35 - 40	45	50	55	60	
Design Pressure (kPa(g)) ¹ :		1400	1400	1300	1200	1100	830	710	
Corrosion Allowance (mm) :		0.0							
Pressure Testing :		Initial service leak testing as per AS 4041. Hydrostatic pressure testing of below ground services only, unless directed by the Engineer. Maximum test pressure 1960kPa(g) at 20°C.							
Non-Destructive Testing :		Visual inspection only							
Description	Size Range	Dimensional Specification		Type	End Preparations		Material Specification		
Pipe ²	OD20 - 900	AS/NZS 4130		SDR11	Plain end		AS/NZS 4131		
Elbow, 90°	OD20 – 90	Manufacturer Std		SDR11	Electrofusion Sockets ¹¹		AS/NZS 4129 AS/NZS 4131		
	Plain end, long spigot								
	Plain end								
Elbow ¹⁰ , 90°, Sweep (fabricated)	OD20 – 900				Plain end				
Elbow ¹⁰ , 60°, Sweep (fabricated)	OD20 - 900	Manufacturer Std		SDR11	Plain end		AS/NZS 4129 AS/NZS 4131		
Elbow, 45°	OD20 – 90	Manufacturer Std		SDR11	Electrofusion Sockets ¹¹		AS/NZS 4129 AS/NZS 4131		
	Plain end, long spigot								
Elbow ¹⁰ , 45°, Sweep (fabricated)	OD20 – 900				Plain end				
Elbow ¹⁰ , 30°, Sweep (fabricated)	OD20 - 900	Manufacturer Std		SDR11	Plain end		AS/NZS 4129 AS/NZS 4131		
Tee, equal ⁵	OD20 – 500	Manufacturer Std		SDR11	Plain end, long spigot		AS/NZS 4129 AS/NZS 4131		
Tee, equal (fabricated) ⁵	OD560 - 900				Plain end				
Tee, reducing ⁵	OD20 – 500				Plain end, long spigot				
Tee, reducing (fabricated) ⁵	OD560 - 900				Plain end				
Y Piece (fabricated) ⁵	OD90 – 900	Manufacturer Std		SDR11	Plain end		AS/NZS 4129 AS/NZS 4131		
Junction (fabricated), 45° ⁵	OD90 – 900								
Junction (fabricated), 60° ⁵	OD90 – 900								
Cap	OD20 – 90	Manufacturer Std		SDR11	Electrofusion Sockets ¹¹		AS/NZS 4129 AS/NZS 4131		
	OD20 – 315				Plain end, long spigot				
	OD355 - 900				Plain end				
Coupling	OD20 – 90	Manufacturer Std		SDR11	Electrofusion Sockets ¹¹		AS/NZS 4129 AS/NZS 4131		
Reducer, eccentric	OD50 – 250	Manufacturer Std		SDR11	Plain end, long spigot		AS/NZS 4129 AS/NZS 4131		
Reducer, concentric	OD20 – 400				Plain end, long spigot				
	OD450 – 900				Plain end, short spigot ⁸				
Adaptor, Female BSP Thread	OD20 - OD63	Manufacturer Std		SDR11	Plain end, long spigot		AS/NZS 4129 AS/NZS 4131		

Stub Flange c/w galvanised backing ring	OD20 – 400	AS/NZS 4129 AS 2129 ⁹	SDR11 Stub, Table E backing ring	Plain end, long spigot	AS/NZS 4129 AS/NZS 4131 AS 3678 Gr 250
	OD450 – 900			Plain end, short spigot ⁸	
Flange, blind	DN15 – 900	AS 2129	Table E, flat face (blind), galvanised		ASTM A105
Tapping Saddle ⁷	OD20 – 315	AS/NZS 4129	SDR11	BSP Female Tapping	AS/NZS 4129
Stud Bolts, Type A	DN15 – 900	AS 2528	Galvanised, Gr. 4.6 bolts		AS 1111.1
Machine Bolts, Metric	DN15 – 900		Galvanised, Hex head, Gr. 4.6 bolts		
Nuts	DN15 – 900		Galvanised, Gr. 5 hex nuts		AS 1112.3
Gaskets ¹⁵	DN15 – 900	AS 2129	Klinger C6327 or approved equivalent, Table E, 2mm thick full face.		
			Table E, Klinger TC2003 or approved equivalent, 1.5mm thick, flat face		
Thread Sealant	DN15 – 50	-	Thread tape or liquid sealant		PTFE
Butterfly Valve Spacer ⁴	OD110 - 630	Manufacturer Std	SDR17	Spacer Ring	AS/NZS 4129 PE100

NOTES	
1	The design pressures stated are permissible for liquids only and are based on temperature deratings for the specified polyethylene pipe material in accordance with ISO 9080 and AS 4130 for the specified design life. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	PE100 Series 1 polyethylene is the minimum material grade for supply of pipe and fittings. All HPDE material shall be 100% virgin. Black polyethylene pipe is permissible for underground service and short length process plant spools where temperature derating is not an issue. White co-extruded is preferred for long above ground service pipeline runs. Continuous coils only to be used underground, use of coil pipe for above ground overland pipelines shall only be used with the approval of the Engineer.
3	Diameters nominated are pipe outer diameter (OD).
4	Polyethylene spacer ring may be required for butterfly or swing check valves to allow disc operation, dependent on pipe diameter.
5	Moulded, machined or formed fittings shall be used as a priority for this piping system. Special fittings requiring fabrication such as reducing tees, equal tees, Y-pieces and laterals may not achieve the design pressure ratings of this pipe system and shall only be permitted with approval by the Engineer.
6	Non-preferred sizes can be used as required to suit slurry settling velocities.
7	Where nominated, the minimum branch size is OD63 to avoid potential mechanical damage. Branch connection to be flanged with reduction added downstream. The use of tapping saddles/threaded metallic transition spigots requires Engineering approval.
8	Long spigot fittings to be used in preference to normal spigot fittings.
9	Thickness to manufacturer standards, drilling to AS 2129 Table D.
10	Sweep Bends only for slurry services. Short Bends are not permitted.
11	Butt welded fittings shall be used as a priority, however electrofusion welded fittings may be used with engineering approval.
12	An equal tee and reducer tee with reducer(s) may be substituted with a moulded reducing tee if available from the Supplier.
13	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343.
14	Use of this Piping System for compressed air service duty is permitted for buried pipelines only where pressure rating has been de-rated accordingly. Above ground service will require further pressure derating and shall only be used where approved by the Engineer.
15	TC2003 to be used on all reagent services. Refer to Project Pipe Materials and Valve Specification for gasket selection

BRANCH CHART																										
Branch Size	900	ET																								
	800	ET+RED	ET																							
	710	ET+RED	ET+RED	ET																						
	630	RT	ET+RED	ET+RED	ET																					
	560	RT	ET+RED	ET+RED	ET+RED	ET																				
	500	RT	RT	ET+RED	ET+RED	ET+RED	ET																			
	450	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET																		
	400	RT	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET																	
	355	RT	RT	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET																
	315	RT	RT	RT	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET															
	280	RT	RT	RT	RT	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET														
	250*	RT	RT	RT	RT	RT	RT	RT	RT	ET+RED	RT	RT	ET													
	225	RT+RED	RT+RED	RT	RT	RT	RT	RT	RT	RT	RT	RT	ET													
	200*	RT+RED	RT+RED	RT	RT	RT	RT	RT	RT	RT	RT	RT	ET+RED	ET												
	180*	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	ET+RED	ET+RED	ET											
	160	RT+RED	RT+RED	RT+RED	RT	RT	RT	RT	RT	RT	RT	RT	ET+RED	ET+RED	RT	ET										
	140*	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	ET+RED	RT	RT	RT	ET									
	125*	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT	RT	RT	RT	ET									
	110	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT	RT	RT	RT	RT	RT	RT	RT	RT	ET									
	90	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT	RT	RT	RT	ET								
	75*	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT+RED	RT	RT+RED	RT	RT	ET							
	63	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT+RED	RT	RT+RED	RT	RT	RT	ET						
	50	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	RT+RED	RT+RED	RT+RED	RT	ET				
	40*	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	RT+RED	RT+RED	RT+RED	RT+RED	ET+RED	ET			
	32	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	RT+RED	RT+RED	RT+RED	RT+RED	ET+RED	ET+RED	ET		
	25	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	Note 7	RT+RED	RT+RED	RT+RED	RT+RED	ET+RED	ET+RED	ET+RED	ET	

LEGEND	
ET	Equal Tee - moulded
RT	Reducing Tee - moulded
ET+RED	Equal Tee + Reducer (+ Reducer) ¹²
RT+RED	Reducing Tee + Reducer (+ Reducer) ¹²

* indicates non-standard pipe size and not preferred where practical ⁶

PE20	Butt Welded HDPE PN20 (SDR 9) piping system with flanges to AS 2129 Table E for high pressure slurry or liquid applications.							
Material :		High Density Polyethylene PE100 PN20 (SDR 9) - Coextruded White Outer/Black Inner Pipe						
Design Codes :		AS 4041, Class 3 (Note 13)						
Design Life (Yr) :		100	100	50	35	22	15	7
Design Temperature (°C) ¹ :		20	30	35 - 40	45	50	55	60
Design Pressure (kPa(g)) ¹ :		1400	1400	1400	1400	1400	1040	900
Corrosion Allowance (mm) :		0.0						
Pressure Testing :		Initial service leak testing as per AS 4041. Hydrostatic pressure testing of below ground services only, unless directed by the Engineer. Maximum test pressure 2,100 kPa(g) at 20°C.						
Non-Destructive Testing :		Visual inspection only						
Description	Size Range	Dimensional Specification		Type	End Preparations		Material Specification	
Pipe ²	OD25 - 630	AS/NZS 4130		SDR9	Plain end		AS/NZS 4131	
Elbow, 90°	OD25 – 315	Manufacturer Std	SDR9	Plain end, long spigot		AS/NZS 4129 AS/NZS 4131		
	OD355 – 500			Plain end				
Elbow ¹⁰ , 90°, Sweep (fabricated)	OD25 – 630			Plain end				
Elbow ¹⁰ , 60°, Sweep (fabricated)	OD25 – 630	Manufacturer Std	SDR9	Plain end		AS/NZS 4129 AS/NZS 4131		
Elbow, 45°	OD25 – 315	Manufacturer Std	SDR9	Plain end, long spigot		AS/NZS 4129 AS/NZS 4131		
Elbow ¹⁰ , 45°, Sweep (fabricated)	OD25 – 630			Plain end				
Elbow ¹⁰ , 30°, Sweep (fabricated)	OD25 – 630	Manufacturer Std	SDR9	Plain end		AS/NZS 4129 AS/NZS 4131		
Tee, equal ⁵	OD25 – 500	Manufacturer Std	SDR9	Plain end, long spigot		AS/NZS 4129 AS/NZS 4131		
Tee, equal (fabricated) ⁵	OD560 - 630			Plain end				
Tee, reducing ⁵	OD25 – 500			Plain end, long spigot				
Tee, reducing (fabricated) ⁵	OD560 - 630			Plain end				
Y Piece (fabricated) ⁵	OD90 – 630	Manufacturer Std	SDR9	Plain end		AS/NZS 4129 AS/NZS 4131		
Junction (fabricated), 45° ⁵	OD90 – 630							
Junction (fabricated), 60° ⁵	OD90 – 630							
Cap	OD25 – 315	Manufacturer Std	SDR9	Plain end, long spigot		AS/NZS 4129 AS/NZS 4131		
	OD355 - 630			Plain end				
Reducer, eccentric	OD50 – 250	Manufacturer Std	SDR9	Plain end, long spigot		AS/NZS 4129 AS/NZS 4131		
Reducer, concentric	OD25 – 400			Plain end, long spigot				
	OD450 – 630			Plain end, short spigot ⁸				
Stub Flange c/w galvanised backing ring	OD90 – 400	AS/NZS 4129 AS 2129 ⁹	SDR9 Stub, Table E backing ring	Plain end, long spigot		AS/NZS 4129 AS/NZS 4131 AS 3678 Gr 250		
	OD450 – 630			Plain end, short spigot ⁸				
Flange, blind	DN80 – 600	AS 2129	Table E, flat face (blind), galvanised			ASTM A105		
Tapping Saddle ⁷	OD90 – 315	AS/NZS 4129	SDR9	BSP Female Tapping		AS/NZS 4129		

Stud Bolts, Type A	DN80 - 600	ISO 965-1	Galvanised, Gr. 4.6 bolts	AS 1111.1
Machine Bolts, Metric ⁴	DN80 - 600		Galvanised, Hex head, Gr. 4.6 bolts	
Nuts	DN80 - 600	ISO 965-1	Galvanised, Gr. 5 hex nuts	AS 1112.3
Gaskets ¹⁵	DN80 - 600	AS 2129	Klinger C6327 or approved equivalent, Table E, 2mm thick full face.	
			Table E, Klinger TC2003 or approved equivalent, 1.5mm thick, flat face	
Thread Sealant	DN15 – 50	-	Thread tape or liquid sealant	PTFE

NOTES	
1	The design pressures stated are permissible for liquids only and are based on temperature deratings for the specified polyethylene pipe material in accordance with ISO 9080 and AS 4130 for the specified design life. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	PE100 Series 1 polyethylene is the minimum material grade for supply of pipe and fittings. All HPDE material shall be 100% virgin. Black polyethylene pipe is permissible for underground service and short length process plant spools where temperature derating is not an issue. White co-extruded is preferred for long above ground service pipeline runs. Continuous coils only to be used underground, use of coil pipe for above ground overland pipelines shall only be used with the approval of the Engineer.
3	Diameters nominated are pipe outer diameter (OD).
4	Not Used
5	Moulded, machined or formed fittings shall be used as a priority for this piping system. Special fittings requiring fabrication such as reducing tees, equal tees, Y-pieces and laterals may not achieve the design pressure ratings of this pipe system and shall only be permitted with approval by the Engineer.
6	Non-preferred sizes can be used as required to suit slurry settling velocities.
7	Where nominated, the minimum branch size is OD63 to avoid potential mechanical damage. Branch connection to be flanged with reduction added downstream. The use of tapping saddles/threaded metallic transition spigots requires Engineering approval.
8	Long spigot fittings to be used in preference to normal spigot fittings.
9	Thickness to manufacturer standards, drilling to AS 2129 Table E.
10	Sweep Bends only for slurry services. Short Bends are not permitted.
11	An equal tee and reducer tee with reducer(s) may be substituted with a moulded reducing tee if available from the Supplier.
12	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343.
14	Use of this Piping System for compressed air service duty is permitted for buried pipelines only where pressure rating has been de-rated accordingly. Above ground service will require further pressure derating and shall only be used where approved by the Engineer.
15	TC2003 to be used on all reagent services. Refer to Project Pipe Materials and Valve Specification for gasket selection

BRANCH CHART																		
Branch Size	630	ET																
	560*	ET+RED	ET															
	500	ET+RED	ET+RED	ET														
	450	ET+RED	ET+RED	ET+RED	ET													
	400	RT	ET+RED	ET+RED	ET+RED	ET												
	355	RT	RT	ET+RED	ET+RED	ET+RED	ET											
	315	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET										
	280	RT	RT	RT	RT	ET+RED	ET+RED	ET+RED	ET									
	250*	RT	RT	RT	RT	RT	ET+RED	RT	RT	ET								
	225	RT	RT	RT	RT	RT	RT	RT	RT	ET								
	200*	RT	RT	RT	RT	RT	RT	RT	RT	ET+RED	ET							
	180*	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	ET+RED	ET+RED	ET						
	160	RT	RT	RT	RT	RT	RT	RT	RT	ET+RED	ET+RED	RT	ET					
	140*	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	ET+RED	RT	RT	RT	ET				
	125*	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT	RT	RT	RT	ET			
	110	RT+RED	RT+RED	RT+RED	RT	RT	RT	RT	RT	RT	RT	RT	RT	RT	RT	ET		
	90	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT+RED	RT	RT	RT	RT	RT	RT	RT	ET	
	630	560*	500	450	400	355	315	280	250*	225	200*	180*	160	140*	125*	110	90	
Header Size																		

LEGEND	
ET	Equal Tee - moulded
RT	Reducing Tee - moulded
ET+RED	Equal Tee + Reducer (+ Reducer) ¹²
RT+RED	Reducing Tee + Reducer (+ Reducer) ¹²

* indicates non-standard pipe size and not preferred where practical ⁶

PEDC1	IR fusion PE100 SDR11 piping system and secondary contained PE100 SDR17 outer piping with AS2129 Table E flanges for low pressure and ambient reagent service applications										
Material :		High Density Polyethylene PE100-RC PN16 (SDR11) - Extruded Black Pipe									
Design Codes :		AS 4041, Class 3									
Design Life (Yr) :		25									
Design Temperature (°C) ¹ :		0	10	20	30	40	45				
Design Pressure (kPa(g)) NaCN ¹ :		1,200	1,200	1,010	860	740	690				
Design Pressure (kPa(g)) HCl ¹ :		1,000	1,000	840	715	615	575				
Corrosion Allowance (mm) :		0 mm									
Pressure Testing :		930 kPa(g) @ 30 °C Water Hydrostatic Pressure Test as per manufacturers recommendation									
Non-Destructive Testing :		100% Visual Inspection									
Description		Size Range		Dimensional Specification		Type		End Preparations		Material Specification	
Pipe ⁵		OD32 - 160 ³		Manufacturer Std ²		Code: 785.1 ²		Plain end		AS/NZS 4131	
Elbow, 90°, Short Radius		OD32 ³		Manufacturer Std ²		Code: 190 ²		Plain end		AS/NZS 4131	
Elbow, 90°, 2x 45° Elbow		OD63 - 110 ³		Manufacturer Std ²		Code: 212 ²		Plain end		AS/NZS 4131	
Elbow, 90°, 4x Segments ⁴		OD110 - 160 ³		Manufacturer Std ²		Code: 211 ²		Plain end		AS/NZS 4131	
Elbow, 45°		OD32 - 110 ³		Manufacturer Std ²		Code: 191 ²		Plain end		AS/NZS 4131	
Elbow, 45°, 2x Segments ⁴		OD110 - 160 ³		Manufacturer Std ²		Code: 213 ²		Plain end		AS/NZS 4131	
Tee, equal ⁵		OD32 - 110 ³		Manufacturer Std ²		Code: 192 ²		Plain end		AS/NZS 4131	
Cap		OD32 - 160 ³		Manufacturer Std ²		Code: 193 ²		Plain end		AS/NZS 4131	
Reducer, concentric		OD32 - 160 ³		Manufacturer Std ²		Code: 198 ²		Plain end		AS/NZS 4131	
Dog Bone		OD32 - 160 ³		Manufacturer Std ²		Code: 195 ²		Plain end		AS/NZS 4131	
Dog Bone, Restraint		OD32 - 160 ³		Manufacturer Std ²		Code: 208 ²		Plain end		AS/NZS 4131	
Dog Bone, Restraint Flow Through		OD32 - 160 ³		Manufacturer Std ²		Code: 208.1 ²		Plain end		AS/NZS 4131	
Dog Bone, Transition to Single Tube ⁸		OD32 - 160 ³		Manufacturer Std ²		Code: 197 ²		Plain end		AS/NZS 4131	
Measuring Fitting, One Side Leak Control		OD32 - 160 ³		Manufacturer Std ²		Code: 196 ²		Plain end		AS/NZS 4131	
Measuring Fitting, Transition to Single Tube ⁸		OD32 - 160 ³		Manufacturer Std ²		Code: 196.1 ²		Plain end		AS/NZS 4131	
Measuring Fitting, Both Side Leak Control		OD32 - 160 ³		Manufacturer Std ²		Code: 196.2 ²		Plain end		AS/NZS 4131	
Induction E-Coupler ⁷		OD32 - 160 ³		Manufacturer Std ²		Code: 199 ²		Plain end		AS/NZS 4131	
Stub Flange c/w groove for O-ring & Galvanised backing ring		OD32 - 110 ³		Manufacturer Std ² AS 21296		Code: 194.2 ² , Table E backing ring		Plain end		AS/NZS 4131 AS 3678 Gr 250	
Stub Flange c/w galvanised backing ring ⁸		OD32 - 160 ³		AS/NZS 4129 AS 2129 ⁶		SDR11 Stub, Table E backing ring		Plain end, long spigot		AS/NZS 4131 AS 3678 Gr 250	
Flange, blind		DN25 – 200		AS 2129 ⁶		Table E, flat face (blind), galvanised				ASTM A105	
Machine Bolts, Metric		DN25 – 200		AS 1111.1		Galvanised, Hex head, Gr. 4.6 bolts				AS 1111.1	
Stud Bolts		DN25 – 200		AS 2528		Galvanised, Gr. 4.6 bolts				AS 2528	
Nuts		DN25 – 200				Galvanised, Gr. 5 hex nuts					

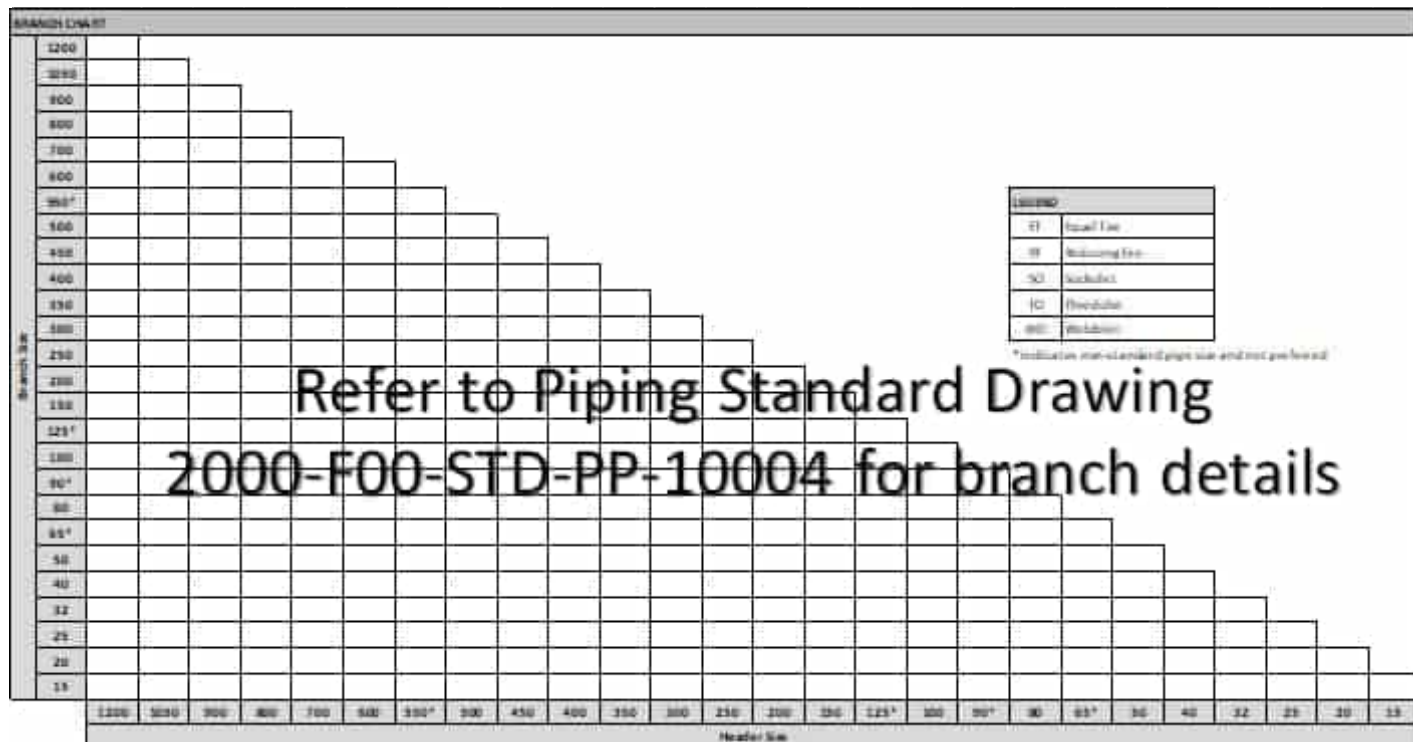
Description	Size Range	Specification	Type	End Preparations	Specification
O-ring/Seal	OD32 - 110	Manufacturer Std ²	FPM O-rings		
Gasket ⁸	DN25 - 200	AS 2129	Table E, Klinger TC2003 or approved equivalent, 1.5mm thick, flat face		
Thread Sealant	DN15 – 50	-	Thread tape or liquid sealant approved to AS4020, WaterMark Certified for Potable Water service.		PTFE

NOTES		
1	The design pressures stated are based on temperature deratings for the specified polyethylene pipe material in accordance with ISO 15494 and derating as per chemical resistance factor according to DVS 2205-1 supplement 5 for hazardous services Sodium Cyanide (NaCN) & Hydrochloric Acid (HCl) for the specified design life. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.	
2	Preferred manufacturer is Agru Poly-Flo	
3	Refer to inner pipe and corresponding outer pipe (containment pipe) below	
	Inner Pipe (OD mm)	Outer Pipe (OD mm)
	32	50
	63	90
	110	160
	160	225
4	Moulded, machined or formed fittings shall be used as a priority for this piping system. Special fittings requiring fabrication such as equal tees and laterals may not achieve the design pressure ratings of this pipe system and shall only be permitted with approval by the Engineer.	
5	Pipe is only available in maximum 5m lengths	
6	Thickness to manufacturer standards, drilling to AS 2129 Table E.	
7	Preferred joint methodology is to simultaneous IR weld joint. Induction weld joint shall only be permitted with approval by the Engineer	
8	To transition from Dual Contained to Single Wall piping systems, use Dog Bone or Measuring Fitting and standard HDPE100 SDR11 stub flange	
9	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343.	

RL1	Fully flanged 6mm thick natural rubber lined steel piping system with flat face flanges to AS2129 Table D for low pressure slurry applications.				
Material :		Carbon Steel – Natural Rubber Lined – 6mm Thick Lining ⁴			
Design Codes :		AS 4041, Class 3 (Note 11)			
Design Temperature (°C) ¹ :		0	70		
Design Pressure (kPa(g)) ¹ :		700	700		
Corrosion Allowance (mm) :		0.0			
Pressure Testing :		Initial service leak testing as per AS 4041. Hydrostatic pressure testing not required unless directed by the Engineer.			
Non-Destructive Testing :		Visual inspection only			

Description	Size Range	Dimensional Specification	Type	End Preparations	Material Specification
Pipe	DN50 ³	ASME B36.10M	STD WT, SMLS	Plain ends	ASTM A53 Gr B / API 5L Grade B
	DN65 ³ – 600		STD WT, ERW		
	DN650 – 1200		STD WT, LSAW		
Induction Bend, 3D or 5D ⁸	DN80 – 900	As per pipe ²	As per pipe	Plain ends	As per pipe
Elbow, 1.5D ⁹	DN80 – 900	ASME B16.9 ²	Butt Weld	Plain ends	ASTM A234 WPB
Lateral, 45°	DN80 – 900	As per pipe ²	As per pipe	Plain ends	As per pipe
Lateral, 60°					
Wye, 90°	DN80 – 900	As per pipe ²	As per pipe	Plain ends	As per pipe
Wye, 60°					
Tees ⁵	DN65 ³ – 900	As per pipe ²	As per pipe	Plain ends	As per pipe
Reducer, eccentric	DN50 ³ – 900	ASME B16.9 ²	As per large end piping	Plain ends	ASTM A234 WPB
Reducer, concentric	DN50 ³ – 900				
Flange, Slip-On	DN50 ³ – 1200	AS 2129 ²	Table D, FF	Plate	ASTM A105
Flange, Spinning	DN50 ³ – 1200			Backing Ring c/w Shoulder Plate	AS/NZS 3678 Gr 250
Flange, Blind	DN50 ³ – 1200			Blank	ASTM A105
Pipe Nipple ³	DN15 – 50	ASTM A733	XS WT, SMLS, 100mm	Threaded, BSP	ASTM A53 Gr B
Plugs ³	DN15 – 50	ASME B16.11	Class 3000	Threaded, BSP	ASTM A105
Stud Bolts, Type A	DN50 – 1200	AS/NZS 1252.1	Galvanised, Gr. 8.8 bolts		AS 4291.1
Machine Bolts, Metric ⁶	DN50 – 1200		Galvanised, Hex head, Gr. 8.8 bolts		
Nuts	DN50 – 1200		Galvanised, Gr. 8 hex nuts		AS 4291.2
Gaskets	DN50 – 1200	AS 2129	Table D, Klinger C4430 or approved equivalent, 1.5mm thick, flat face		
Rubber Lining	DN50 – 1200	6mm thick Natural Rubber lining, IRHD 38. Linatex Premium Red rubber or Reglin R-Line 38 Premium or approved equivalent. Refer to Note 4 for lining thickness of fixed flanges.			

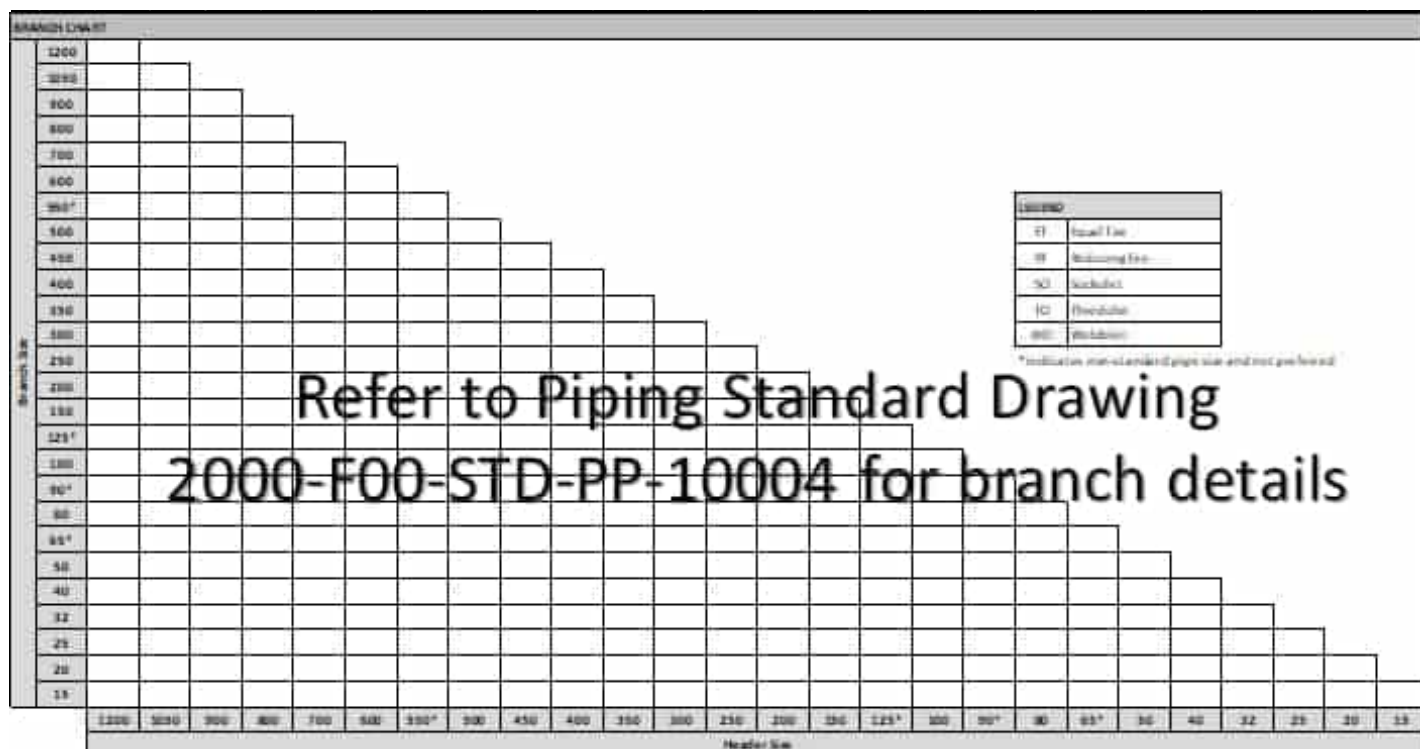
1	The design pressure and temperature conditions are based on plate flange design, STD WT fittings with no branch reinforcement and the maximum temperature limitation of the specified rubber lining. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	Refer to design drawings and the Standard Piping Drawings for dimensional details for fittings and plate flanges.
3	Pipe diameters less than DN80 shall not be used for main pipe runs and shall only be permitted for branch connections for drains, flushing points or instrumentation connections unless approved by the engineer.
4	All rubber lining to be 3mm thick lining on flange faces, IRHD 60. For details of rubber lining at flanges, joints etc., refer to the Standard Piping Drawings.
5	For preferred branch details refer to the Standard Piping Drawings. Use only laterals or Y-piece fittings for branches except for flushing connections, vents or instrument connections.
6	Not Used
7	Length of flanged spools shall be 3m maximum for DN80 and 6m maximum for DN100 and above.
8	Induction 3D bends to be used as priority. Induction 5D bends shall only be used with the approval by the Engineer.
9	1.5D elbows shall only to be used with approval by the Engineer.
10	For external surface treatment requirements refer to the Project Pipe Materials and Valve Specification.
11	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343.



RL2	Fully flanged 6mm thick natural rubber lined steel piping system with flat face flanges to AS2129 Table E for medium pressure slurry applications.				
Material :		Carbon Steel - Natural Rubber Lined – 6mm Thick Lining ⁴			
Design Codes :		AS 4041, Class 3 (Note 11)			
Design Temperature (°C) ¹ :		0	70		
Design Pressure (kPa(g)) ¹ :		1400	1400		
Corrosion Allowance (mm) :		0.0			
Pressure Testing :		Initial service leak testing as per AS 4041. Hydrostatic pressure testing not required unless directed by the Engineer.			
Non-Destructive Testing :		Visual inspection only			

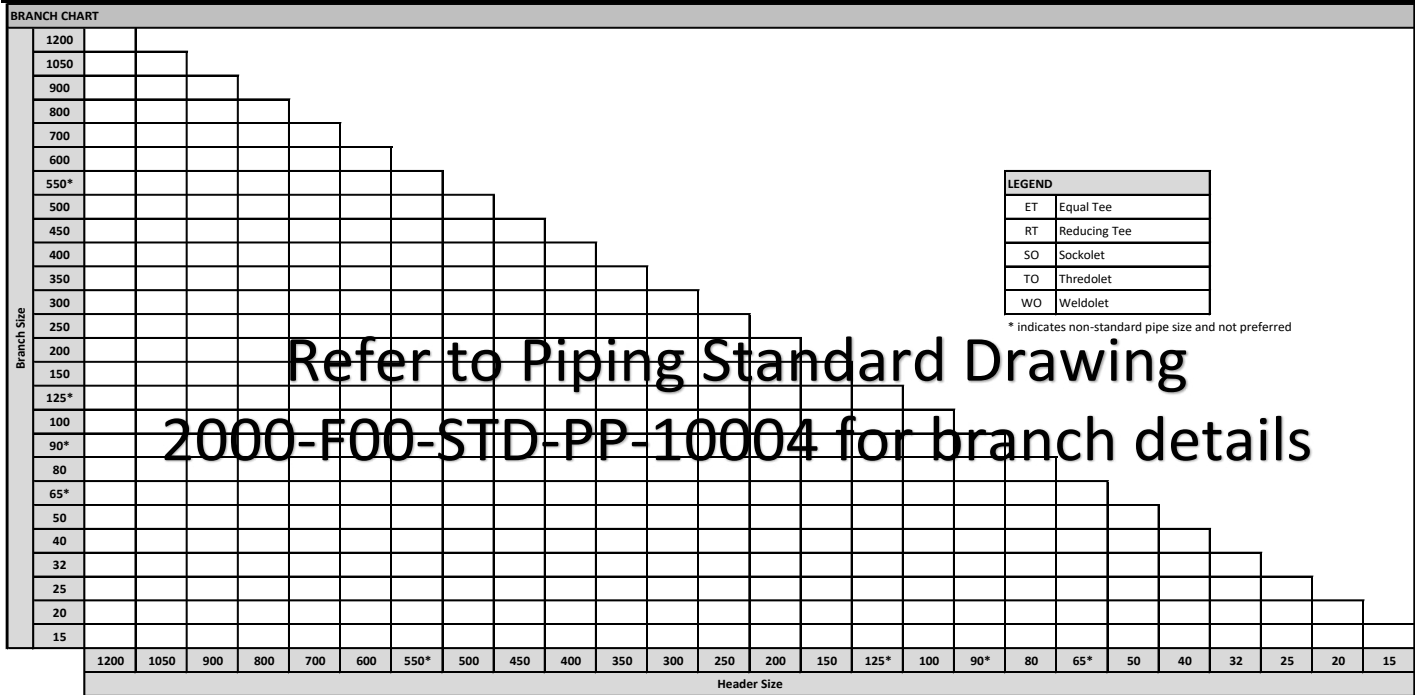
Description	Size Range	Dimensional Specification	Type	End Preparations	Material Specification
Pipe	DN50 ³	ASME B36.10M	STD WT, SMLS	Plain ends	ASTM A53 Gr B / API 5L Grade B
	DN65 – 600 ³		STD WT, ERW		
	DN650 – 900		STD WT, LSAW		
Induction Bend, 3D or 5D ⁸	DN80 – 600	As per pipe ²	As per pipe	Plain ends	As per pipe
Elbow, 1.5D ⁹	DN80 – 600	ASME B16.9 ²	Butt Weld	Plain ends	ASTM A234 WPB
Tees ⁵	DN65 ³ – 600	As per pipe ²	As per pipe	Plain ends	As per pipe
Reducer, eccentric	DN50 – 600	ASME B16.9 ²	As per large end piping	Plain ends	ASTM A234 WPB
Reducer, concentric	DN50 – 600				
Flange, Slip-On	DN50 – 900	AS 2129 ²	Table E, FF	Plate	ASTM A105
Flange, Spinning	DN50 – 900			Backing Ring c/w Shoulder Plate	AS/NZS 3678 Gr 250
Flange, Blind	DN50 – 900			Blank	ASTM A105
Pipe Nipple ³	DN15 – 50	ASTM A733	XS WT, SMLS, 100mm	Threaded, BSP	ASTM A53 Gr B
Plugs ³	DN15 – 50	ASME B16.11	Class 3000	Threaded, BSP	ASTM A105
Stud Bolts, Type A	DN50 – 900	AS/NZS 1252.1	Galvanised, Gr. 8.8 bolts		AS 4291.1
Machine Bolts, Metric ⁶	DN50 – 900		Galvanised, Hex head, Gr. 8.8 bolts		
Nuts	DN50 – 900		Galvanised, Gr. 8 hex nuts		AS 4291.2
Gaskets	DN50 – 900	AS 2129	Table E, Klinger C4430 or approved equivalent, 1.5mm thick, flat face		
Rubber Lining	DN50 – 900	6mm thick Natural Rubber lining, IRHD 38. Linatex Premium Red rubber or Reglin R-Line 38 Premium or approved equivalent. Refer to Note 4 for lining thickness of fixed flanges.			

1	The design pressure and temperature conditions are based on plate flange design, STD WT fittings with no branch reinforcement and the maximum temperature limitation of the specified rubber lining. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	Refer to design drawings and the Standard Piping Drawings for dimensional details for fittings and plate flanges.
3	Pipe diameters less than DN80 shall not be used for main pipe runs and shall only be permitted for branch connections for drains, flushing points or instrumentation connections unless approved by the engineer.
4	All rubber lining to be 3mm thick lining on flange faces, IRHD 60. For details of rubber lining at flanges, joints etc., refer to the Standard Piping Drawings.
5	For preferred branch details refer to the Standard Piping Drawings. Tees shall be avoided except for flushing connections, vents or instrument connections.
6	Not Used
7	Length of flanged spools shall be 3m maximum for DN80 and 6m maximum for DN100 and above.
8	Induction 3D bends to be used as priority. Induction 5D bends shall only be used with the approval by the Engineer.
9	1.5D elbows shall only to be used with approval by the Engineer.
10	For external surface treatment requirements refer to the Project Pipe Materials and Valve Specification.
11	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343.



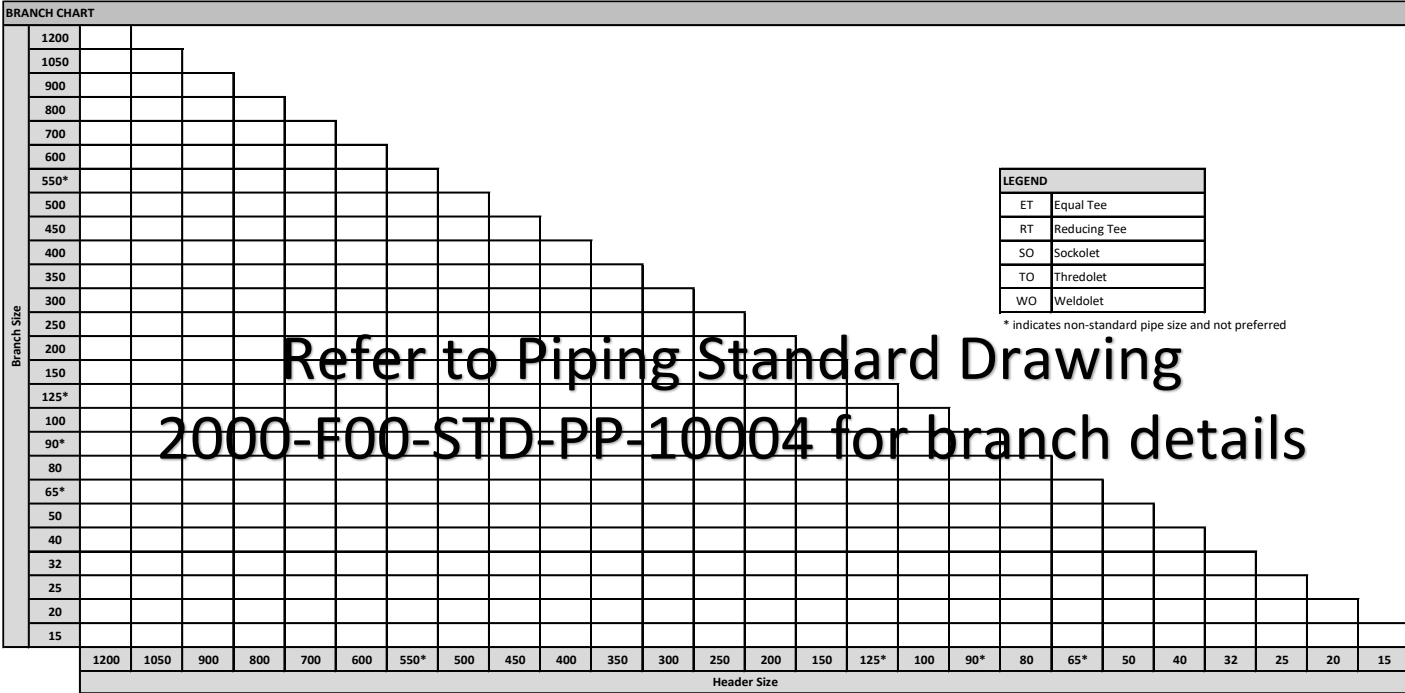
RL3	Fully flanged 12mm thick natural rubber lined steel piping system with flat face flanges to AS2129 Table D for low pressure slurry applications.				
Material :		Carbon Steel - Natural Rubber Lined – 12mm Thick Lining ⁴			
Design Codes :		AS 4041, Class 3 (Note 11)			
Design Temperature (°C) ¹ :		0	70		
Design Pressure (kPa(g)) ¹ :		700	700		
Corrosion Allowance (mm) :		0.0			
Pressure Testing :		Initial service leak testing as per AS 4041. Hydrostatic pressure testing not required unless directed by the Engineer.			
Non-Destructive Testing :		Visual inspection only			
Description	Size Range	Dimensional Specification	Type	End Preparations	Material Specification
Pipe	DN50 ³	ASME B36.10M	STD WT, SMLS	Plain ends	ASTM A53 Gr B / API 5L Grade B
	DN65 ³ – 600		STD WT, ERW		
	DN650 – 1200		STD WT, LSAW		
Induction Bend, 3D or 5D ⁸	DN80 – 900	As per pipe ²	As per pipe	Plain ends	As per pipe
Elbow, 1.5D ⁹	DN80 – 900	ASME B16.9 ²	Butt Weld	Plain ends	ASTM A234 WPB
Lateral, 45°	DN80 – 900	As per pipe ²	As per pipe	Plain ends	As per pipe
Lateral, 60°					
Wye, 90°	DN80 – 900	As per pipe ²	As per pipe	Plain ends	As per pipe
Wye, 60°					
Tees ⁵	DN65 ³ – 900	As per pipe ²	As per pipe	Plain ends	As per pipe
Reducer, eccentric	DN50 ³ – 900	ASME B16.9 ²	As per large end piping	Plain ends	ASTM A234 WPB
Reducer, concentric	DN50 ³ – 900				
Flange, Slip-On	DN50 ³ – 1200	AS 2129 ²	Table D, FF	Plate	ASTM A105
Flange, Spinning	DN50 ³ – 1200			Backing Ring c/w Shoulder Plate	AS/NZS 3678 Gr 250
Flange, Blind	DN50 ³ – 1200			Blank	ASTM A105
Pipe Nipple ³	DN15 – 50	ASTM A733	XS WT, SMLS, 100mm	Threaded, BSP	ASTM A53 Gr B
Plugs ³	DN15 – 50	ASME B16.11	Class 3000	Threaded, BSP	ASTM A105
Stud Bolts, Type A	DN50 – 1200	AS/NZS 1252.1	Galvanised, Gr. 8.8 bolts		AS 4291.1
Machine Bolts, Metric ⁶	DN50 – 1200		Galvanised, Hex head, Gr. 8.8 bolts		
Nuts	DN50 – 1200		Galvanised, Gr. 8 hex nuts		AS 4291.2
Gaskets	DN50 – 1200	AS 2129	Table D, Klinger C4430 or approved equivalent, 1.5mm thick, flat face		
Rubber Lining	DN50 – 1200	12mm thick Natural Rubber lining, IRHD 38. Linatex Premium Red rubber or Reglin R-Line 38 Premium or approved equivalent. Refer to Note 4 for lining thickness of fixed flanges.			

1	The design pressure and temperature conditions are based on plate flange design, STD WT fittings with no branch reinforcement and the maximum temperature limitation of the specified rubber lining. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	Refer to design drawings and the Standard Piping Drawings for dimensional details for fittings and plate flanges.
3	Pipe diameters less than DN80 shall not be used for main pipe runs and shall only be permitted for branch connections for drains, flushing points or instrumentation connections unless approved by the engineer. Lining for DN80 and under to be 6mm
4	All rubber lining to be 3mm thick lining on flange faces, IRHD 60. For details of rubber lining at flanges, joints etc., refer to the Standard Piping Drawings.
5	For preferred branch details refer to the Standard Piping Drawings. Use only laterals or Y-piece fittings for branches except for flushing connections, vents or instrument connections.
6	Not Used
7	Length of flanged spools shall be 3m maximum for DN80 and 6m maximum for DN100 and above.
8	Induction 3D bends to be used as priority. Induction 5D bends shall only be used with the approval by the Engineer.
9	1.5D elbows shall only to be used with approval by the Engineer.
10	For external surface treatment requirements refer to the Project Pipe Materials and Valve Specification.
11	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343.



RL5	Fully flanged 6mm thick Butyl lined steel piping system with flat face flanges to AS2129 Table D for low pressure slurry applications.				
Material :		Carbon Steel - Butyl Lined – 6mm Thick Lining ⁴			
Design Codes :		AS 4041, Class 3 (Note 11)			
Design Temperature (°C) ¹ :		0	70		
Design Pressure (kPa(g)) ¹ :		700	700		
Corrosion Allowance (mm) :		0.0			
Pressure Testing :		Initial service leak testing as per AS 4041. Hydrostatic pressure testing not required unless directed by the Engineer.			
Non-Destructive Testing :		Visual inspection only			
Description	Size Range	Dimensional Specification	Type	End Preparations	Material Specification
Pipe	DN50 ³	ASME B36.10M	STD WT, SMLS	Plain ends	ASTM A53 Gr B / API 5L Grade B
	DN65 ³ – 600		STD WT, ERW		
	DN650 – 1200		STD WT, LSAW		
Induction Bend, 3D or 5D ⁸	DN80 – 900	As per pipe ²	As per pipe	Plain ends	As per pipe
Elbow, 1.5D ⁹	DN80 – 900	ASME B16.9 ²	Butt Weld	Plain ends	ASTM A234 WPB
Lateral, 45°	DN80 – 900	As per pipe ²	As per pipe	Plain ends	As per pipe
Lateral, 60°					
Wye, 90°	DN80 – 900	As per pipe ²	As per pipe	Plain ends	As per pipe
Wye, 60°					
Tees ⁵	DN65 ³ – 900	As per pipe ²	As per pipe	Plain ends	As per pipe
Reducer, eccentric	DN50 ³ – 900	ASME B16.9 ²	As per large end piping	Plain ends	ASTM A234 WPB
Reducer, concentric	DN50 ³ – 900				
Flange, Slip-On	DN50 ³ – 1200	AS 2129 ²	Table D, FF	Plate	ASTM A105
Flange, Spinning	DN50 ³ – 1200			Backing Ring c/w Shoulder Plate	AS/NZS 3678 Gr 250
Flange, Blind	DN50 ³ – 1200			Blank	ASTM A105
Pipe Nipple ³	DN15 – 50	ASTM A733	XS WT, SMLS, 100mm	Threaded, BSP	ASTM A53 Gr B
Plugs ³	DN15 – 50	ASME B16.11	Class 3000	Threaded, BSP	ASTM A105
Stud Bolts, Type A	DN50 – 1200	AS/NZS 1252.1	Galvanised, Gr. 8.8 bolts		AS 4291.1
Machine Bolts, Metric ⁶	DN50 – 1200		Galvanised, Hex head, Gr. 8.8 bolts		
Nuts	DN50 – 1200		Galvanised, Gr. 8 hex nuts		AS 4291.2
Gaskets	DN50 – 1200	AS 2129	Table D, Klinger C4430 or approved equivalent, 1.5mm thick, flat face		
Rubber Lining	DN50 – 1200	6mm thick Butyl lining. Refer to Note 4 for lining thickness of fixed flanges.			

1	The design pressure and temperature conditions are based on plate flange design, STD WT fittings with no branch reinforcement and the maximum temperature limitation of the specified rubber lining. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	Refer to design drawings and the Standard Piping Drawings for dimensional details for fittings and plate flanges.
3	Pipe diameters less than DN80 shall not be used for main pipe runs and shall only be permitted for branch connections for drains, flushing points or instrumentation connections unless approved by the engineer.
4	All rubber lining to be 3mm thick lining on flange faces. For details of rubber lining at flanges, joints etc., refer to the Standard Piping Drawings.
5	For preferred branch details refer to the Standard Piping Drawings. Use only laterals or Y-piece fittings for branches except for flushing connections, vents or instrument connections.
6	Not Used
7	Length of flanged spools shall be 3m maximum for DN80 and 6m maximum for DN100 and above.
8	Induction 3D bends to be used as priority. Induction 5D bends shall only be used with the approval by the Engineer.
9	1.5D elbows shall only to be used with approval by the Engineer.
10	For external surface treatment requirements refer to the Project Pipe Materials and Valve Specification.
11	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343.



SS1	Welded Gr316 stainless steel piping system, Schedule 10S, with flanges to AS 2129 Table D for low pressure liquid or gas applications.					
Material:		Stainless Steel 316				
Codes:		AS 4041, Class 3 (Note 8)				
Design Temperature (°C) ¹ :	- 50	232	250	275	300	
Design Pressure (kPa(g)) ¹ :	700	700	650	600	570	
Corrosion Allowance (mm) :	1.0					
Pressure Testing :	Initial service leak testing as per AS 4041. Hydrostatic pressure testing not required unless directed by the Engineer.					
Non-Destructive Testing :	Visual inspection only					
Description	Size Range	Dimensional Specification		Type	End Preparations	Material Specification
Pipe	DN15 – 50	ASME B36.19M		SCH 40S, SMLS	Plain Ends	ASTM A312 GR. 316
	SCH 10S, ERW			Bevelled Ends		
	DN750				SCH 10S, LSAW	
Elbow, 90°	DN15 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A182 GR F316	
	DN65 – 750	ASME B16.9	SCH 10S	Bevelled Ends	ASTM A403 GR WP S 316	
Elbow, 45°	DN15 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A182 GR F316	
	DN65 – 750	ASME B16.9	SCH 10S	Bevelled Ends	ASTM A403 GR WP S 316	
Tee, equal	DN15 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A182 GR F316	
	DN65 – 750	ASME B16.9	SCH 10S	Bevelled Ends	ASTM A403 GR WP S 316	
Tee, reducing	DN20 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A182 GR F316	
	DN65 – 750	ASME B16.9	SCH 10S	Bevelled Ends	ASTM A403 GR WP S 316	
Cap	DN15 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A182 GR F316	
	DN65 – 750	ASME B16.9	SCH 10S	Bevelled Ends	ASTM A403 GR WP S 316	
Coupling, reducing	DN20 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A182 GR F316	
Reducer, eccentric ⁴	DN65 - 750	ASME B16.9	SCH 10S	Bevelled Ends	ASTM A403 GR WP S 316	
Reducer, concentric ⁴	DN65 - 750					
Flange, Plate, Slip-On	DN15 - 750	AS 2129	Table D, FF	Plate	ASTM A240 GR 316L	
Flange, Blind	DN15 - 750			Blank		
Flange, Plate, Slip-On	DN15 - 750	AS 2129	Table D, RF ⁷	Plate	ASTM A240 GR 316L	
Flange, Blind	DN15 - 750			Blank		
Thredoilet ⁵	DN15 – 20	MSS-SP-97	Class 3000	Bevelled End x BSP Female	ASTM A182 GR F316	
Sockolet	DN25 – 50	MSS-SP-97	Class 3000	Socket Weld	ASTM A182 GR F316	
Weldolet	DN65 – 350	MSS-SP-97	STD WT	Bevelled	ASTM A182 GR F316	
Pipe Nipple ⁵	DN15 – 50	ASTM A733	SCH 80S SMLS, 100mm	BSP Threaded	ASTM A312 GR. 316	
Hex Nipple ⁵	DN15 – 50	ASME B16.11	Class 3000	BSP Threaded	ASTM A182 GR F316	
Hex Head Bushing ⁵	DN15 – 50	ASME B16.11	Class 3000	BSP Threaded	ASTM A182 GR F316	
Hex Head Plug ⁵	DN15 – 50	ASME B16.11	Class 3000	BSP Threaded	ASTM A182 GR F316	

Stud Bolts	DN15 – 750	ISO 965-1	304SS, Gr. A2-70 bolts	ISO 3506-1
Machine Bolt, Metric ⁶			304SS, Hex head, Gr. A2-70 bolts	
Nuts			304SS, Gr. A2-70 hex nuts	ISO 3506-2
Gaskets ⁹	DN15 – 750	AS 2129	Table D, Klinger C4430 or approved equivalent, 1.5mm thick, flat face	
			Table D, Klinger TC2003 or approved equivalent, 1.5mm thick, flat face	

NOTES	
1	The design pressure and temperature conditions are based on AS2129 flange ratings. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	Refer to Piping Line List for test pressure requirements for each line.
3	The maximum length of flanged spools shall be 6 m for DN80 and 12 m maximum for DN100 and above.
4	For Non-Destructive Examination requirements refer to Piping, Fabrication, Installation and Testing Specification.
5	This specification is strictly to be welded fittings only. Threaded fittings only to be used for instrumentation or tie-in for mechanical packages where required. PTFE Thread tape or liquid sealant to be used.
6	Not Used
7	Flat Face is preferred. Raised Face to be used only as required e.g. Valves, Instrumentation, Mechanical Packages. Raised Face connecting to Flat Face is not permitted.
8	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343. Pipe system not suitable for Potable Water.
9	TC2003 to be used on all reagent services. Refer to Project Pipe Materials and Valve Specification for gasket selection

BRANCH CHART																									
Branch Size	750	ET																							
	600	RT	ET																						
	550*	RT	RT	ET																					
	500	RT	RT	RT	ET																				
	450	RT	RT	RT	RT	ET																			
	400	RT	RT	RT	RT	RT	ET																		
	350	RT	RT	RT	RT	RT	RT	ET																	
	300	RT	RT	RT	RT	RT	RT	RT	ET																
	250	RT	RT	RT	RT	RT	RT	RT	RT	ET															
	200	WO	WO	WO	RT	RT	RT	RT	RT	RT	ET														
Branch Size	150	WO	WO	WO	WO	WO	RT	RT	RT	RT	RT	ET													
	125*	WO	WO	WO	WO	WO	WO	WO	RT	RT	RT	RT	ET												
	100	WO	WO	WO	WO	WO	WO	WO	WO	RT	RT	RT	RT	ET											
	90*	WO	WO	WO	WO	WO	WO	WO	WO	WO	RT	RT	RT	RT	ET										
	80	WO	WO	WO	WO	WO	WO	WO	WO	WO	WO	RT	RT	RT	RT	ET									
	65*	WO	WO	WO	WO	WO	WO	WO	WO	WO	WO	RT	RT	RT	RT	RT	ET								
	50	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	RT	RT	RT	RT	RT	ET							
	40	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	RT	RT	RT	RT	RT	ET						
	32*	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	RT	RT	RT	RT	RT	ET				
	25	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	RT	RT	RT	RT	RT	ET			
Branch Size	20	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	RT	ET		
	15	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	RT	RT	ET
	750	600	550*	500	450	400	350	300	250	200	150	125*	100	90*	80	65*	50	40	32*	25	20	15			
Header Size																									

LEGEND	
ET	Equal Tee
RT	Reducing Tee
SO	Sockolet
TO	Thredolet
WO	Weldolet

* indicates non-standard pipe size and not preferred

SS2	Welded Gr316 stainless steel piping system, Schedule 10S, with flanges to AS 2129 Table E for medium pressure liquid or gas applications.					
Material:	Stainless Steel 316					
Codes:	AS 4041, Class 3 (Note 8)					
Design Temperature (°C) ¹ :	- 50	232	250	275	300	
Design Pressure (kPa(g)) ¹ :	1400	1400	1300	1200	1100	
Corrosion Allowance (mm) :	1.0					
Pressure Testing :	Initial service leak testing as per AS 4041. Hydrostatic pressure testing not required unless directed by the Engineer.					
Non-Destructive Testing :	Visual inspection only					

Description	Size Range	Dimensional Specification	Type	End Preparations	Material Specification
Pipe	DN25 – 50	ASME B36.19M	SCH 40S, SMLS	Plain Ends	ASTM A312 GR. 316
	DN65 – 600		SCH 10S, ERW	Bevelled Ends	
	DN750		SCH 10S, LSAW		
Elbow, 90°	DN25 – 50	ASME B16.11	Class 3000	Socket weld	ASTM A182 GR F316
	DN65 – 750	ASME B16.9	SCH 10S	Bevelled Ends	ASTM A403 GR WP S 316
Elbow, 45°	DN25 – 50	ASME B16.11	Class 3000	Socket weld	ASTM A182 GR F316
	DN65 – 750	ASME B16.9	SCH 10S	Bevelled Ends	ASTM A403 GR WP S 316
Tee, equal	DN25 – 50	ASME B16.11	Class 3000	Socket weld	ASTM A182 GR F316
	DN65 – 750	ASME B16.9	SCH 10S	Bevelled Ends	ASTM A403 GR WP S 316
Tee, reducing	DN32 – 50	ASME B16.11	Class 3000	Socket weld	ASTM A182 GR F316
	DN65 – 750	ASME B16.9	SCH 10S	Bevelled Ends	ASTM A403 GR WP S 316
Cap	DN25 – 50	ASME B16.11	Class 3000	Socket weld	ASTM A182 GR F316
	DN65 – 750	ASME B16.9	SCH 10S	Bevelled Ends	ASTM A403 GR WP S 316
Coupling, reducing	DN32 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A182 GR F316
Reducer, eccentric ⁴	DN65 - 750	ASME B16.9	SCH 10S	Bevelled Ends	ASTM A403 GR WP S 316
Reducer, concentric ⁴	DN65 - 750				
Flange, Plate, Slip-On	DN25 - 750	AS 2129	Table E, FF	Plate	ASTM A240 GR 316L
Flange, Blind	DN25 - 750			Blank	
Flange, Plate, Slip-On	DN25 - 750	AS 2129	Table E, RF ⁷	Plate	ASTM A240 GR 316L
Flange, Blind	DN25 - 750			Blank	
Thredolet ⁵	DN15 – 20	MSS-SP-97	Class 3000	Bevelled End x BSP Female	ASTM A182 GR F316
Sockolet	DN25 – 50	MSS-SP-97	Class 3000	Socket Weld	ASTM A182 GR F316
Weldolet	DN65 – 350	MSS-SP-97	STD WT	Bevelled	ASTM A182 GR F316
Pipe Nipple ⁵	DN15 – 50	ASTM A733	SCH 80S SMLS, 100mm	BSP Threaded	ASTM A312 GR. 316
Hex Nipple ⁵	DN15 – 50	ASME B16.11	Class 3000	BSP Threaded	ASTM A182 GR F316
Hex Head Bushing ⁵	DN15 – 50	ASME B16.11	Class 3000	BSP Threaded	ASTM A182 GR F316
Hex Head Plug ⁵	DN15 – 50	ASME B16.11	Class 3000	BSP Threaded	ASTM A182 GR F316

Stud Bolts	DN25 – 750	ISO 965-1	304SS, Gr. A2-70 bolts	ISO 3506-1
Machine Bolt, Metric ⁶			304SS, Hex head, Gr. A2-70 bolts	
Nuts			304SS, Gr. A2-70 hex nuts	ISO 3506-2
Gaskets ⁹	DN25 – 750	AS 2129	Table E, Klinger C4430 or approved equivalent, 1.5mm thick, flat face	
			Table E, Klinger TC2003 or approved equivalent, 1.5mm thick, flat face	

NOTES	
1	The design pressure and temperature conditions are based on AS2129 flange ratings. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	Refer to Piping Line List for test pressure requirements for each line.
3	The maximum length of flanged spools shall be 6 m for DN80 and 12 m maximum for DN100 and above.
4	For Non-Destructive Examination requirements refer to Piping, Fabrication, Installation and Testing Specification.
5	This specification is strictly to be welded fittings only. Threaded fittings only to be used for instrumentation or tie-in for mechanical packages where required. PTFE Thread tape or liquid sealant to be used.
6	Not Used
7	Flat Face is preferred. Raised Face to be used only as required e.g. Valves, Instrumentation, Mechanical Packages. Raised Face connecting to Flat Face is not permitted.
8	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343. Pipe system not suitable for Potable Water.
9	TC2003 to be used on all reagent services. Refer to Project Pipe Materials and Valve Specification for gasket selection

BRANCH CHART																								
Branch Size	750	ET																						
	600	RT	ET																					
	550*	RT	RT	ET																				
	500	RT	RT	RT	ET																			
	450	RT	RT	RT	RT	ET																		
	400	RT	RT	RT	RT	RT	ET																	
	350	RT	RT	RT	RT	RT	RT	ET																
	300	RT	RT	RT	RT	RT	RT	RT	ET															
	250	RT	RT	RT	RT	RT	RT	RT	RT	ET														
	200	WO	WO	WO	RT	RT	RT	RT	RT	RT	ET													
Branch Size	150	WO	WO	WO	WO	WO	RT	RT	RT	RT	RT	ET												
	125*	WO	WO	WO	WO	WO	WO	WO	RT	RT	RT	RT	ET											
	100	WO	WO	WO	WO	WO	WO	WO	WO	RT	RT	RT	RT	ET										
	90*	WO	WO	WO	WO	WO	WO	WO	WO	WO	RT	RT	RT	RT	ET									
	80	WO	WO	WO	WO	WO	WO	WO	WO	WO	WO	RT	RT	RT	RT	ET								
	65*	WO	WO	WO	WO	WO	WO	WO	WO	WO	WO	WO	RT	RT	RT	RT	RT	ET						
	50	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	RT	RT	RT	RT	RT	ET					
	40	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	RT	RT	RT	RT	RT	ET				
	32*	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	RT	RT	RT	RT	RT	ET		
	25	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	RT	RT	RT	RT	RT	ET	
Branch Size	20	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	RT	ET		
	15	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	RT	RT	ET
		750	600	550*	500	450	400	350	300	250	200	150	125*	100	90*	80	65*	50	40	32*	25	20	15	
Header Size																								

LEGEND	
ET	Equal Tee
RT	Reducing Tee
SO	Sockolet
TO	Thredolet
WO	Weldolet

* indicates non-standard pipe size and not preferred

SS3	Viega Megapress (cold press) Gr316 stainless steel piping system with flanges to AS 2129 Table E for medium pressure liquid or gas applications.					
Material :		Stainless Steel 316				
Design Codes :		AS 4041, Class 3 (Note 8)				
Design Temperature (°C) ¹ :		-5	140			
Design Pressure (kPa(g)) ¹ :		1400	1400			
Corrosion Allowance (mm) :		0.0				
Pressure Testing :		Initial service leak testing as per AS 4041 ⁷ . Hydrostatic pressure testing not required unless directed by the Engineer.				
Non-Destructive Testing :		Visual inspection only				
Description	Size Range	Dimensional Specification	Type	End Preparations	Material Specification	
Pipe	DN15 – 50	ASME B36.19M	Sch 40S, SMLS	Plain Ends	ASTM A312 GR. 316	
Elbow, 90°	DN15 – 50	Manufacturer Std	Model 6816	Press Connector	Manufacturer Std	
			Model 6816.1	Plain End x Press Connector		
Elbow, 45°	DN15 – 50	Manufacturer Std	Model 6826	Press Connector	Manufacturer Std	
			Model 6826.1	Plain End x Press Connector		
Tee, equal ⁴	DN15 – 50	Manufacturer Std	Model 6818	Press Connector	Manufacturer Std	
			Model 6817.2	Plain End x Threaded, NPT ¹⁰		
Tee, reducing ⁴	DN15 – 50	Manufacturer Std	Model 6818	Press Connector	Manufacturer Std	
			Model 6817.2	Plain End x Threaded, NPT ¹⁰		
Cap	DN15 – 50	Manufacturer Std	Model 6856	Press Connector	Manufacturer Std	
Union, Screw Fitting ⁴	DN15 – 50	Manufacturer Std	Model 6865	Press Connector x Threaded, NPT ¹⁰ Male	Manufacturer Std	
			Model 6863	Press Connector x Threaded, NPT ¹⁰		
Coupling	DN15 – 50	Manufacturers Standard	Model 6815	Press Connector	Manufacturer Std	
Coupling, Sliding	DN15 – 50	Manufacturers Standard	Model 6815.5	Press Connector		
Reducer	DN20 - 50	Manufacturer Std	Model 6815.1	Plain End x Press Connector	Manufacturer Std	
Adapter, threaded ⁴	DN15 – 50	Manufacturer Std	Model 6811	Press Connector x Threaded, NPT ¹⁰ Male	Manufacturer Std	
			Model 6812	Press Connector x Threaded, NPT ¹⁰		
Flange	DN15 - 25	Manufacturers Standard & AS 2129	6359, Table E FF	Press Connector	Manufacturer Std	
			6359, Table E RF ⁶			
	DN32 - 50		6359.1, Table E FF			
			6359.1, Table E RF ⁶			
Flange, Blind	DN25 - 50	AS 2129	Table E, FF	Blank	ASTM A240 GR 316L	
Hex Head Bushing ^{4, 10}	DN15 – 50	ASME B16.11	Class 3000	Threaded, NPT ¹⁰	ASTM A182 GR F316	

Hex Head Nipple ^{4, 10}	DN15 – 50	ASME B16.11	Class 3000	Threaded, NPT ¹⁰	ASTM A182 GR F316
				Threaded, NPT x BSP ¹⁰	ASTM A182 GR F316
Plugs ⁴	DN15 – 50	ASME B16.11	Class 3000	Threaded, NPT ¹⁰	ASTM A182 GR F316
Stud Bolts	DN32 - 50	ISO 965-1	304SS, Gr. A2-70 bolts		ISO 3506-1
Machine Bolts, Metric ⁵	DN32 - 50		304SS, Hex head, Gr. A2-70 bolts		
Nuts	DN32 - 50		304SS, Gr. A2-70 hex nuts		ISO 3506-2
Gaskets ¹¹	DN32 - 50	AS 2129	Table E, Klinger C4430 or approved equivalent, 1.5mm thick, flat face		
			Table E, Klinger TC2003 or approved equivalent, 1.5mm thick, flat face		
Sealing element	DN15 – 50	Megapress S	Model 4386		FKM

NOTES

1	The design pressure and temperature conditions are based on AS2129 flange ratings and are within Viega Megapress joints and fitting pressure and temperature ratings. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	Not Used
3	The maximum length of spools shall be 6 m. Couplings preferred over flanges for spool connections.
4	This specification is strictly to be press connection fittings only. Threaded fittings only to be used for instrumentation or mechanical package tie-in where required. BSP thread can be used subject to Engineer approval. PTFE Thread tape or liquid sealant to be used.
5	Not Used
6	Flat Face is preferred. Raised Face to be used only as required e.g. Valves, Instrumentation, Mechanical Packages. Raised Face connecting to Flat Face is not allowed.
7	All fittings have Viega Smart Connect Feature that assure leakage of sealing element of an unpressed connection during the leak test.
8	Piping system not suitable for Lethal Fluids and limited to DN50 maximum size for Very Harmful Fluids as defined by AS 4343.
9	The maximum length between pipe clamps shall be 4 m.
10	BSP Thread not available in MegaPress Stainless Steel. Use BSP x NPT adaptor accordingly.
11	TC2003 to be used on all reagent services. Refer to Project Pipe Materials and Valve Specification for gasket selection

BRANCH CHART

Branch Size											
	50	ET									
	40	ET+RED	ET								
	32	RT	ET+RED	ET							
	25	RT+RED	RT	ET+RED	ET						
	20	RT	RT+RED	RT	ET+RED	ET					
15	RT+RED	RT	RT+RED	RT	ET+RED	ET					
	50	40	32	25	20	15					
Header Size											

LEGEND

ET

Equal Tee Press Connection

ET+RED

Equal Tee Press Conn. + Reducer Press Conn.

RT

Reducing Tee Press Connection

RT+RED

Reducing Tee Press Conn. + Reducer Press Conn.

SS4	Vic-Press Grade 316L stainless steel piping system with flanges to AS 2129 Table E for low pressure potable water applications.				
Material :		Stainless Steel 316			
Design Codes :		AS 4041, Class 3 (Note 7)			
Design Temperature (°C) ¹ :		-5	85		
Design Pressure (kPa(g)) ¹ :		700	700		
Corrosion Allowance (mm) :		0.0			
Pressure Testing :		Initial service leak testing as per AS 5200.053 ⁶ . Hydrostatic pressure testing not required unless directed by the Engineer.			
Non-Destructive Testing :		Visual inspection only			
Description	Size Range	Dimensional Specification	Type	End Preparations	Material Specification
Pipe	DN15 – 50	ASME B36.19M	Schedule 10S SMLS	Plain Ends	ASTM A312 TP316L
Elbow, 90°	DN15 – 50	Victaulic 18.11	Style P568 (P x P)	Manufacturer Std	ASTM A312 TP316L
Street Elbow, 90°			Style P562 (P x T)	Manufacturer Std	
Elbow, 45°			Style P571 (P x P)	Manufacturer Std	
Street Elbow, 45°			Style P563 (P x T)	Manufacturer Std	
Tee, equal			Style P572 (P x P x P)	Manufacturer Std	
Tee, reducing ³			Style P573 (P x P x P)	Manufacturer Std	
Tee, threaded			Style P578 (P x P x F)	Threaded, BSP	
Reducer, concentric			Style P574 (P x P)	Manufacturer Std	
Adaptor, male thread			Style P576 (P x M)	Threaded, BSP	
Adaptor, female thread			Style P579 (P x F)		
Coupling, standard			Style P507 (P x P)	Manufacturer Std	
Union, threaded			Style P585 (P x P)		
End Plug			Style P560 (T)		
Flange	DN15 – 50	AS 2129 Table E	Flange adaptor Style P575 (P x L)	Raised Face	Type 316L
Flange, blind	DN15 – 50		Plate, blind		ASTM A182 F316L
Stud Bolts	DN15 – 50	ISO 965-1	304SS, Gr. A2-70 bolts		ISO 3506-1
Machine Bolts, Metric ⁷			304SS, Hex head, Gr. A2-70 bolts		
Nuts			304SS, Gr. A2-70 hex nuts		ISO 3506-2
Coupling O-rings	DN15 – 50	Victaulic 18.11	EPDM, Grade “E”, green stripe colour code.		
Gaskets		AS 2129	Klinger C4430 or approved equivalent, Table E, 1.5mm thick raised-face		
Thread Sealant	DN15 – 50	-	Thread tape or liquid sealant approved to AS4020, Watermarked for Potable Water service.		PTFE (Watermarked)

1	The design pressure and temperature conditions are based on AS2129 flange ratings and are within Victaulic Vic-Press joint and fitting pressure ratings. Permissible pipe system temperatures and pressures are subject to O-ring, gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	M (Male) - Male Thread, F (Female) - Female Thread
3	Press fit end preparations are grooved manufacturer prepared components, complete with O-ring suitable for the fluid service being conveyed through the piping system.
4	Lubricate threads and ensure minimum one thread exposed after installation. Loctite “Anti-Seize” or approved equivalent to be used on bolted connections.
5	Vic-Press pipe end preparation to comply with the latest Vic-Press I-P500 Field Installation Handbook.
6	Not Used
7	Not Used
8	For external surface treatment requirements refer to the Project Pipe Materials and Valve Specification.
9	All valves and instrumentation shall be approved to AS 4020 for Potable Water service.
10	Piping system not suitable for Lethal Fluids as defined by AS 4343.

BRANCH CHART											
Branch Size											
	50	ET									
	40	RT	ET								
	32	RT	RT	ET							
	25	RT	RT	RT	ET						
	20	RT	RT	RT	RT	ET					
	15	RT+RED	RT+RED	RT	RT	RT	ET				
		50	40	32	25	20	15				
Header Size											
Header Size											

LEGEND	
ET	Equal Tee Press Connection
ET+RED	Equal Tee Press Conn. + Reducer Press Conn.
RT	Reducing Tee Press Connection
RT+RED	Reducing Tee Press Conn. + Reducer Press Conn.

SS5	Butt welded Gr316 stainless steel piping system, Schedule 10S, with flanges to AS 2129 Table E for hydrogen peroxide duty.					
Material:		Stainless Steel 316				
Codes:		AS 4041, Class 3 (Note 8)				
Design Temperature (°C) ¹ :		- 50	232	250	275	300
Design Pressure (kPa(g)) ¹ :		1400	1400	1300	1200	1100
Corrosion Allowance (mm) :		1.0				
Pressure Testing :		Hydrostatic Test as per AS 4041				
Non-Destructive Testing :		100% Visual, 10% RT and 10% PT				
Description		Size Range	Dimensional Specification	Type	End Preparations	Material Specification
Pipe		DN25 – 50	ASME B36.19M	SCH 40S, SMLS	Bevelled Ends	ASTM A312 GR. 316
		DN65 – 150		SCH 10S, SMLS		
Elbow, 90°		DN25 – 150	ASME B16.9	As per pipe WT	Bevelled Ends	ASTM A403 GR WP S 316
Elbow, 45°		DN25 – 150	ASME B16.9	As per pipe WT	Bevelled Ends	ASTM A403 GR WP S 316
Tee, equal		DN25 – 150	ASME B16.9	As per pipe WT	Bevelled Ends	ASTM A403 GR WP S 316
Tee, reducing		DN25 – 150	ASME B16.9	As per pipe WT	Bevelled Ends	ASTM A403 GR WP S 316
Cap		DN25 – 150	ASME B16.9	As per pipe WT	Bevelled Ends	ASTM A403 GR WP S 316
Union		DN25 – 150	ASME B16.9	Class 3000	Bevelled Ends	ASTM A182 GR F316
Reducer, eccentric ⁴		DN32 - 150	ASME B16.9	As per pipe WT	Bevelled Ends	ASTM A403 GR WP S 316
Reducer, concentric ⁴						
Flange, Weld Neck		DN25 – 150	AS 2129	Table E, FF	Bevelled End	ASTM A182 GR F316
Flange, Blind		DN25 – 150			Blank	
Flange, Weld Neck		DN25 – 150	AS 2129	Table E, RF ⁷	Bevelled End	ASTM A182 GR F316
Flange, Blind		DN25 – 150			Blank	
Threadolet ⁵		DN15 – 20	MSS-SP-97	Class 3000	Bevelled End x BSP Female	ASTM A182 GR F316
Weldolet		DN15 – 150	MSS-SP-97	Class 3000	Bevelled	ASTM A182 GR F316
Pipe Nipple ⁵		DN15 – 50	ASME B16.11	Class 3000	BSP Threaded	ASTM A182 GR F316
Hex Nipple ⁵		DN15 – 50	ASME B16.11	Class 3000	BSP Threaded	ASTM A182 GR F316
Hex Head Bushing ⁵		DN15 – 50	ASME B16.11	Class 3000	BSP Threaded	ASTM A182 GR F316
Stud Bolts		DN25 – 150	ISO 965-1	304SS, Gr. A2-80 bolts		ISO 3506-1
Machine Bolt, Metric ⁶				304SS, Hex head, Gr. A2-80 bolts		
Nuts				304SS, Gr. A2-80 hex nuts		ISO 3506-2
Gasket		DN25 – 150	AS 2129	Table E, Klinger top-chem 2003 or approved equivalent, 1.5mm thick, flat face		

1	The design pressure and temperature conditions are based on AS2129 flange ratings. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	Refer to Piping Line List for test pressure requirements for each line.
3	The maximum length of flanged spools shall be 6 m for DN80 and 12 m maximum for DN100 and above.
4	For Non-Destructive Examination requirements refer to Piping Fabrication, Installation and Testing Specification.
5	This specification is strictly to be welded fittings only. Threaded fittings only to be used for instrumentation or tie-in for mechanical packages where required. PTFE Thread tape or liquid sealant to be used.
6	Not Used
7	Flat Face is preferred. Raised Face to be used only as required e.g. Valves, Instrumentation, Mechanical Packages. Raised Face connecting to Flat Face is not allowed.
8	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343. Pipe system not suitable for Potable Water.
9	All piping, fittings, valves and instruments to be cleaned, pickled and passivated in accordance to ASTM A380/A380M and ASTM A967/A967M prior commissioning for hydrogen peroxide service. Piping fabricator to test spools with diluted hydrogen peroxide test solution and observe for any hydrogen peroxide to decompose and emit bubbles. If no bubbles are seen, drain the test solution and rinse with demineralised water. Seal all openings to avoid contamination. If bubble are seen, remove the source and repeat pickling and passivation procedures.

BRANCH CHART

Branch Size		<table><tr><th colspan="2">LEGEND</th></tr><tr><td>ET</td><td>Equal Tee</td></tr><tr><td>RT</td><td>Reducing Tee</td></tr><tr><td>WO</td><td>Weldolet</td></tr></table>										LEGEND		ET	Equal Tee	RT	Reducing Tee	WO	Weldolet
	LEGEND																		
	ET	Equal Tee																	
	RT	Reducing Tee																	
	WO	Weldolet																	
	150	ET																	
	125*	RT	ET																
	100	RT	RT	ET															
	90*	RT	RT	RT	ET														
	80	RT	RT	RT	RT	ET													
	65*	RT	RT	RT	RT	RT	ET												
	50	WO	WO	WO	WO	WO	WO	ET											
40	WO	WO	WO	WO	WO	WO	RT	ET											
32*	WO	WO	WO	WO	WO	WO	RT	RT	ET										
25	WO	WO	WO	WO	WO	WO	RT	RT	RT	ET									
20	WO	WO	WO	WO	WO	WO	RT	RT	RT	RT									
15	WO	WO	WO	WO	WO	WO	WO	RT	RT	RT									
	150	125*	100	90*	80	65*	50	40	32*	25									
Header Size																			

* indicates non-standard pipe size and not preferred

SS6	Butt welded Gr316 stainless steel piping system, Schedule 10S, with flanges to AS 2129 Table E for oxygen duty.					
Material:		Stainless Steel 316				
Codes:		AS 4041, Class 3 (Note 8)				
Design Temperature (°C) ¹ :	- 50	232	250	275	300	
Design Pressure (kPa(g)) ¹ :	1400	1400	1300	1200	1100	
Corrosion Allowance (mm) :	1.0					
Pressure Testing :	Pneumatic Test as per AS 4041					
Non-Destructive Testing :	100% Visual, 10% RT and 10% PT					
Description	Size Range	Dimensional Specification	Type	End Preparations	Material Specification	
Pipe	DN25 – 50	ASME B36.19M	SCH 40S, SMLS	Bevelled Ends	ASTM A312 GR. 316	
	DN65 – 150		SCH 10S, SMLS			
Elbow, 90°	DN25 – 150	ASME B16.9	As per pipe WT	Bevelled Ends	ASTM A403 GR WP S 316	
Elbow, 45°	DN25 – 150	ASME B16.9	As per pipe WT	Bevelled Ends	ASTM A403 GR WP S 316	
Tee, equal	DN25 – 150	ASME B16.9	As per pipe WT	Bevelled Ends	ASTM A403 GR WP S 316	
Tee, reducing	DN25 – 150	ASME B16.9	As per pipe WT	Bevelled Ends	ASTM A403 GR WP S 316	
Cap	DN25 – 150	ASME B16.9	As per pipe WT	Bevelled Ends	ASTM A403 GR WP S 316	
Union	DN25 – 150	ASME B16.9	Class 3000	Bevelled Ends	ASTM A182 GR F316	
Reducer, eccentric ⁴	DN32 - 150	ASME B16.9	As per pipe WT	Bevelled Ends	ASTM A403 GR WP S 316	
Reducer, concentric ⁴						
Flange, Weld Neck	DN25 – 150	AS 2129	Table E, FF	Bevelled End	ASTM A182 GR F316	
Flange, Blind	DN25 – 150			Blank		
Flange, Weld Neck	DN25 – 150	AS 2129	Table E, RF ⁷	Bevelled End	ASTM A182 GR F316	
Flange, Blind	DN25 – 150			Blank		
Thredolet ⁵	DN15 – 20	MSS-SP-97	Class 3000	Bevelled End x BSP Female	ASTM A182 GR F316	
Weldolet	DN15 – 150	MSS-SP-97	As per pipe WT	Bevelled	ASTM A182 GR F316	
Pipe Nipple ⁵	DN15 – 50	ASME B16.11	Class 3000	BSP Threaded	ASTM A182 GR F316	
Hex Nipple ⁵	DN15 – 50	ASME B16.11	Class 3000	BSP Threaded	ASTM A182 GR F316	
Hex Head Bushing ⁵	DN15 – 50	ASME B16.11	Class 3000	BSP Threaded	ASTM A182 GR F316	
Stud Bolts	DN25 – 150	ISO 965-1	304SS, Gr. A2-70 bolts		ISO 3506-1	
Machine Bolt, Metric ⁶			304SS, Hex head, Gr. A2-70 bolts			
Nuts			304SS, Gr. A2-70 hex nuts		ISO 3506-2	
Gasket	DN25 – 150	AS 2129	Table E, Klinger TC2003 or approved equivalent, 1.5mm thick, flat face			

1	The design pressure and temperature conditions are based on AS2129 flange ratings. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	Refer to Piping Line List for test pressure requirements for each line.
3	The maximum length of flanged spools shall be 6 m for DN80 and 12 m maximum for DN100 and above.
4	For Non-Destructive Examination requirements refer to Piping Fabrication, Installation and Testing Specification.
5	This specification is strictly to be welded fittings only. Threaded fittings only to be used for instrumentation or tie-in for mechanical packages where required. PTFE Thread tape or liquid sealant to be used.
6	Machine bolts shall be used with lugged valves that have tapped blind holes.
7	Flat Face is preferred. Raised Face to be used only as required e.g. Valves, Instrumentation, Mechanical Packages. Raised Face connecting to Flat Face is not allowed.
8	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343. Pipe system not suitable for Potable Water.
9	All piping, fittings, valves and instruments to be cleaned, pickled and passivated in accordance to Piping Fabrication, Installation and Testing Specification prior to commissioning for oxygen service. Seal all openings to avoid contamination. Following cleaning process, components shall be labelled "SUITABLE FOR OXYGEN SERVICE"
10	Supplier shall submit their cleaning, drying and packaging procedure for Approval. Inspection and record keeping shall be in accordance with EIGA "Oxygen Pipeline and Piping System"

BRANCH CHART

Branch Size		<table><tr><th colspan="2">LEGEND</th></tr><tr><td>ET</td><td>Equal Tee</td></tr><tr><td>RT</td><td>Reducing Tee</td></tr><tr><td>WO</td><td>Weldolet</td></tr></table>										LEGEND		ET	Equal Tee	RT	Reducing Tee	WO	Weldolet
	LEGEND																		
	ET	Equal Tee																	
	RT	Reducing Tee																	
	WO	Weldolet																	
	150	ET																	
	125*	RT	ET																
	100	RT	RT	ET															
	90*	RT	RT	RT	ET														
	80	RT	RT	RT	RT	ET													
65*	RT	RT	RT	RT	RT	ET													
50	WO	WO	WO	WO	WO	WO	ET												
40	WO	WO	WO	WO	WO	WO	RT	ET											
32*	WO	WO	WO	WO	WO	WO	RT	RT	ET										
25	WO	WO	WO	WO	WO	WO	RT	RT	RT	ET									
20	WO	WO	WO	WO	WO	WO	RT	RT	RT	RT									
15	WO	WO	WO	WO	WO	WO	WO	RT	RT	RT									
		150	125*	100	90*	80	65*	50	40	32*	25								
		Header Size																	

* indicates non-standard pipe size and not preferred

SS7	Welded 2205 Duplex Stainless Steel piping system, Schedule 10S, with flanges to AS 2129 Table E for hot pregnant eluate solution or hydrochloric acid applications.				
Material:		2205 Duplex Stainless Steel			
Codes:		AS 4041, Class 3 (Note 8)			
Design Temperature (°C) ¹ :		- 50	200	250	300
Design Pressure (kPa(g)) ¹ :		1,400	1,380	1,210	1,020
Corrosion Allowance (mm) :		1.6			
Pressure Testing :		Hydrostatic Test as per AS 4041			
Non-Destructive Testing :		100% Visual, 10% UT and 10% PT			
Description	Size Range	Dimensional Specification	Type	End Preparations	Material Specification
Pipe	DN15 – 50	ASME B36.19M	SCH 80S, SMLS	Plain Ends	ASTM A790-S31803/S32205
	DN65 – 200		SCH 10S, SMLS	Bevelled Ends	
Elbow, 90°	DN15 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A182- F60
	DN65 – 200	ASME B16.9	SCH 10S	Bevelled Ends	ASTM A815-WP S31803-S/S32205-S
Elbow, 45°	DN15 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A182- F60
	DN65 – 200	ASME B16.9	SCH 10S	Bevelled Ends	ASTM A815-WP S31803-S/S32205-S
Tee, equal	DN15 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A182- F60
	DN65 – 200	ASME B16.9	SCH 10S	Bevelled Ends	ASTM A815-WP S31803-W/S32205-S
Tee, reducing	DN20 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A182- F60
	DN65 – 200	ASME B16.9	SCH 10S	Bevelled Ends	ASTM A815-WP S31803-W/S32205-S
Cap	DN15 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A182- F60
	DN65 – 200	ASME B16.9	SCH 10S	Bevelled Ends	ASTM A815-WP S31803-S/S32205-S
Coupling, reducing	DN20 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A182- F60
Reducer, eccentric ⁴	DN65 – 200	ASME B16.9	SCH 10S	Bevelled Ends	ASTM A815-WP S31803-S/S32205-S
Reducer, concentric ⁴	DN65 – 200				
Flange, Plate, Slip-On	DN15 - 200	AS 2129	Table E, FF	Plate	ASTM A182- F60
Flange, Blind	DN15 - 200			Blank	
Flange, Plate, Slip-On	DN15 - 200	AS 2129	Table E, RF ⁷	Plate	ASTM A182- F60
Flange, Blind	DN15 - 200			Blank	
Thredoilet ⁵	DN15 – 20	MSS-SP-97	Class 3000	Bevelled End x BSP Female	ASTM A182- F60
Sockolet	DN25 – 50	MSS-SP-97	Class 3000	Socket Weld	ASTM A182- F60
Pipe Nipple ⁵	DN15 – 50	ASTM A733	SCH 80S, SMLS, 100mm	Threaded, BSP	ASTM A790-S31803/S32205
Hex Nipple ⁵	DN15 – 50	ASME B16.11	Class 3000	Threaded, BSP	ASTM A182- F60
Hex Head Bushing ⁵	DN15 – 50	ASME B16.11	Class 3000	Threaded, BSP	ASTM A182- F60
Hex Head Plug ⁵	DN15 – 50	ASME B16.11	Class 3000	Threaded, BSP	ASTM A182- F60

Stud Bolts	DN15 - 200	ISO 965-1	304SS, Gr. A2-70 bolts	ISO 3506-1
Machine Bolt, Metric ⁶			304SS, Hex head, Gr. A2-70 bolts	
Nuts			304SS, Gr. A2-70 hex nuts	ISO 3506-2
Gaskets	DN15 - 200	AS 2129	Table E, Klinger TC2003 or approved equivalent, 1.5mm thick, flat face	

NOTES	
1	The design pressure and temperature conditions are based on AS2129 Table E flange ratings. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	Refer to Piping Line List for test pressure requirements for each line.
3	The maximum length of flanged spools shall be 6 m for DN80 and 12 m maximum for DN100 and above.
4	For Non-Destructive Examination requirements refer to Piping, Fabrication, Installation and Testing Specification.
5	This specification is strictly to be welded fittings only. Threaded fittings only to be used for instrumentation or tie-in for mechanical packages where required. PTFE Thread tape or liquid sealant to be used.
6	Not Used
7	Flat Face is preferred. Raised Face to be used only as required e.g. Valves, Instrumentation, Mechanical Packages. Raised Face connecting to Flat Face is not permitted.
8	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343. Pipe system not suitable for Potable Water.

BRANCH CHART												
Branch Size		LEGEND ET Equal Tee RT Reducing Tee SO Sockolet WO Weldolet										
	200	ET										
	150	ET	ET									
	125*	RT	RT	ET								
	100	RT	RT	RT	ET							
	90*	RT	RT	RT	RT	ET						
	80	RT	RT	RT	RT	RT	ET					
	65*	RT	RT	RT	RT	RT	RT	ET				
	50	SO	SO	SO	SO	SO	SO	SO	ET			
	40	SO	SO	SO	SO	SO	SO	SO	RT	ET		
	32*	SO	SO	SO	SO	SO	SO	SO	RT	RT	ET	
	25	SO	SO	SO	SO	SO	SO	SO	RT	RT	RT	ET
	20	SO	SO	SO	SO	SO	SO	SO	RT	RT	RT	RT
15	SO	SO	SO	SO	SO	SO	SO	SO	RT	RT	RT	
		200	150	125*	100	90*	80	65*	50	40	32*	25
		Header Size										

* indicates non-standard pipe size and not preferred

SS8	Swagelok fitting Gr316 stainless steel tube system, with SAE flanges for standard pressure hydraulic/lubrication applications.									
Material:		Stainless Steel 316								
Codes:		AS 4041, Class 1								
Design Temperature (°C) ¹ :		-20 °C to 93 °C								
Design Pressure (kPa(g) [psi(g)]) ^{1,7,8} :		27,579 [4,000]								
Pipe OD (inch [mm]) ⁷ :		5/16 [8]	3/8 [10]	1/2 [12]	5/8 [16]	3/4 [20] ⁸				
Pipe Wall Thick. (inch) ⁷ :		0.035	0.049	0.065	0.065	0.109				
Corrosion Allowance (mm) :		0								
Pressure Testing :		Hydrostatic pressure test with service fluid to test pressure on Line List for 15 minutes								
Non-Destructive Testing :		100% Visual inspection as per AS 4037								

Description	Size Range	Dimensional Specification	Type	End Preparations	Material Specification
Tube	OD8 - 22	Manufacturers Std ^{5,7}	Cold drawn, annealed SMLS tube	Plain Ends	ASTM A213M/A269M
Elbow, 90°	OD8 - 22	Manufacturers Std ⁵	Imperial	Tube Fitting End	316 SS ⁵
Tee, equal	OD8 - 22	Manufacturers Std ⁵	Imperial	Tube Fitting End	316 SS ⁵
Tee, reducing	OD8 - 22	Manufacturers Std ⁵	Imperial	Tube Fitting End	316 SS ⁵
Union, equal	OD8 - 22	Manufacturers Std ⁵	Imperial	Tube Fitting End	316 SS ⁵
Union, reducing	OD8 - 22	Manufacturers Std ⁵	Imperial	Tube Fitting End	316 SS ⁵
Adapter, threaded	OD8 - 22	Manufacturers Std ⁵	Imperial	Tube Fitting End x Threaded, BSPP Male ³	316 SS ⁵
Adapter, threaded	OD8 - 22	Manufacturers Std ⁵	Imperial	Tube Fitting End x Threaded, BSPP Female	316 SS ⁵
Cap	OD8 - 22	Manufacturers Std ⁵	Imperial	Plain End	316 SS ⁵
Flange ⁶	DN15 - 20	ISO 6162-2 (SAE Code 62) ³	Single-Part Flange	Threaded, BSPP Female, Flange	EN 1.4404 - V4A (316L)
				Threaded, BSPP Female, Counterflange	
Flange, Blind ⁶	DN15 - 20	ISO 6162-2 (SAE Code 62) ³	Single-Part Flange	Threaded, BSPP Female, Flange	EN 1.4404 - V4A (316L)
				Threaded, BSPP Female, Counterflange	
Machine Bolt, Metric	DN15 - 20	ISO 4762	Hex. socket head cap machined steel bolts, Gr. 10.9		ISO 898-1
Spring Ring, Metric		DIN 7980	Spring rings for metric bolts		ISO 3506-1 A2
Seal	DN15 - 20	ISO 3601-1	NBR O-ring, 90 IRHD		

Datasheet Title :	Piping System SS8	
Document Number :	2000-F00-DSH-PP-SS8	
EPC	Project Number :	

NOTES	
1	The Design pressure and temperature conditions are based on Swagelok 316 SS fittings with their recommended 4:1 design factor based on hydraulic fluid leakage. Permissible pipe system temperatures and pressures are subject to valve & instrumentation selection by the Equipment Package Vendor for the oil/hydraulic system being supplied at detail design.
2	All tubes should be supported on both sides of a bend, as close to the bend radius as possible. All tubes shall be supported on both sides of a valve or fitting, the maximum spacing between supports either side of a valve or fitting shall be 0.9 m for OD10 and below and 1.5 m for OD15 to OD22.
3	ISO 6162-1 (SAE Code 61) to be used to match vendor connection if required. Engineer approval required as design pressure of piping system is to be derated to suit.
4	Tube supports shall be single or twin bolted plastic clamp supports or equivalent, complete with mounting rail and rail nuts mounting kit.
5	Swagelok or Hy-Lok tube fittings or equivalent as approved by the Engineer.
6	Single-part flanges with metric bolts, spring ring and O-ring shall be matched with with single-part counterflange.
7	Imperial tubing shall be used. The use of metric tubing is permitted only with prior approval from the Engineer, and must meet the specified pressure rating requirements.
8	3/4" Tubing Design Pressure is suitable up to 5,800 psi(g)

SS9	Swagelok fitting Gr316 stainless steel tube system, with SAE flanges for standard pressure hydraulic/lubrication applications.										
Material:		Stainless Steel 316									
Codes:		AS 4041, Class 1									
Design Temperature (°C) ¹ :		-20 °C to 93 °C									
Des. Press. (kPa(g) [psi(g)]) ^{1, 7} :		15,838 [2,300]			15,168 [2,200]						
Pipe OD (inch [mm]) ⁷ :		1 [25]	1.25 [32]	1.5 [38]	2 [50]						
Pipe Wall Thick. (inch) ⁷ :		0.065	0.083	0.095	0.12						
Corrosion Allowance (mm) :		0									
Pressure Testing :		Hydrostatic pressure test with service fluid to test pressure on Line List for 15 minutes									
Non-Destructive Testing :		100% Visual inspection and sensitive leak test as per AS 4037.									
Description		Size Range		Dimensional Specification		Type		End Preparations		Material Specification	
Tube		OD25 - 50		Manufacturers Std ^{5, 7}		Cold drawn, annealed SMLS tube		Plain Ends		ASTM A213M/A269M	
Elbow, 90°		OD25 - 50		Manufacturers Std ⁵		Imperial		Tube Fitting End		316 SS ⁵	
Tee, equal		OD25 - 50		Manufacturers Std ⁵		Imperial		Tube Fitting End		316 SS ⁵	
Tee, reducing		OD25 - 50		Manufacturers Std ⁵		Imperial		Tube Fitting End		316 SS ⁵	
Union, equal		OD25 - 50		Manufacturers Std ⁵		Imperial		Tube Fitting End		316 SS ⁵	
Union, reducing		OD25 - 50		Manufacturers Std ⁵		Imperial		Tube Fitting End		316 SS ⁵	
Adapter, threaded		OD25 - 50		Manufacturers Std ⁵		Imperial		Tube Fitting End x Threaded, BSPP Male ³		316 SS ⁵	
Adapter, threaded		OD25 - 50		Manufacturers Std ⁵		Imperial		Tube Fitting End x Threaded, BSPP Female ³		316 SS ⁵	
Cap		OD25 - 50		Manufacturers Std ⁵		Imperial		Plain End		316 SS ⁵	
Flange ⁶		DN25 - 50		ISO 6162-1 (SAE Code 61) ³		Single-Part Flange		Threaded, BSPP Female, Flange		EN 1.4404 - V4A (316L)	
								Threaded, BSPP Female, Counterflange			
Flange, Blind ⁶		DN25 - 50		ISO 6162-1 (SAE Code 61) ³		Single-Part Flange		Threaded, BSPP Female, Flange		EN 1.4404 - V4A (316L)	
								Threaded, BSPP Female, Counterflange			
Machine Bolt, Metric		DN25 - 50		ISO 4762		Hex. socket head cap machined steel bolts, Gr. 10.9		ISO 898-1			
Spring Ring, Metric				DIN 7980		Spring rings for metric bolts		ISO 3506-1 A2			
Seal		DN25 - 50		ISO 3601-1		NBR O-ring, 90 IRHD					

1	The Design pressure and temperature conditions are based on Swagelok 316 SS fittings with their recommended 4:1 design factor based on hydraulic fluid leakage. Permissible pipe system temperatures and pressures are subject to valve & instrumentation selection by the Equipment Package Vendor for the oil/hydraulic system being supplied at detail design.
2	All tubes should be supported on both sides of a bend, as close to the bend radius as possible. All tubes shall be supported on both sides of a valve or fitting, the maximum spacing between supports either side of a valve or fitting shall be 2.1 m.
3	ISO 6162-2 (SAE Code 62) to be used to match vendor connection if required. Engineer approval required as design pressure of piping system is to be derated to suit.
4	Tube supports shall be single or twin bolted plastic clamp supports or equivalent, complete with mounting rail and rail nuts mounting kit.
5	Swagelok or Hy-Lok tube fittings or equivalent as approved by the Engineer.
6	Single-part flanges with metric bolts, spring ring and O-ring shall be matched with with single-part counterflange.
7	Imperial tubing shall be used. The use of metric tubing is permitted only with prior approval from the Engineer, and must meet the specified pressure rating requirements.

SS10	Welded Gr316 stainless steel piping system, Schedule 40S, with flanges to AS 2129 Table H for high pressure liquid or gas applications.					
Material:		Stainless Steel 316				
Codes:		AS 4041, Class 3 (Note 8)				
Design Temperature (°C) ¹:		- 50	232	250	275	300
Design Pressure (kPa(g)) ¹:		3500	3500	3300	3100	2900
Corrosion Allowance (mm) :		1.0				
Pressure Testing :		Initial service leak testing as per AS 4041. Hydrostatic pressure testing not required unless directed by the Engineer.				
Non-Destructive Testing :		Visual inspection only				

Description	Size Range	Dimensional Specification	Type	End Preparations	Material Specification
Pipe	DN25 – 50	ASME B36.19M	SCH 40S, SMLS	Plain Ends	ASTM A312 GR. 316
	DN65 – 300		SCH 40S, ERW	Bevelled Ends	
Elbow, 90°	DN25 – 50	ASME B16.11	Class 3000	Socket weld	ASTM A182 GR F316
	DN65 – 300	ASME B16.9	SCH 40S	Bevelled Ends	ASTM A403 GR WP S 316
Elbow, 45°	DN25 – 50	ASME B16.11	Class 3000	Socket weld	ASTM A182 GR F316
	DN65 – 300	ASME B16.9	SCH 40S	Bevelled Ends	ASTM A403 GR WP S 316
Tee, equal	DN25 – 50	ASME B16.11	Class 3000	Socket weld	ASTM A182 GR F316
	DN65 – 300	ASME B16.9	SCH 40S	Bevelled Ends	ASTM A403 GR WP S 316
Tee, reducing	DN32 – 50	ASME B16.11	Class 3000	Socket weld	ASTM A182 GR F316
	DN65 – 300	ASME B16.9	SCH 40S	Bevelled Ends	ASTM A403 GR WP S 316
Cap	DN25 – 50	ASME B16.11	Class 3000	Socket weld	ASTM A182 GR F316
	DN65 – 300	ASME B16.9	SCH 40S	Bevelled Ends	ASTM A403 GR WP S 316
Coupling, reducing	DN32 – 50	ASME B16.11	Class 3000	Socket Weld	ASTM A182 GR F316
Reducer, eccentric ⁴	DN65 – 300	ASME B16.9	SCH 40S	Bevelled Ends	ASTM A403 GR WP S 316
Reducer, concentric ⁴	DN65 – 300				
Flange, Plate, Slip-On	DN25 – 300	AS 2129	Table H, FF	Plate	ASTM A240 GR 316L
Flange, Blind	DN25 – 300			Blank	
Flange, Plate, Slip-On	DN25 – 300	AS 2129	Table H, RF ⁷	Plate	ASTM A240 GR 316L
Flange, Blind	DN25 – 300			Blank	
Threadolet ⁵	DN15 – 20	MSS-SP-97	Class 3000	Bevelled End x BSP Female	ASTM A182 GR F316
Sockolet	DN25 – 50	MSS-SP-97	Class 3000	Socket Weld	ASTM A182 GR F316
Weldolet	DN65 – 350	MSS-SP-97	SCH 40S	Bevelled	ASTM A182 GR F316
Pipe Nipple ⁵	DN15 – 50	ASTM A733	SCH 80S	BSP Threaded	ASTM A182 GR F316
Hex Nipple ⁵	DN15 – 50	ASME B16.11	Class 3000	BSP Threaded	ASTM A182 GR F316
Hex Head Bushing ⁵	DN15 – 50	ASME B16.11	Class 3000	BSP Threaded	ASTM A182 GR F316
Hex Head Plug ⁵	DN15 – 50	ASME B16.11	Class 3000	BSP Threaded	ASTM A182 GR F316

Stud Bolts	DN25 – 300	ISO 965-1	304SS, Gr. A2-70 bolts	ISO 3506-1
Machine Bolt, Metric ⁶			304SS, Hex head, Gr. A2-70 bolts	
Nuts			304SS, Gr. A2-70 hex nuts	ISO 3506-2
Gaskets	DN25 – 300	AS 2129	Table H, Klinger C4430 or approved equivalent, 1.5mm thick, flat face	

NOTES	
1	The design pressure and temperature conditions are based on AS2129 flange ratings. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service
2	Refer to Piping Line List for test pressure requirements for each line.
3	The maximum length of flanged spools shall be 6 m for DN80 and 12 m maximum for DN100 and above.
4	For Non-Destructive Examination requirements refer to Piping, Fabrication, Installation and Testing Specification.
5	This specification is strictly to be welded fittings only. Threaded fittings only to be used for instrumentation or tie-in for mechanical packages where required. PTFE Thread tape or liquid sealant to be used.
6	Not Used
7	Flat Face is preferred. Raised Face to be used only as required e.g. Valves, Instrumentation, Mechanical Packages. Raised Face connecting to Flat Face is not permitted.
8	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343. Pipe system not suitable for Potable Water.

BRANCH CHART

Branch Size	300	ET														
	250	RT	ET													
	200	RT	RT	ET												
	150	RT	RT	RT	ET											
	125*	RT	RT	RT	RT	ET										
	100	WO	RT	RT	RT	RT	ET									
	90*	WO	WO	RT	RT	RT	RT	ET								
	80	WO	WO	WO	RT	RT	RT	RT	ET							
	65*	WO	WO	WO	RT	RT	RT	RT	RT	ET						
	50	SO	SO	SO	SO	RT	RT	RT	RT	RT	ET					
	40	SO	SO	SO	SO	SO	RT	RT	RT	RT	RT	ET				
	32*	SO	SO	SO	SO	SO	SO	SO	RT	RT	RT	RT	ET			
	25	SO	SO	SO	SO	SO	SO	SO	SO	RT	RT	RT	RT	ET		
	20	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	RT	ET	
	15	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	RT	RT	ET
	300	250	200	150	125*	100	90*	80	65*	50	40	32*	25	20	15	

LEGEND	
ET	Equal Tee
RT	Reducing Tee
SO	Sockolet
TO	Thredolet
WO	Weldolet

* indicates non-standard pipe size and not preferred

XS	Welded carbon steel piping system XS schedule, with flanges to AS2129 Table D for low pressure, low abrasion slurry applications.					
Material :		Carbon Steel				
Design Codes :		AS 4041, Class 3 (Note 7)				
Design Temperature (°C) ¹ :		- 50	232	250	275	300
Design Pressure (kPa(g)) ¹ :		700	700	650	600	570
Corrosion Allowance (mm) :		1.5				
Pressure Testing :		Initial service leak testing as per AS 4041. Hydrostatic pressure testing not required unless directed by the Engineer.				
Non-Destructive Testing :		Visual inspection only				
Description	Size Range	Dimensional Specification		Type	End Preparations	Material Specification
Pipe	DN25 – 50	ASME B36.10		XS WT, SMLS	Plain Ends	ASTM A53 Gr B / API 5L Gr B
	DN65 – 600			XS WT, ERW	Bevelled Ends	
Elbow, 90°	DN25 – 50	ASME B16.11		Class 3000	Socket Weld	ASTM A105
	DN65 – 600	ASME B16.9		XS WT	Bevelled Ends	ASTM A234 Gr WPB
Elbow, 45°	DN25 – 50	ASME B16.11		Class 3000	Socket Weld	ASTM A105
	DN65 – 600	ASME B16.9		XS WT	Bevelled Ends	ASTM A234 Gr WPB
Tee, equal	DN25 – 50	ASME B16.11		Class 3000	Socket Weld	ASTM A105
	DN65 – 600	ASME B16.9		XS WT	Bevelled Ends	ASTM A234 Gr WPB
Tee, reducing	DN25 – 50	ASME B16.11		Class 3000	Socket Weld	ASTM A105
	DN65 – 600	ASME B16.9		XS WT	Bevelled Ends	ASTM A234 Gr WPB
Cap	DN25 – 50	ASME B16.11		Class 3000	Socket Weld	ASTM A105
	DN65 – 600	ASME B16.9		XS WT	Bevelled Ends	ASTM A234 Gr WPB
Coupling, reducing	DN32 – 50	ASME B16.11		Class 3000	Socket Weld	ASTM A105
Reducer, eccentric	DN65 – 600	ASME B16.9		XS WT	Bevelled Ends	ASTM A234 Gr WPB
Reducer, concentric	DN65 – 600	ASME B16.9		XS WT	Bevelled Ends	ASTM A234 Gr WPB
Flange, Plate, Slip-On	DN25 – 600	AS 2129		Table D, FF	Plate	AS 3678 GR250
Flange, Blind	DN25 – 600				Blank	
Flange, Slip-On	DN25 – 600	AS 2129		Table D, RF ⁷	Plate	AS 3678 GR250
Flange, Blind	DN25 – 600				Blank	
Thredolet ⁵	DN15 – 20 ⁴	MSS-SP-97		Class 3000	Threaded, BSP	ASTM A105
Sockolet	DN25 – 50 ⁴	MSS-SP-97		Class 3000	Socket Weld	ASTM A105
Weldolet	DN25 – 200 ⁴	MSS SP-97		XS WT	Bevelled	ASTM A105
Pipe Nipple ⁵	DN15 – 50	ASTM A733		XS WT, SMLS, 100mm	Threaded, BSP	ASTM A53 Gr B
Hex Nipple ⁵	DN15 – 50	ASME B16.11		Class 3000	Threaded, BSP	ASTM A105
Hex Head Bushing ⁵	DN15 – 50	ASME B16.11		Class 3000	Threaded, BSP	ASTM A105
Hex Head Plug ⁵	DN15 – 50	ASME B16.11		Class 3000	Threaded, BSP	ASTM A105
Stud Bolts, Type A	DN25 – 600	AS/NZS 1252.1		Galvanised, Gr. 8.8 bolts		AS 4291.1
Machine Bolts, Metric ⁶	DN25 – 600			Galvanised, Hex head, Gr. 8.8 bolts		
Nuts	DN25 – 600			Galvanised, Gr. 8 hex nuts		AS 4291.2
Gaskets ⁹	DN25 – 600	AS 2129		Table D, Klinger C4430 or approved equivalent, 1.5mm thick, flat face		
				Table D, Klinger TC2300 or approved equivalent, 1.5mm thick, flat face		

1	The design pressure and temperature conditions are based on AS2129 flange ratings. Permissible pipe system temperatures and pressures are subject to gasket and valve selection. Refer to the Project Pipe Materials and Valve Specification for valve selection to suit the specific fluid service requirements.
2	For external surface treatment requirements refer to the Project Pipe Materials and Valve Specification.
3	The maximum length of flanged spools shall be 6 m for DN80 and 12 m maximum for DN100 and above.
4	Sizing refers to maximum branching size allowed. Refer to branching chart for more detail
5	This specification is strictly to be welded fittings only. Threaded fittings only to be used for instrumentation or tie-in for mechanical packages where required. PTFE Thread tape or liquid sealant to be used.
6	Not Used
7	Flat Face is preferred. Raised Face to be used only as required e.g. Valves, Instrumentation, Mechanical Packages. Raised Face connecting to Flat Face is not allowed.
8	Piping system not suitable for Lethal Fluids and limited to DN150 maximum size for Very Harmful Fluids as defined by AS 4343. Pipe system not suitable for Potable Water.
9	TC2003 to be used on all reagent services. Refer to Project Pipe Materials and Valve Specification for gasket selection

BRANCH CHART																				
Branch Size	600	ET																		
	550*	RT	ET																	
	500	RT	RT	ET																
	450	RT	RT	RT	ET															
	400	RT	RT	RT	RT	ET														
	350	RT	RT	RT	RT	RT	ET													
	300	RT	RT	RT	RT	RT	RT	ET												
	250	RT	RT	RT	RT	RT	RT	RT	ET											
	200	WO	WO	RT	RT	RT	RT	RT	RT	ET										
	150	WO	WO	WO	WO	RT	RT	RT	RT	RT	ET									
	125*	WO	WO	WO	WO	WO	WO	RT	RT	RT	RT	ET								
	100	WO	WO	WO	WO	WO	WO	WO	RT	RT	RT	RT	ET							
	90*	WO	WO	WO	WO	WO	WO	WO	WO	RT	RT	RT	RT	ET						
	80	WO	WO	WO	WO	WO	WO	WO	WO	WO	RT	RT	RT	RT	ET					
	65*	WO	WO	WO	WO	WO	WO	WO	WO	WO	RT	RT	RT	RT	RT	ET				
	50	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	RT	RT	RT	RT	RT	ET			
	40	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	RT	RT	RT	RT	RT	ET		
	32	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	RT	RT	RT	RT	ET	
	25	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	SO	RT	RT	RT	ET
	20	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO
	15	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO
Header Size																				

LEGEND	
ET	Equal Tee
RT	Reducing Tee
SO	Socketlet
TO	Thredolet
WO	Weldolet

* indicates non-standard pipe size

APPENDIX B – VALVE DATASHEETS

VB03N1	Ball Valve – Full bore, three piece body, Hastelloy, BSP threaded for isolation duty in elution and corrosive duties at elevated temperatures.
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Item	EPC Data	Supplier Data	Notes
Type	Ball Valve - Full Bore, 3 piece body		1
Service	Elution service & corrosive duties at elevated temperatures		1
Size Range (mm)	DN15 to DN50		1
End Connections	Threaded, BSP Female		1
Actuation	Lever		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Zetco	2
Model		Series 4098	2
Valve Standard		tbc	2
Dimensional Standard		tbc	2
Body Material	Hastelloy C276	Hastelloy C276	3
Ball Material	Hastelloy C276	Hastelloy C276	3
Stem/Trim Material	Hastelloy C276	Hastelloy C276	3
Not-wetted Components Material	316 Stainless Steel	304 Stainless Steel	3
Seat Material	PTFE	TFM 1600	3
Stem Packing	PTFE	PTFE	3
Body Seal	PTFE	PTFE	3
Handle Sleeve	316 Stainless Steel	316 Stainless Steel + Vinyl	3
Design Pressure	6,900kPa(g)	700kPa(g) at 180°C 2,000kPa(g) at 170°C 10,000kPa(g) at 70°C	3
Design Temperature	-10 to 200°C	-40 to 200°C	3
Surface Protection	Manufacturer's Standard	Manufacturer's Standard	2

NOTES		
1	Data supplied by EPC The Supplier shall comply with this data.	
	Data shall be provided by Supplier.	
2	Data proposed by EPC The Supplier shall confirm data or state alternative.	

VB03S1	Ball Valve – Full bore, three piece body, stainless steel, BSP threaded for general service and non-hazardous reagent applications.
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Item	EPC Data	Supplier Data Manual Valve	Supplier Data Actuated Valve	Notes
Type	Ball Valve - Full Bore, 3 piece body, floating ball			1
Service	General Service and Non-Hazardous Reagent Applications			1
Size Range (mm)	DN15 to DN65			1
End Connections	Threaded, BSPT Female			1
Actuation	Lever			1
Lockable	Yes			1
Supplier Name		Emerson	Emerson	2
Manufacturer		K Ball	K Ball	2
Model		F155	F182	2
Valve Standard		Manufacturer Standard	Manufacturer Standard	2
Dimensional Standard		Manufacturer Standard	Manufacturer Standard	2
Body Material	316 Stainless Steel	ASTM A351 CF8M	ASTM A351 CF8M	3
Ball Material	316 Stainless Steel	ASTM A351 CF8M	ASTM A351 CF8M	3
Stem Material	316 Stainless Steel	AISI 316	AISI 316	3
Not-wetted Components Material	316 Stainless Steel	AISI 304	AISI 304	3
Seat Material	PTFE	PTFE	TFM 1600	3
Stem Packing	PTFE	PTFE	PTFE	3
Body Seal	PTFE	PTFE	PTFE	3
Handle Sleeve	316 Stainless Steel	Handle: AISI 304 Sleeve: Plastic	Handle: AISI 304 Handle Nut: AISI 304	3
Design Pressure	6,900kPa(g)	<40°C: 6,900 kPa(g) 150°C: 1,380 kPa(g)	DN8-50 (NPS ¼-2) ASME600 DN65-100 (NPS 2½-4) ASME300	3
Design Temperature	-10 to 200°C	-10 to 170°C	-10 to 170°C	3
Surface Protection	Manufacturer's Standard	Nil	Nil	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.

VB03S3	Ball Valve – Full bore, three piece body, stainless steel, BSP threaded, fire-safe/anti-static for hydrocarbon applications.
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Item	EPC Data	Supplier Data	Notes
Type	Ball Valve - Full Bore, 3 piece body, fire safe to API 607 and fitted with anti-static device		1
Service	Hydrocarbons		1
Size Range (mm)	DN15 to DN50		1
End Connections	Threaded, BSP Female		1
Actuation	Lever		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Valtec	2
Model		V3WMH-FS	2
Valve Standard		EN 10226-2	2
Dimensional Standard		DIN 3202 standard M3 length	2
Body Material	316 Stainless Steel	CF8M Stainless Steel	3
Ball Material	316 Stainless Steel	CF8M Stainless Steel	3
Stem/Trim Material	316 Stainless Steel	316 Stainless Steel	3
Not-wetted Components Material	316 Stainless Steel	304 Stainless Steel	3
Seat Material	PTFE	PTFE	3
Stem Packing	PTFE	Graphite	3
Body Seal	PTFE	Graphite	3
Handle	316 Stainless Steel	PVC coated 304 Stainless Steel	3
Design Pressure	3,300kPa(g)	600kPa(g) at 150°C 3,300kPa(g) at 100°C 6,100kPa(g) at 50°C 6,900Pa(g) at 35°C	3
Design Temperature	0 to 100°C	0 to 150°C	3
Surface Protection	Manufacturer's Standard	None	2

NOTES		
1	Data supplied by EPC	The Supplier shall comply with this data.
	Data shall be provided by Supplier.	
2	Data proposed by EPC	The Supplier shall confirm data or state alternative.

VB03S4	Ball Valve – Full bore, three piece body, stainless steel, BSP threaded, fire-safe/anti-static for oxygen service.
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Item	EPC Data	Supplier Data	Notes
Type	Ball Valve - Full Bore, 3 piece body, fire safe to API 607 and fitted with anti-static device		1
Service	Oxygen		1
Size Range (mm)	DN15 to DN50		1
End Connections	Threaded, BSP		1
Actuation	Lever		1
Lockable	Yes		1
Supplier Name			2
Manufacturer			2
Model			2
Valve Standard			2
Dimensional Standard			2
Body Material	316 Stainless Steel		3
Ball Material	316 Stainless Steel		3
Stem/Trim Material	316 Stainless Steel		3
Not-wetted Components Material	316 Stainless Steel		3
Seat Material	PTFE		3
Stem Packing	PTFE		3
Body Seal	PTFE		3
Handle Sleeve	316 Stainless Steel		3
Design Pressure	6,900kPa(g)		3
Design Temperature	-10 to 200°C		3
Surface Protection	Manufacturer's Standard		2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.
4	Valves to be cleaned and passivated in accordance to ASTM G93-03 and supplied in vacuum packed containers with dessicant and marked as cleaned for oxygen service.
5	Inspection and record keeping of valves to be in accordance with EIGA "Oxygen Pipeline and Piping Systems"

VB03S5	Ball Valve – Full bore, three piece body, stainless steel, BSP threaded, WaterMark Certified for potable water service.
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Item	EPC Data	Supplier Data	Notes
Type	Ball Valve - Full Bore, 3 piece body, WaterMark Certified	WaterMark approved to WMTS-012	1
Service	Potable Water		1
Size Range (mm)	DN15 to DN50		1
End Connections	Threaded, BSPT Female		1
Actuation	Lever		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Zetco	2
Model		Series 4419	2
Valve Standard		-	2
Dimensional Standard		-	2
Body Material	316 Stainless Steel	316 Stainless Steel	3
Ball Material	316 Stainless Steel	316 Stainless Steel	3
Stem/Trim Material	316 Stainless Steel	316 Stainless Steel	3
Not-wetted Components Material	316 Stainless Steel	304 Stainless Steel	3
Seat Material	PTFE	PTFE	3
Stem Packing	PTFE	PTFE	3
Body Seal	PTFE	PTFE	3
Handle Sleeve	316 Stainless Steel	304 Stainless Steel w/ Plastic Sleeve	3
Design Pressure	6,900kPa(g)	-10 to 75°C: 4,000kPag 140: 1,000 kPag	3
Design Temperature	-10 to 200°C	-10 to 140°C	3
Surface Protection	Manufacturer's Standard	N/A	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.
4	For use of this piping system for potable water service, all pipe, fittings, valves and instrumentation shall be WaterMark Certified and approved to AS 4020.

VB03S6	Double Block and Bleed Valve - Two-valve single outlet instrumentation gauge manifold, stainless steel, BSP threaded for reagent applications.
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Item	EPC Data	Supplier Data	Notes
Type	Double Block and Bleed Valve - Two-valve single outlet gauge valve		1
Service	Reagents		1
Size Range (mm)	DN6 to DN25		1
End Connections	Threaded, BSP Female		1
Actuation	Tee Handle		1
Lockable	Yes		1
Supplier Name			2
Manufacturer			2
Model			2
Valve Standard			2
Dimensional Standard			2
Body Material	316 Stainless Steel		3
Ball Material	316 Stainless Steel		3
Stem/Trim Material	316 Stainless Steel		3
Not-wetted Components Material	316 Stainless Steel		3
Seat Material	316 Stainless Steel		3
Stem Packing	PTFE		3
Body Seal	PTFE		3
Handle Sleeve	316 Stainless Steel		3
Design Pressure	1,400kPa(g)		3
Design Temperature	-10 to 200°C		3
Surface Protection	Manufacturer's Standard		2

NOTES		
1	Data supplied by EPC	The Supplier shall comply with this data.
2	Data shall be provided by Supplier.	
3	Data proposed by EPC	The Supplier shall confirm data or state alternative.

VB03S7	Ball Valve – Full bore, two piece body, stainless steel, BSP threaded, vented for Hydrogen Peroxide service.
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Item	EPC Data	Supplier Data	Notes
Type	Ball Valve - Full Bore, 2 piece body, upstream pressure relief in "Open" or "Closed" position.		1
Service	Hydrogen peroxide		1
Size Range (mm)	DN15 to DN50		1
End Connections	Threaded, BSP Female		1
Actuation	Lever		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		K Ball	2
Model		F182	2
Valve Standard		Manufacturer Standard	2
Dimensional Standard		Manufacturer Standard	2
Body Material	316 Stainless Steel	CF8M	3
Ball Material	316 Stainless Steel	CF8M	3
Stem/Trim Material	316 Stainless Steel	AISI 316	3
Not-wetted Components Material	316 Stainless Steel	AISI 304	3
Seat Material	PTFE	PTFE	3
Stem Packing	PTFE	PTFE	3
Body Seal	PTFE	PTFE	3
Handle Sleeve	316 Stainless Steel	Handle: AISI 304 Sleeve: Plastic	3
Design Pressure	1,400kPa(g)	<40°C: 10.3 MPa(g) 150°C: 2,400 kPa(g)	3
Design Temperature	-10 to 200°C	-10 to 170°C	3
Surface Protection	Manufacturer's Standard	Nil	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.
4	Valves to be cleaned and passivated in accordance to ASTM A380/A380M and ASTM A967/A967M and supplied in vacuum packed containers with dessicant and marked as cleaned for hydrogen peroxide service.

VB09S1	Ball Valve – Full bore, block body, stainless steel, BSP threaded for high pressure scheme water
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Item	EPC Data	Supplier Data	Notes
Type	Ball Valve - Full Bore, block body		1
Service	High Pressure Scheme Water		1
Size Range (mm)	DN15		1
End Connections	Threaded, Female, BSP		1
Actuation	Lever		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Process Systems	2
Model		BLSH4PB-L-D	2
Valve Standard		Manufacturer Standard	2
Dimensional Standard		Manufacturer Standard	2
Body Material	316 Stainless Steel	CF8M	3
Ball Material	316 Stainless Steel	SS316	3
Stem/Trim Material	316 Stainless Steel	SS316	3
Not-wetted Components Material	316 Stainless Steel	SS304	3
Seat Material	FKM	Delrin	3
Stem Packing	PTFE	R-PTFE	3
Body Seal	PTFE	R-PTFE	3
Handle Sleeve	316 Stainless Steel	CF8 w/ Plastic Sleeve	3
Design Pressure	15,000kPa(g)	20,700 kPa(g)	3
Design Temperature	-10 to 100°C	-25 to 40°C	3
Surface Protection	Manufacturer's Standard	N/A	2

NOTES		
1	Data supplied by EPC The Supplier shall comply with this data.	
2	Data shall be provided by Supplier.	
3	Data proposed by EPC The Supplier shall confirm data or state alternative.	

VBD1C1	Ball Valve – Full bore, two piece body, carbon steel, flanged to AS 2129 Table D for cyanide and caustic service.
---------------	--

Item	EPC Data	Supplier Data	Notes
Type	Ball Valve - Full Bore, 2 piece body		1
Service	Cyanide and Caustic		1
Size Range (mm)	DN15 to DN200		1
End Connections	Flanged, FF to AS 2129 Table D	Flanged, FF to AS 2129 Table D	1
Actuation	SDN80 – Lever ≥DN100 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		KTM	2
Model		EB11	2
Valve Standard	Manufactuer Standard	Manufacturer Standard	2
Dimensional Standard	ASME B16.10 Short Pattern	ASME B16.10	2
Body Material	Mild Carbon Steel	WCB	3
Ball Material	Mild Carbon Steel	SS316	3
Stem/Trim Material	Vendor to advise	SS316	3
Not-wetted Components Material	Vendor to advise	SS316	3
Seat Material	PTFE	PTFE	3
Stem Packing	PTFE	PTFE	3
Body Seal	PTFE	PTFE	3
Handle Sleeve	Vendor to advise	Handle - Carbon steel Zn-plating / Painting	3
Design Pressure	700kPa(g)	ASME CL 150	3
Design Temperature	-10 to 200°C	-29 to 200°C	3
Surface Protection	Manufacturer's Standard	Manufacturer's Standard	2

NOTES		
1	Data supplied by EPC	The Supplier shall comply with this data.
2	Data shall be provided by Supplier.	
3	Data proposed by EPC	The Supplier shall confirm data or state alternative.

VBD1N1	Ball Valve – Full bore, two piece body, Hastelloy, flanged to AS 2129 Table D for isolation duty in elution and corrosive duties at elevated temperatures.
---------------	---

Item	EPC Data	Supplier Data	Notes
Type	Ball Valve - Full Bore, 2 piece body		1
Service	Elution service & corrosive duties at elevated temperatures		1
Size Range (mm)	DN15 to DN200		1
End Connections	Flanged, FF to AS 2129 Table D		1
Actuation	≤DN80 – Lever ≥DN100 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Australian Pipeline Valve	2
Model		SLFSBV01	2
Valve Standard	Manufacturer Standard	API 608 / API 6D	2
Dimensional Standard	ASME B16.10 Short Pattern	Vendor to advise	2
Body Material	Hastelloy C276	Hastelloy C	3
Ball Material	Hastelloy C276	Hastelloy C	3
Stem/Trim Material	Hastelloy C276	Hastelloy C	3
Not-wetted Components Material	316 Stainless Steel	Vendor to advise	3
Seat Material	PTFE	PTFE	3
Stem Packing	PTFE	Vendor to advise	3
Body Seal	PTFE	Vendor to advise	3
Handle Sleeve	316 Stainless Steel	Vendor to advise	3
Design Pressure	700kPa(g)	Vendor to advise	3
Design Temperature	-10 to 200°C	Vendor to advise	3
Surface Protection	Manufacturer's Standard	Vendor to advise	2

NOTES		
1	Data supplied by EPC The Supplier shall comply with this data.	
	Data shall be provided by Supplier.	
2	Data proposed by EPC The Supplier shall confirm data or state alternative.	

VBD1S1	Ball Valve – Full bore, two piece body, stainless steel, flanged to AS 2129 Table D for general service and non-hazardous reagent applications.
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Item	EPC Data	Manual Valve Supplier Data	Control Valve Supplier Data	Notes
Type	Ball Valve - Full Bore, 2 piece body, floating ball			1
Service	General Service and Non-Hazardous Reagent Applications			1
Size Range (mm)	DN15 to DN200			1
End Connections	Flanged, FF to AS 2129 Table D	≤DN80 - Flanged, RF to AS 2129 Table D/E ≥DN200 - Flanged, RF to AS 2129 Table D	≤DN80 - Flanged, RF to AS 2129 Table D/E ≥DN200 - Flanged, RF to AS 2129 Table D	1
Actuation	≤DN80 – Lever ≥DN100 – Worm gear with hand wheel	≤DN80 – Lever ≥DN100 – Worm gear with hand wheel	Pneumatic Actuator as per Control Valve Detail	1
Lockable	Yes			1
Supplier Name		Emerson	Emerson	2
Manufacturer		Arita - Unimech	Arita - Unimech	2
Model		ARTTBV-TE	ARTTBV-TE	2
Valve Standard	Manufactuer Standard	ANSI B16.34	ANSI B16.34	2
Dimensional Standard	ASME B16.10 Short Pattern	ASME B16.10	ASME B16.10	2
Body Material	316 Stainless Steel	ASTM A351 CF8M	ASTM A351 Gr. CF8M	3
Ball Material	316 Stainless Steel	316 Stainless Steel	316 Stainless Steel	3
Stem Material	316 Stainless Steel	ASTM A276 316	ASTM A276 316	3
Not-wetted Components Material	316 Stainless Steel	304 Stainless Steel	316 Stainless Steel	3
Seat Material	PTFE	RPTFE	RPTFE	3
Stem Packing	PTFE	RPTFE	RPTFE	3
Body Seal	PTFE	RPTFE	RPTFE	3
Handle Sleeve	316 Stainless Steel	201 Stainless Steel	N/A	3
Design Pressure	700kPa(g)	700 kPa(g)	700 kPa(g)	3
Design Temperature	-10 to 200°C	-10 to 200°C	-10 to 180°C	3
Surface Protection	Manufacturer's Standard	Noted & Confirm	NIL	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.

VBD1S2	Ball Valve – Full bore, two piece body, stainless steel, flanged to AS 2129 Table D, fire-safe/anti-static for hydrocarbon applications.
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Item	EPC Data	Supplier Data	Notes
Type	Ball Valve - Full Bore, 2 piece body, fire safe to API 607 and fitted with anti-static device		1
Service	Hydrocarbons		1
Size Range (mm)	DN15 to DN200		1
End Connections	Flanged, FF to AS 2129 Table D		1
Actuation	≤DN80 – Lever ≥DN100 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Emerson	2
Model		KTM EB11	2
Valve Standard	Manufactuer Standard	Manufactuer Standard	2
Dimensional Standard	ASME B16.10 Short Pattern	ASME B16.10 Short Pattern	2
Body Material	316 Stainless Steel	CF8M Stainless Steel	3
Ball Material	316 Stainless Steel	A182 Stainless Steel	3
Stem/Trim Material	316 Stainless Steel	A276 Stainless Steel	3
Not-wetted Components Material	316 Stainless Steel	304 Stainless Steel	3
Seat Material	PTFE	PTFE	3
Stem Packing	PTFE	GRAPHITE	3
Body Seal	PTFE	GRAPHITE	3
Handle Sleeve	316 Stainless Steel	Zinc Plated A216 Carbon Steel	3
Design Pressure	700kPa(g)	700kPa(g)	3
Design Temperature	-10 to 200°C	-10 to 200°C	3
Surface Protection	Manufacturer's Standard	Manufacturer's Standard	2

NOTES		
1	Data supplied by EPC	The Supplier shall comply with this data.
2	Data shall be provided by Supplier.	
3	Data proposed by EPC	The Supplier shall confirm data or state alternative.

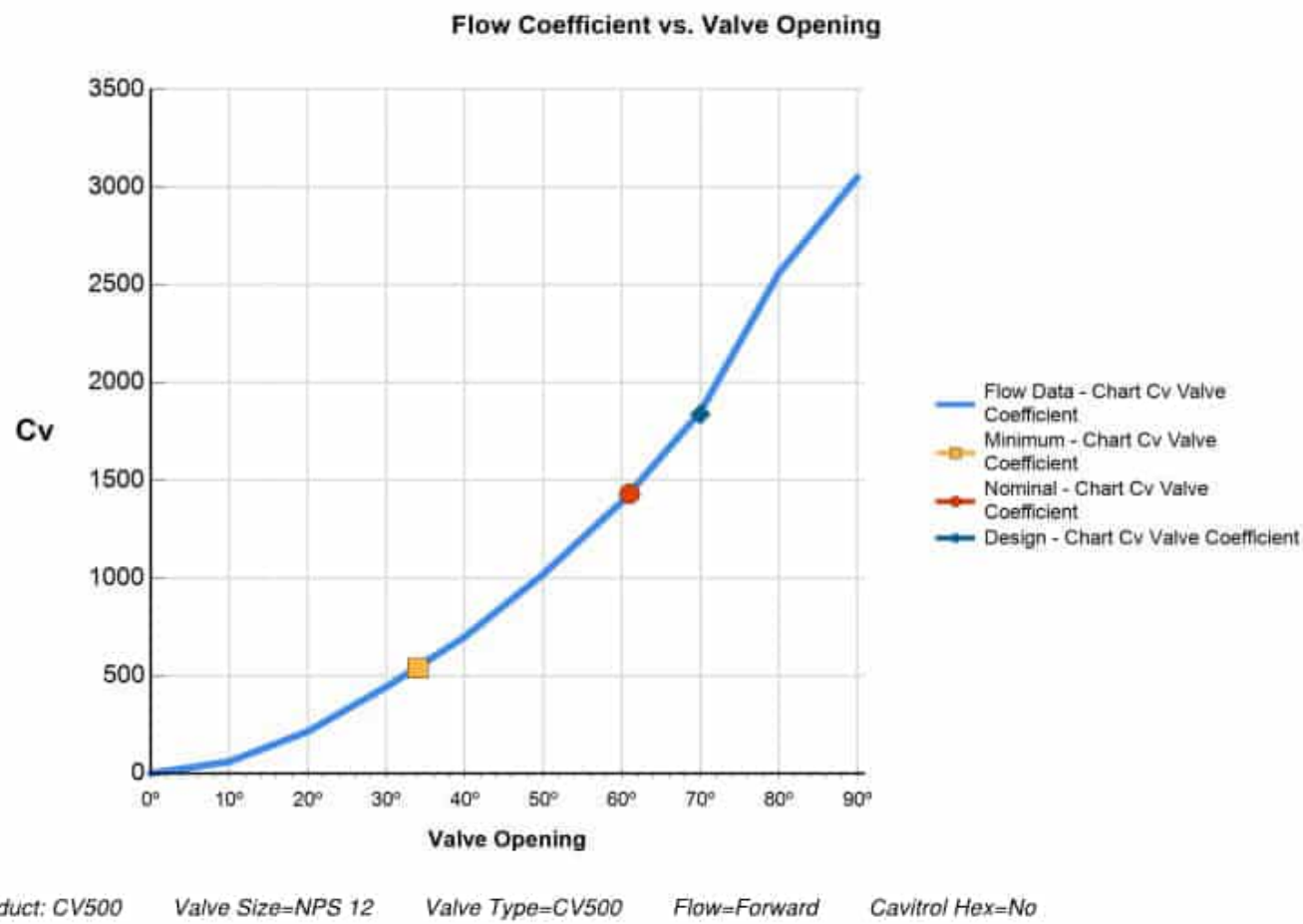
VBD1S3	Ball Valve – Full bore, two piece body, stainless steel, flanged to AS 2129 Table D, vented, for Reagents service, with extended stem.
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Item	EPC Data	Supplier Data	Notes
Type	Ball Valve - Full Bore, 2 piece body, upstream pressure relief in "Open" or "Closed" position, with extended stem		1
Service	Reagents		1
Size Range (mm)	DN25 to DN200		1
End Connections	Flanged, FF to AS 2129 Table D	Flanged, RF to AS 2129 Table D	1
Actuation	≤DN80 – Lever, with extended stem ≥DN100 – Worm gear with hand wheel, with extended stem		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Emerson	2
Model		KTM Series EB1	2
Valve Standard	Manufactuer Standard	ASME B16.34, API 608, API 6D, ISO 17292	2
Dimensional Standard	ASME B16.10 Long Pattern	ASME B16.10 Long Pattern	2
Body Material	316 Stainless Steel	ASTM A351 Gr. CF8M Stainless Steel	3
Ball Material	316 Stainless Steel	ASTM A351 Gr. CF8M Stainless Steel	3
Stem/Trim Material	316 Stainless Steel	ASTM A182 Gr. F51 Stainless Steel	3
Not-wetted Components Material	316 Stainless Steel	A276 (TP) 316 Stainless Steel	3
Seat Material	PTFE	PTFE	3
Stem Packing	PTFE	PTFE	3
Body Seal	PTFE	PTFE	3
Handle	316 Stainless Steel	Carbon steel painted	3
Design Pressure	1,400kPa(g)	1,900kPa(g)	3
Design Temperature	-10 to 200°C	-29 to 180°C	3
Surface Protection	Manufacturer's Standard	NIL	2

NOTES		
1	Data supplied by EPC	The Supplier shall comply with this data.
2	Data shall be provided by Supplier.	
3	Data proposed by EPC	The Supplier shall confirm data or state alternative.

Input Signal		4 to 20 mA DC	2
Valve Standard	Manufacturer Standard	Manufacturer Standard	3
Dimensional Standard	Manufacturer Standard	ISA S75.04 Face-to-face	3
Body Material	316 Stainless Steel	CF8M	3
Ball Material	316 Stainless Steel	Chrome plated CF3M	3
Stem/Trim Material	316 Stainless Steel	S17400 (17-4PH SST)	3
Not-wetted Components Material	316 Stainless Steel	316 Stainless Steel	3
Seat Material	PTFE	CF8M	3
Stem Packing	PTFE	PTFE	3
Body Seal	PTFE	CF8M	3
Weight (kg)		341	2
Handle	Not applicable	Not applicable	3
Design Pressure	700 kPa(g)	1,400 kPa(g)	3
Design Temperature	-10 to 200°C	-10 to 200°C	3
Surface Protection	Manufacturer's Standard	Manufacturer's Standard	3
NOTES			
1	Data supplied by EPC The Supplier shall comply with this data.		
2	Data shall be provided by Supplier.		
3	Data proposed by EPC The Supplier shall confirm data or state alternative.		

Datasheet Title :		Cyclone Feed Hopper Make-Up Water Valve
Document Number :		2000-F00-DSH-PP-11039
EPC	Project Number :	



VBD1T1	Ball Valve – Full bore, flanged to AS 2129 Table D for Hydrochloric acid service.
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Item	EPC Data	Manual Valve Supplier Data	Actuated Valve Supplier Data	Notes
Type	Ball Valve - Full Bore, 3 piece body			1
Service	Hydrochloric Acid			1
Size Range (mm)	DN15 to DN150			1
End Connections	Flanged, FF to AS 2129 Table D	Flanged, RF to AS 2129 Table D	Flanged, FF to AS 2129 Table D	1
Actuation	≤DN80 – Lever ≥DN100 – Worm gear with hand wheel			1
Lockable	Yes			1
Supplier Name		Emerson	Emerson	2
Manufacturer		KDV	GF Piping Systems	2
Model		Series 60 Ball Valve	Type 546 Pro	2
Valve Standard	ISO 16135	Manufacturers Standard	ISO 16135	2
Dimensional Standard	ISO 16135	Manufacturers Standard	ISO 16135	2
Body Material	PP-H (Polypropylene Homopolymer)	PFA lined A351-CF8 Stainless Steel	PP-H	3
Ball Material	PP-H	PFA lined A351-CF8 Stainless Steel	PP-H	3
Stem/Trim Material	PP-H/Inox	PFA lined A351-CF8 Stainless Steel	PP-H	3
Not-wetted Components Material	PP-H	304 Stainless Steel	PP-H	3
Seat Material	PP-H	PTFE	PP-H	3
Stem Packing	EPDM	PTFE	EPDM	3
Body Seal	PP-H	PTFE	PP-H	3
Handle Sleeve	PVC	Zinc plated carbon steel	N/A	3
Design Pressure	700kPa(g)	1,000kPa(g)	550kPa(g) at 50°C 700kPa(g) at 40°C	3
Design Temperature	0 to 80°C	-29 to 180°C	0 to 80°C	3
Surface Protection	Manufacturers Standard	None	None	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
	Data shall be provided by Supplier.
2	Data proposed by EPC The Supplier shall confirm data or state alternative.

VBE1C1	Ball Valve – Full bore, two piece body, carbon steel, flanged to AS 2129 Table E for cyanide and air applications.
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Item	EPC Data	Supplier Data	Notes
Type	Ball Valve - Full Bore, 2 piece body, floating ball		1
Service	Cyanide and air Applications		1
Size Range (mm)	DN15 to DN200		1
End Connections	Flanged, FF to AS 2129 Table E	Flanged, RF to AS 2129 Table E	1
Actuation	≤DN80 – Lever ≥DN100 – Worm gear with hand wheel		1
Lockable	Yes (provision for locking)		1
Supplier Name		Emerson	2
Manufacturer		KTM	2
Model		EB11	2
Valve Standard	Manufactuer Standard	Manufacturer Standard	2
Dimensional Standard	ASME B16.10 Long Pattern	ASME B16.10 Long Pattern	2
Body Material	Carbon Steel	WCB	3
Ball Material	SS316	SS316	3
Stem/Trim Material	SS316	SS316	3
Not-wetted Components Material	SS316	SS316	3
Seat Material	PTFE	PTFE	3
Stem Packing	PTFE	PTFE	3
Body Seal	PTFE	PTFE	3
Handle Sleeve	Carbon Steel	Handle - Carbon steel Zn-plating / Painting	3
Design Pressure	1,400kPa(g)	ASME CL 150	3
Design Temperature	-10 to 200°C	-29 to 200°C	3
Surface Protection	Manufacturer's Standard	Manufacturer's Standard	2

NOTES		
1	Data supplied by EPC	The Supplier shall comply with this data.
2	Data shall be provided by Supplier.	
3	Data proposed by EPC	The Supplier shall confirm data or state alternative.

VBE1N1 Ball Valve – Full bore, two piece body, Hastelloy, flanged to AS 2129 Table E for isolation duty for Acid Wash & Elution column			
Item	EPC Data	Supplier Data	Notes
Type	Ball Valve - Full Bore, 2 piece body		1
Service	Elution service at elevated temperatures & HCl		1
Size Range (mm)	DN15 to DN200		1
End Connections	Flanged, FF to AS 2129 Table E		1
Actuation	≤DN80 – Lever ≥DN100 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Emerson	2
Model		KTM EB11	2
Valve Standard	Manufacturer Standard	Manufacturer Standard	2
Dimensional Standard	ASME B16.10 Short Pattern	ASME B16.10 Short Pattern	2
Body Material	Hastelloy C276	Hastelloy A494	3
Ball Material	Hastelloy C276	Hastelloy A494	3
Stem/Trim Material	Hastelloy C276	Hastelloy B574	3
Not-wetted Components Material	316 Stainless Steel	304 Stainless Steel	3
Seat Material	PTFE	Reinforced PTFE	3
Stem Packing	PTFE	PTFE	3
Body Seal	PTFE	Reinforced PTFE	3
Handle Sleeve	316 Stainless Steel	Zinc plated carbon steel	3
Design Pressure	1,400kPa(g)	1,400kPa(g)	3
Design Temperature	-10 to 200°C	-10 to 200°C	3
Surface Protection	Manufacturer's Standard	Manufacturer's Standard	2
NOTES			
1	Data supplied by EPC The Supplier shall comply with this data.		
	Data shall be provided by Supplier.		
2	Data proposed by EPC The Supplier shall confirm data or state alternative.		

VBE1S1	Ball Valve – Full bore, two piece body, stainless steel, flanged to AS 2129 Table E for general service and non-hazardous reagent applications.
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Item	EPC Data	Supplier Data Manual Valve	Supplier Data Actuated Valve	Notes
Type	Ball Valve - Full Bore, 2 piece body, floating ball			1
Service	General Service and Non-Hazardous Reagent Applications			1
Size Range (mm)	DN15 to DN200			1
End Connections	Flanged, FF to AS 2129 Table E	Flanged, RF to AS 2129 Table E	Flanged, RF to AS 2129 Table E	1
Actuation	≤DN80 – Lever ≥DN100 – Worm gear with hand wheel			1
Lockable	Yes			1
Supplier Name		Emerson	Emerson	2
Manufacturer		Arita - Unimech	K-BALL	2
Model		ARTTBV-TE	F194	2
Valve Standard	Manufactuer Standard	ANSI B16.34	Manufacturer Standard	2
Dimensional Standard	ASME B16.10 Short Pattern	ASME B16.10 Short Pattern	ASME B16.10	2
Body Material	316 Stainless Steel	ASTM A351 CF8M	ASTM A351 Gr. CF8M	3
Ball Material	316 Stainless Steel	316 Stainless Steel	ASTM A351 Gr. CF8M	3
Stem/Trim Material	316 Stainless Steel	ASTM A276 316	ASTM A182 Gr. F51	3
Not-wetted Components Material	316 Stainless Steel	304 Stainless Steel	316 Stainless Steel	3
Seat Material	PTFE	RPTFE	TFM 1600	3
Stem Packing	PTFE	RPTFE	PTFE	3
Body Seal	PTFE	RPTFE	PTFE	3
Handle Sleeve	316 Stainless Steel	201 Stainless Steel	SS304 with PVC handle sleeve	3
Design Pressure	1,400kPa(g)	1,400 kPa(g)	ASME CL 150	3
Design Temperature	-10 to 200°C	-10 to 200°C	-29 to 200°C	3
Surface Protection	Manufacturer's Standard	Noted & Confirm	NIL	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.

VBE1S3	Ball Valve – Full bore, two piece body, stainless steel, flanged to AS 2129 Table E, Fire-safe/Anti-static for Oxygen Service.
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Item	EPC Data	Supplier Data	Notes
Type	Ball Valve - Full Bore, 2 piece body, fire safe to API 607 and fitted with anti-static device		1
Service	Oxygen		1
Size Range (mm)	DN15 to DN150		1
End Connections	Flanged, FF to AS 2129 Table E		1
Actuation	≤DN80 – Lever ≥DN100 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Emerson	2
Model		KTM EB11	2
Valve Standard	Manufacturer Standard	Manufacturer Standard	2
Dimensional Standard	ASME B16.10 Short Pattern	ASME B16.10 Short Pattern	2
Body Material	316 Stainless Steel	A351 CF8M Stainless Steel	3
Ball Material	316 Stainless Steel	A182 Stainless Steel	3
Stem/Trim Material	316 Stainless Steel	A276 Stainless Steel	3
Not-wetted Components Material	316 Stainless Steel	304 Stainless Steel	3
Seat Material	PTFE	Reinforced PTFE	3
Stem Packing	PTFE	PTFE	3
Body Seal	PTFE	Graphite	3
Handle Sleeve	316 Stainless Steel	Zinc Plated Carbon Steel	3
Design Pressure	1,400kPa(g)	1,400kPa(g)	3
Design Temperature	-10 to 200°C	-10 to 200°C	3
Surface Protection	Manufacturer's Standard	Manufacturer's Standard	2

NOTES		
1	Data supplied by EPC	The Supplier shall comply with this data.
	Data shall be provided by Supplier.	
2	Data proposed by EPC	The Supplier shall confirm data or state alternative.
3		

- 4 Valves to be cleaned and passivated in accordance to ASTM G93-03 and supplied in vacuum packed containers with dessicant and marked as cleaned for oxygen service.
- 5 Inspection and record keeping of valves to be in accordance with EIGA "Oxygen Pipeline and Piping Systems"

VBE1S4	Ball Valve – Full bore, two piece body, stainless steel, flanged to AS 2129 Table E, vented for Hydrogen Peroxide service.
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Item	EPC Data	Supplier Data	Notes
Type	Ball Valve - Full Bore, 2 piece body, upstream pressure relief in "Open" or "Closed" position.		1
Service	Hydrogen peroxide		1
Size Range (mm)	DN25 to DN100		1
End Connections	Flanged, FF to AS 2129 Table E	Flanged RF to AS2129 Table E	1
Actuation	≤DN80 – Lever ≥DN100 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		K Ball	2
Model		F194	2
Valve Standard	Manufactuer Standard	Manufacturer Standard	2
Dimensional Standard	ASME B16.10 Short Pattern	ASME B16.10 Short Pattern	2
Body Material	316 Stainless Steel	ASTM A351 Gr. CF8M	3
Ball Material	316 Stainless Steel	ASTM A351 Gr. CF8M	3
Stem/Trim Material	316 Stainless Steel	ASTM A182 Gr. F51	3
Not-wetted Components Material	316 Stainless Steel	316 Stainless Steel	3
Seat Material	PTFE	TFM 1600	3
Stem Packing	PTFE	PTFE	3
Body Seal	PTFE	PTFE	3
Handle Sleeve	316 Stainless Steel	SS304 with PVC handle sleeve	3
Design Pressure	1,400kPa(g)	1,400kPa(g)	3
Design Temperature	-10 to 200°C	-29 to 200°C	3
Surface Protection	Manufacturer's Standard	NIL	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.
4	Valves to be cleaned and passivated in accordance to ASTM A380/A380M and ASTM A967/A967M and supplied in vacuum packed containers with dessicant and marked as cleaned for hydrogen peroxide service.

VBE1S5	Ball Valve – Full bore, two piece body, stainless steel, flanged to AS 2129 Table E, vented, for Hydrogen Peroxide service, with extended stem.
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Item	EPC Data	Supplier Data	Notes
Type	Ball Valve - Full Bore, 2 piece body, upstream pressure relief in "Open" or "Closed" position, with extended stem		1
Service	Hydrogen Peroxide		1
Size Range (mm)	DN25 to DN200		1
End Connections	Flanged, RF to AS 2129 Table E	Flanged, RF to AS 2129 Table E	1
Actuation	≤DN80 – Lever, with extended stem ≥DN100 – Worm gear with hand wheel, with extended stem		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Emerson	2
Model		KTM Series EB1	2
Valve Standard	Manufacturer Standard	ASME B16.34, API 608, API 6D, ISO 17292	2
Dimensional Standard	ASME B16.10 Long Pattern	ASME B16.10 Long Pattern	2
Body Material	316 Stainless Steel	ASTM A351 Gr. CF8M Stainless Steel	3
Ball Material	316 Stainless Steel	ASTM A351 Gr. CF8M Stainless Steel	3
Stem/Trim Material	316 Stainless Steel	ASTM A182 Gr. F51 Stainless Steel	3
Not-wetted Components Material	316 Stainless Steel	A276 (TP) 316 Stainless Steel	3
Seat Material	PTFE	PTFE	3
Stem Packing	PTFE	PTFE	3
Body Seal	PTFE	PTFE	3
Handle	316 Stainless Steel	Carbon steel painted	3
Design Pressure	1,400kPa(g)	ASME CL 150	3
Design Temperature	-10 to 200°C	-29 to 180°C	3
Surface Protection	Manufacturer's Standard	NIL	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.
4	Valves to be cleaned and passivated in accordance to ASTM A380/A380M and ASTM A967/A967M and supplied in vacuum packed containers with dessicant and marked as cleaned for hydrogen peroxide service.

VBFD1L1	Butterfly Valve - Ductile iron, EPDM liner & seat, wafer to AS2129 Table D for isolation and flow control duty for general service and non-hazardous reagent applications.
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Item	EPC Data	Supplier Data	Notes
Type	Butterfly Valve, Wafer Type		1
Service	General Service and Non-Hazardous Reagent Applications - Isolation & Flow Control		1
Size Range (mm)	DN50 to DN900		1
End Connections	Wafer body to suit AS 2129 Table D		1
Actuation	≤DN200 – Trigger operated lever with multi-position notch plate ≥DN250 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Keystone	2
Model		GRW	2
Valve Standard		Designed according to EN 593 and API 609	2
Dimensional Standard		According to EN 558 Series 20 and API 609	2
Body Material	Ductile Iron - EPDM Lined	Ductile Iron ASTM A536	3
Disc Material	316 Stainless Steel	316 Stainless Steel	3
Shaft Material	316 Stainless Steel	431 stainless steel	3
Seat Material	EPDM	EPDM	3
Design Pressure	700kPa(g) - Tight Shut-off	1600 kPag - DN50-DN300 - Tight Shut off 1000 kPag - DN350-DN900 - Tight Shut off	3
Design Temperature	0 to 120°C	-28 to 160°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2

NOTES		
1	Data supplied by EPC The Supplier shall comply with this data.	
2	Data shall be provided by Supplier.	
3	Data proposed by EPC The Supplier shall confirm data or state alternative.	

VBFD1L2	Butterfly Valve - Ductile iron, EPDM liner & seat, lugged to AS2129 Table D for tank nozzles or end of line isolation duty for general service and non-hazardous reagent applications.
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Item	EPC Data	Supplier Data	Notes
Type	Butterfly Valve, Lug Type		1
Service	General, Caustic and Cyanide - Tank Nozzle & End of Line Isolation		1
Size Range (mm)	DN50 to DN600		1
End Connections	Lug body to suit AS 2129 Table D, Metric body lugs tapping		1
Actuation	≤DN200 – Trigger operated lever with multi-position notch plate ≥DN250 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Keystone	2
Model		GRL	2
Valve Standard		Designed according to EN 593 and API 609	2
Dimensional Standard		According to EN 558 Series 20 and API 609	2
Body Material	Ductile Iron - EPDM Lined	Ductile Iron ASTM A536	3
Disc Material	316 stainless steel	316 Stainless Steel	3
Shaft Material	316 stainless steel	431 stainless steel	3
Seat Material	EPDM	EPDM	3
Design Pressure	700kPa(g) - Tight Shut-off	1600 kPag - DN50-DN300 - Tight Shut off 1000 kPag - DN350-DN900 - Tight Shut off	3
Design Temperature	0 to 120°C	-28 to 160°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2

NOTES		
1	Data supplied by EPC	The Supplier shall comply with this data.
2	Data shall be provided by Supplier.	
3	Data proposed by EPC	The Supplier shall confirm data or state alternative.

VBFD1L3	Butterfly Valve - Ductile iron, FKM liner & seat, wafer to AS2129 Table D for isolation and flow control duty for low pressure air applications.		
Item	EPC Data	Supplier Data	Notes
Type	Butterfly Valve, Wafer Type		1
Service	Low Pressure Air - Isolation & Flow Control		1
Size Range (mm)	DN50 to DN900		1
End Connections	Wafer body to suit AS 2129 Table D		1
Actuation	≤DN200 – Trigger operated lever with multi-position notch plate ≥DN250 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Keystone	2
Model		GRW	2
Valve Standard		Designed according to EN 593 and API 609	2
Dimensional Standard		According to EN 558 Series 20 and API 609	2
Body Material	Ductile Iron - FKM Lined	Ductile Iron - FKM Lined	3
Disc Material	316 stainless steel	316 Stainless Steel	3
Shaft Material	316 stainless steel	431 stainless steel	3
Seat Material	FKM	FKM	3
Design Pressure	700kPa(g)	700kPa(g)	3
Design Temperature	0 to 200°C	-28 to 160°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2
NOTES			
1	Data supplied by EPC The Supplier shall comply with this data.		
2	Data shall be provided by Supplier.		
3	Data proposed by EPC The Supplier shall confirm data or state alternative.		

VBFD1L4	Butterfly Valve - Ductile iron, FKM liner & seat, lugged to AS2129 Table D for end of line isolation duty for low pressure air applications.
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Item	EPC Data	Supplier Data	Notes
Type	Butterfly Valve, Lug Type		1
Service	Low Pressure Air - Tank Nozzle & End of Line Isolation		1
Size Range (mm)	DN50 to DN900		1
End Connections	Lug body to suit AS 2129 Table D, Metric body lugs tapping		1
Actuation	≤DN200 – Trigger operated lever with multi-position notch plate ≥DN250 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name			2
Manufacturer			2
Model			2
Valve Standard			2
Dimensional Standard			2
Body Material	Ductile Iron - FKM Lined		3
Disc Material	316 stainless steel		3
Shaft Material	316 stainless steel		3
Seat Material	FKM		3
Design Pressure	700kPa(g)		3
Design Temperature	0 to 200°C		3
Surface Protection	Manufacturers Standard		2

NOTES		
1	Data supplied by EPC	The Supplier shall comply with this data.
2	Data shall be provided by Supplier.	
3	Data proposed by EPC	The Supplier shall confirm data or state alternative.

VBFD1L5	Butterfly Valve - Ductile iron, PTFE lined seat and disc, lugged to AS2129 Table D for Hydrochloric acid service.		
Item	EPC Data	Supplier Data	Notes
Type	Butterfly Valve, Lug Type		1
Service	Hydrochloric Acid ⁴		1
Size Range (mm)	DN50 to DN600		1
End Connections	Lug body to suit AS 2129 Table D, Metric body lugs tapping		1
Actuation	≤DN200 – Trigger operated lever with multi-position notch plate ≥DN250 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Emerson	2
Model		Keystone F920	2
Valve Standard		Vendor to confirm	2
Dimensional Standard		Vendor to confirm	2
Body Material	Ductile Iron - PTFE Lined	Ductile Iron - PTFE Lined	3
Disc Material	316 stainless steel - PTFE Lined	Carbon steel - PTFE Lined	3
Shaft Material	316 stainless steel	316 stainless steel	3
Seat Material	PTFE	PTFE/EPDM green	3
Design Pressure	700kPa(g) - Tight Shut-off	700kPa(g)	3
Design Temperature	0 to 200°C	0 to 120°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2
NOTES			
1	Data supplied by EPC The Supplier shall comply with this data.		
	Data shall be provided by Supplier.		
2	Data proposed by EPC The Supplier shall confirm data or state alternative.		
	Valves shall be used for piping that is sharing Hydrochloric Acid and Pregnant Eluate Service.		

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VBFD1L6	Butterfly Valve – Ductile Iron, EPDM liner & seat, flanged to AS2129 Table D for tank nozzles or end of line isolation duty for general service and non-hazardous reagent applications.
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Item	EPC Data	Supplier Data	Notes
Type	Butterfly Valve		1
Service	General Service and Non-Hazardous Reagent Applications - Tank Nozzle & End of Line Isolation		1
Size Range (mm)	DN600 to DN900		1
End Connections	Flanged to suit AS 2129 Table D		1
Actuation	Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Keystone	2
Model		GRF	2
Valve Standard		Designed according to EN 593 and API 609	2
Dimensional Standard		According to EN 558 Series 20, API 609 and AWWA C504 short	2
Body Material	Ductile Iron - EPDM Lined	Ductile Iron ASTM A536	3
Disc Material	316 stainless Steel	316 Stainless Steel	3
Shaft Material	316 stainless steel integral with disc	431 stainless steel (Not integral with disc)	3
Seat Material	EPDM	EPDM	3
Design Pressure	700kPa(g) - Tight Shut-off	1000 kPag - Tight Shut-off	3
Design Temperature	0 to 120°C	-40°C to +160°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2

NOTES		
1	Data supplied by EPC The Supplier shall comply with this data.	
2	Data shall be provided by Supplier.	
3	Data proposed by EPC The Supplier shall confirm data or state alternative.	

VBFD1S1	Butterfly Valve – Stainless steel, EPDM liner & seat, wafer to AS2129 Table D for isolation and flow control duty for general service and non-hazardous reagent applications.
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Item	EPC Data	Supplier Data	Notes
Type	Butterfly Valve, Wafer Type		1
Service	General Service and Non-Hazardous Reagent Applications - Isolation & Control		1
Size Range (mm)	DN50 to DN900		1
End Connections	Wafer body to suit AS 2129 Table D		1
Actuation	≤DN200 – Trigger operated lever with multi-position notch plate ≥DN250 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Keystone	2
Model		GRW	2
Valve Standard		Designed according to EN 593 and API 609	2
Dimensional Standard		According to EN 558 Series 20 and API 609	2
Body Material	316 Stainless Steel	ASTM A351 Gr CF8M	3
Disc Material	316 stainless Steel	316 stainless Steel	3
Shaft Material	316 stainless steel integral with disc	431 stainless steel (Not integral with disc)	3
Seat Material	EPDM	EPDM	3
Design Pressure	700kPa(g) - Tight Shut-off	1600 kPag - DN50-DN300 - Tight Shut off 1000 kPag - DN350-DN900 - Tight Shut off	3
Design Temperature	0 to 120°C	-28 to 160°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.

VBFD1S2	Butterfly Valve – Stainless steel, EPDM liner & seat, lugged to AS2129 Table D for tank nozzles or end of line isolation duty for general service and non-hazardous reagent applications.
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Item	EPC Data	Supplier Data	Notes
Type	Butterfly Valve, Lug Type		1
Service	General Service and Non-Hazardous Reagent Applications - Tank Nozzle & End of Line Isolation		1
Size Range (mm)	DN50 to DN600		1
End Connections	Lug body to suit AS 2129 Table D, Metric body lugs tapping		1
Actuation	≤DN200 – Trigger operated lever with multi-position notch plate ≥DN250 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Keystone	2
Model		GRL	2
Valve Standard		Designed according to EN 593 and API 609	2
Dimensional Standard		According to EN 558 Series 20 and API 609	2
Body Material	316 Stainless Steel	ASTM A351 Gr CF8M	3
Disc Material	316 stainless Steel	316 stainless Steel	3
Shaft Material	316 stainless steel integral with disc	431 stainless steel (Not integral with disc)	3
Seat Material	EPDM	EPDM	3
Design Pressure	700kPa(g) - Tight Shut-off	1600 kPag - DN50-DN300 - Tight Shut off 1000 kPag - DN350-DN900 - Tight Shut off	3
Design Temperature	0 to 120°C	-28 to 160°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2

NOTES		
1	Data supplied by EPC The Supplier shall comply with this data.	
2	Data shall be provided by Supplier.	
3	Data proposed by EPC The Supplier shall confirm data or state alternative.	

VBFD1S3	Butterfly Valve – Stainless steel, FKM liner & seat, wafer to AS2129 Table D for isolation and flow control duty for general service and non-hazardous reagent applications.
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Item	EPC Data	Supplier Data	Notes
Type	Butterfly Valve, Wafer Type		1
Service	General Service and Non-Hazardous Reagent Applications - Isolation & Control		1
Size Range (mm)	DN50 to DN900		1
End Connections	Wafer body to suit AS 2129 Table D		1
Actuation	≤DN200 – Trigger operated lever with multi-position notch plate ≥DN250 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name			2
Manufacturer			2
Model			2
Valve Standard			2
Dimensional Standard			2
Body Material	316 Stainless Steel		3
Disc Material	316 stainless Steel		3
Shaft Material	316 stainless steel integral with disc		3
Seat Material	FKM		3
Design Pressure	700kPa(g) - Tight Shut-off		3
Design Temperature	0 to 200°C		3
Surface Protection	Manufacturers Standard		2

NOTES		
1	Data supplied by EPC	The Supplier shall comply with this data.
2	Data shall be provided by Supplier.	
3	Data proposed by EPC	The Supplier shall confirm data or state alternative.

VBFD1S4	Butterfly Valve – Stainless steel, FKM liner & seat, lugged to AS2129 Table D for tank nozzles or end of line isolation duty for general service and non-hazardous reagent applications.
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Item	EPC Data	Supplier Data	Notes
Type	Butterfly Valve, Lug Type		1
Service	General Service and Non-Hazardous Reagent Applications - Tank Nozzle & End of Line Isolation		1
Size Range (mm)	DN50 to DN900		1
End Connections	Lug body to suit AS 2129 Table D, Metric body lugs tapping		1
Actuation	≤DN200 – Trigger operated lever with multi-position notch plate ≥DN250 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name			2
Manufacturer			2
Model			2
Valve Standard			2
Dimensional Standard			2
Body Material	316 Stainless Steel		3
Disc Material	316 stainless Steel		3
Shaft Material	316 stainless steel integral with disc		3
Seat Material	FKM		3
Design Pressure	700kPa(g) - Tight Shut-off		3
Design Temperature	0 to 200°C		3
Surface Protection	Manufacturers Standard		2

NOTES		
1	Data supplied by EPC	The Supplier shall comply with this data.
2	Data shall be provided by Supplier.	
3	Data proposed by EPC	The Supplier shall confirm data or state alternative.

VBFE1L1	Butterfly Valve - Ductile iron, EPDM liner & seat, wafer to AS2129 Table E for isolation and flow control duty for general service.
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Item	EPC Data	Supplier Data	Notes
Type	Butterfly Valve, Wafer Type		1
Service	General Service - Isolation & Flow Control		1
Size Range (mm)	DN50 to DN900		1
End Connections	Wafer body to suit AS 2129 Table E		1
Actuation	≤DN200 – Trigger operated lever with multi-position notch plate ≥DN250 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Keystone	2
Model		GRW	2
Valve Standard		Designed according to EN 593 and API 609	2
Dimensional Standard		According to EN 558 Series 20 and API 609	2
Body Material	Ductile Iron - EPDM Lined	Ductile Iron A536	3
Disc Material	316 stainless steel	316 Stainless Steel	3
Shaft Material	316 stainless steel	431 stainless steel	3
Seat Material	EPDM	EPDM	3
Design Pressure	1,400kPa(g) - Tight Shut-off	1600 kPag - DN50-DN300 - Tight Shut off 1000 kPag - DN350-DN900 - Tight Shut off	3
Design Temperature	0 to 120°C	-28 to 160°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2

NOTES		
1	Data supplied by EPC The Supplier shall comply with this data.	
2	Data shall be provided by Supplier.	
3	Data proposed by EPC The Supplier shall confirm data or state alternative.	

VBFE1L2	Butterfly Valve - Ductile iron, EPDM liner & seat, lugged to AS2129 Table E for tank nozzles or end of line isolation duty for general service.			
Item	EPC Data	Supplier Data	Supplier Data	Notes
Type	Butterfly Valve, Lug Type			1
Service	General Service - Tank Nozzle & End of Line Isolation			1
Size Range (mm)	DN50 to DN900			1
End Connections	Lug body to suit AS 2129 Table E, Metric body lugs tapping			1
Actuation	≤DN200 – Trigger operated lever with multi-position notch plate ≥DN250 – Worm gear with hand wheel			1
Lockable	Yes			1
Supplier Name		Emerson	Bray Controls	2
Manufacturer		Keystone	Bray Controls	2
Model		GRL	Series 3L-3JF	2
Valve Standard		Designed according to EN 593 and API 609	MSS SP-67	2
Dimensional Standard		According to EN 558 Series 20 and API 609	API 609	2
Body Material	Ductile Iron - EPDM Lined	Ductile Iron A536	Ductile Iron ASTM A536	3
Disc Material	316 stainless steel	316 Stainless Steel	316 Stainless Steel	3
Shaft Material	316 stainless steel	431 stainless steel	431 Stainless Steel	3
Seat Material	EPDM	EPDM Reinforced	EPDM	3
Design Pressure	1,400kPa(g) - Tight Shut-off	1600kPag - Tight Shut Off	1,720 kPag Bubble tight Shut-off	3
Design Temperature	0 to 120°C	-28 to 160°C	-29 to 121°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	Manufacturers Standard	3
NOTES				
1	Data supplied by EPC The Supplier shall comply with this data.			
2	Data shall be provided by Supplier.			
3	Data proposed by EPC The Supplier shall confirm data or state alternative.			

VBFE1L3	Butterfly Valve - Ductile iron, FKM liner & seat, wafer to AS2129 Table E for isolation and flow control duty for compressed air service applications.
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Item	EPC Data	Supplier Data	Notes
Type	Butterfly Valve, Wafer Type		1
Service	Compressed Air - Isolation & Flow Control		1
Size Range (mm)	DN50 to DN900		1
End Connections	Wafer body to suit AS 2129 Table E		1
Actuation	≤DN200 – Trigger operated lever with multi-position notch plate ≥DN250 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Keystone	2
Model		GRW	2
Valve Standard		Designed according to EN 593 and API 609	2
Dimensional Standard		According to EN 558 Series 20 and API 609	2
Body Material	Ductile Iron - FKM Lined	Ductile Iron - FKM Lined	3
Disc Material	316 stainless steel	316 stainless steel	3
Shaft Material	316 stainless steel	431 stainless steel	3
Seat Material	FKM	FKM	3
Design Pressure	1,400kPa(g) - Tight Shut-off	1,400 kPa(g)	3
Design Temperature	0 to 200°C	-28 to 160°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2

NOTES		
1	Data supplied by EPC The Supplier shall comply with this data.	
2	Data shall be provided by Supplier.	
3	Data proposed by EPC The Supplier shall confirm data or state alternative.	

VBFE1L4	Butterfly Valve - Ductile iron, FKM liner & seat, lugged to AS2129 Table E for end of line isolation duty for compressed air service applications.
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Item	EPC Data	Supplier Data	Notes
Type	Butterfly Valve, Lug Type		1
Service	Compressed Air - Tank Nozzle & End of Line Isolation		1
Size Range (mm)	DN50 to DN900		1
End Connections	Lug body to suit AS 2129 Table E, Metric body lugs tapping		1
Actuation	≤DN200 – Trigger operated lever with multi-position notch plate ≥DN250 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Keystone	2
Model		GRL	2
Valve Standard		Designed according to EN 593 and API 609	2
Dimensional Standard		According to EN 558 Series 20 and API 609	2
Body Material	Ductile Iron - FKM Lined	Ductile Iron - FKM Lined	3
Disc Material	316 stainless steel	316 stainless steel	3
Shaft Material	316 stainless steel	431 stainless steel	3
Seat Material	FKM	FKM	3
Design Pressure	1,400kPa(g) - Tight Shut-off	End of Line: 1,000 kPa(g)	3
Design Temperature	0 to 200°C	-28 to 160°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2

NOTES		
1	Data supplied by EPC	The Supplier shall comply with this data.
2	Data shall be provided by Supplier.	
3	Data proposed by EPC	The Supplier shall confirm data or state alternative.

VBFE1S1	Butterfly Valve – Stainless steel, EPDM liner & seat, wafer to AS2129 Table E for isolation and flow control duty for general service.
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Item	EPC Data	Supplier Data	Notes
Type	Butterfly Valve, Wafer Type		1
Service	General Service - Isolation & Control		1
Size Range (mm)	DN50 to DN600		1
End Connections	Wafer body to suit AS 2129 Table E		1
Actuation	≤DN200 – Trigger operated lever with multi-position notch plate ≥DN250 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Keystone	2
Model		GRW	2
Valve Standard		Designed according to EN 593 and API 609	2
Dimensional Standard		According to EN 558 Series 20 and API 609	2
Body Material	316 Stainless Steel	ASTM A351 Gr CF8M	3
Disc Material	316 stainless Steel	316 Stainless Steel	3
Shaft Material	316 stainless steel integral with disc	431 stainless steel (Not integral with disc)	3
Seat Material	EPDM	EPDM	3
Design Pressure	1,400kPa(g) - Tight Shut-off	1600 kPag - DN50-DN300 - Tight Shut off 1000 kPag - DN350-DN900 - Tight Shut off	3
Design Temperature	0 to 120°C	-28 to 160°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2

NOTES		
1	Data supplied by EPC	The Supplier shall comply with this data.
2	Data shall be provided by Supplier.	
3	Data proposed by EPC	The Supplier shall confirm data or state alternative.

VBFE1S2	Butterfly Valve – Stainless steel, EPDM liner & seat, lugged to AS2129 Table E for tank nozzles or end of line isolation duty for general service.		
Item	EPC Data	Supplier Data	Notes
Type	Butterfly Valve, Lug Type		1
Service	General Service - Tank Nozzle & End of Line Isolation		1
Size Range (mm)	DN50 to DN600		1
End Connections	Lug body to suit AS 2129 Table E, Metric body lugs tapping		1
Actuation	≤DN200 – Trigger operated lever with multi-position notch plate ≥DN250 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Keystone	2
Model		GRL	2
Valve Standard		Designed according to EN 593 and API 609	2
Dimensional Standard		According to EN 558 Series 20 and API 609	2
Body Material	316 Stainless Steel	A351 CF8M stainless steel	3
Disc Material	316 stainless Steel	A351 CF8M stainless steel	3
Shaft Material	316 stainless steel integral with disc	A479 431 stainless steel	3
Seat Material	EPDM	<DN350: EPDM Else: Reinforced EPDM	3
Design Pressure	1,400kPa(g) - Tight Shut-off	End of Line: 1,000 kPa(g)	3
Design Temperature	0 to 120°C	-28 to 160°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2
NOTES			
1	Data supplied by EPC The Supplier shall comply with this data.		
	Data shall be provided by Supplier.		
2	Data proposed by EPC The Supplier shall confirm data or state alternative.		

VBFE1S4	Butterfly Valve – Stainless steel, FKM liner & seat, lugged to AS2129 Table E for tank nozzles or end of line isolation duty for general service and non-hazardous reagent applications.
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Item	EPC Data	Supplier Data	Notes
Type	Butterfly Valve, Lug Type		1
Service	General Service and Non-Hazardous Reagent Applications - Tank Nozzle & End of Line Isolation		1
Size Range (mm)	DN50 to DN900		1
End Connections	Lug body to suit AS 2129 Table E, Metric body lugs tapping		1
Actuation	≤DN200 – Trigger operated lever with multi-position notch plate ≥DN250 – Worm gear with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Keystone	2
Model		GRL	2
Valve Standard		Designed according to EN 593 and API 609	2
Dimensional Standard		According to EN 558 Series 20 and API 609	2
Body Material	316 Stainless Steel	ASTM A351 Gr CF8M	3
Disc Material	316 stainless Steel	316 Stainless Steel	3
Shaft Material	316 stainless steel integral with disc	431 stainless steel (Not integral with disc)	3
Seat Material	FKM	FKM	3
Design Pressure	700kPa(g) - Tight Shut-off	1600 kPag - DN50-DN300 - Tight Shut off 1000 kPag - DN350-DN900 - Tight Shut off	3
Design Temperature	0 to 200°C	-28 to 160°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2

NOTES		
1	Data supplied by EPC	The Supplier shall comply with this data.
2	Data shall be provided by Supplier.	
3	Data proposed by EPC	The Supplier shall confirm data or state alternative.

VC01P1	Swing Check Valve – HDPE, flanged for general service
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Item	EPC Data	Supplier Data	Notes
Type	Swing Check Valve		1
Service	General Service		1
Size Range (mm)	DN80 to DN400		1
End Connections	Flanged ASME B16.5 CL150		1
Actuation	Swing Close (gravity)		1
Lockable	Not Applicable		1
Supplier Name		Emerson	2
Manufacturer		Termopro S.P.A.	2
Model		Termopro PolyCheck Valve	2
Valve Standard		Manufacturer's Standard	2
Dimensional Standard		Manufacturer's Standard	2
Body Material	HDPE	HDPE	3
Disc Material	HDPE	HDPE	3
Seat Material	Integral with Body	Integral with Body	3
Spring/Pin Material	316 Stainless Steel	316 Stainless Steel	3
Gasket/Seal Material	EPDM	EPDM	3
Design Pressure (kPa)	1,000kPa(g)	1,000kPa(g)	3
Design Temperature (°C)	-65 to 70°C	-65 to 70°C	3
Surface Protection	None	None	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.

VC01S1	Swing Check Valve – Stainless steel, BSP threaded for general service and non-hazardous reagent applications.		
Item	EPC Data	Supplier Data	Notes
Type	Swing Check Valve, Bolted Bonnet		1
Service	General Service and Non-Hazardous Reagent Applications		1
Size Range (mm)	DN15 to DN50		1
End Connections	Threaded, BSP Female		1
Actuation	Swing Close (gravity)		1
Lockable	Not Applicable		1
Supplier Name		Emerson	2
Manufacturer		Arita - Unimech	2
Model		SCVS36	2
Valve Standard		Manufacturer's Standard	2
Dimensional Standard		Manufacturer's Standard	2
Body Material	316 Stainless Steel CF8M	316 Stainless Steel CF8M	3
Disc Material	316 Stainless Steel CF8M	316 Stainless Steel CF8M	3
Seat Material	Integral with Body	Integral with Body	3
Spring/Pin Material	Not Applicable	Not Applicable	3
Gasket/Seal Material	PTFE	PTFE	3
Design Pressure (kPa)	1,400kPa(g)	1,400kPa(g)	3
Design Temperature (°C)	-10 to 200°C	-10 to 200°C	3
Surface Protection	Manufacturer's Standard	Manufacturer's Standard	2
NOTES			
1	Data supplied by EPC The Supplier shall comply with this data.		
2	Data shall be provided by Supplier.		
3	Data proposed by EPC The Supplier shall confirm data or state alternative.		

VC01S2	Swing Check Valve – Stainless steel, BSP threaded, WaterMark certified for potable water service.
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Item	EPC Data	Supplier Data	Notes
Type	Swing Check Valve, Bolted Bonnet		1
Service	Potable Water		1
Size Range (mm)	DN15 to DN50		1
End Connections	Threaded, BSP Female		1
Actuation	Swing Close (gravity)		1
Lockable	Not Applicable		1
Supplier Name			2
Manufacturer			2
Model			2
Valve Standard			2
Dimensional Standard			2
Body Material	316 Stainless Steel CF8M		3
Disc Material	316 Stainless Steel CF8M		3
Seat Material	Integral with Body		3
Spring/Pin Material	Not Applicable		3
Gasket/Seal Material	PTFE		3
Design Pressure (kPa)	1,400kPa(g)		3
Design Temperature (°C)	-10 to 200°C		3
Surface Protection	Manufacturer's Standard		2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.
4	For use of this piping system for potable water service, all pipe, fittings, valves and instrumentation shall be WaterMark Certified and approved to AS 4020.

VCD1C1	Dual Disc Check Valve – Carbon steel, wafer to AS2129 Table D for general service applications.
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Item	EPC Data	Supplier Data	Notes
Type	Dual Disc Check Valve		1
Service	General Service and Non-Hazardous Reagent Applications		1
Size Range (mm)	DN50 to DN600		1
End Connections	Wafer type to suit AS 2129 Table D		1
Actuation	Spring Close		1
Lockable	Not Applicable		1
Supplier Name		Emerson	2
Manufacturer		APV	2
Model		Super Check	2
Valve Standard		API 594	2
Dimensional Standard		ASME B16.40	2
Body Material	Carbon Steel	ASTM A216 WCB	3
Disc Material	316 Stainless Steel CF8M	Stainless Steel A351 CF8M	3
Seat Material	NBR	VITON	3
Spring/Pin Material	316 Stainless Steel	INCONEL X750 / A276 316 SS	3
Gasket/Seal Material	PTFE	VITON	3
Design Pressure (kPa)	700kPa(g)	1380kPa(g)	3
Design Temperature (°C)	-10 to 200°C	-10 to 200°C	3
Surface Protection	Manufacturer's Standard	Manufacturer's Standard	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
	Data shall be provided by Supplier.
2	Data proposed by EPC The Supplier shall confirm data or state alternative.

VCD1S1	Dual Disc Check Valve – Stainless steel, wafer to AS2129 Table D for general service and non-hazardous reagent applications.
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Item	EPC Data	Supplier Data	Notes
Type	Dual Disc Check Valve		1
Service	General Service and Non-Hazardous Reagent Applications		1
Size Range (mm)	DN40 to DN900		1
End Connections	Wafer type to AS 2129 Table D		1
Actuation	Spring Close		1
Lockable	Not Applicable		1
Supplier Name		Emerson	2
Manufacturer		Arita - Unimech	2
Model		SCV-W-SUS	2
Valve Standard		DIN3202	2
Dimensional Standard		DIN3202	2
Body Material	316 Stainless Steel CF8M	316 Stainless Steel CF8M	3
Disc Material	316 Stainless Steel CF8M	316 Stainless Steel CF8M	3
Seat Material	FKM	FKM	3
Spring/Pin Material	316 Stainless Steel	316 Stainless Steel	3
Gasket/Seal Material	PTFE	Shim - PTFE	3
Design Pressure (kPa)	700kPa(g)	700kPa(g)	3
Design Temperature (°C)	-10 to 200°C	-10 to 200°C	3
Surface Protection	Manufacturer's Standard	Manufacturer's Standard	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.

VCD1S2	Swing Check Valve – Stainless steel, flanged to AS2129 Table D, fire-safe/anti-static for hydrocarbon applications.
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Item	EPC Data	Supplier Data	Notes
Type	Swing Check Valve, Bolted Bonnet, fire safe to API 607 and fitted with anti-static device		1
Service	Hydrocarbons		1
Size Range (mm)	DN50 to DN600		1
End Connections	Flanged, FF to AS 2129 Table D		1
Actuation	Swing Close (gravity)		1
Lockable	Not Applicable		1
Supplier Name			2
Manufacturer			2
Model			2
Valve Standard			2
Dimensional Standard			2
Body Material	316 Stainless Steel CF8M		3
Disc Material	316 Stainless Steel CF8M		3
Seat Material	Buna-N		3
Spring/Pin Material	316 Stainless Steel		3
Gasket/Seal Material	PTFE		3
Design Pressure (kPa)	700kPa(g)		3
Design Temperature (°C)	-10 to 200°C		3
Surface Protection	Manufacturer's Standard		2

NOTES		
1	Data supplied by EPC The Supplier shall comply with this data.	
2	Data shall be provided by Supplier.	
3	Data proposed by EPC The Supplier shall confirm data or state alternative.	

VCD1S3	Dual Disc Check Valve – Stainless steel, wafer to AS2129 Table D for potable water applications.
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Item	EPC Data	Supplier Data	Notes
Type	Dual Disc Check Valve		1
Service	Potable Water		1
Size Range (mm)	DN50 to DN900		1
End Connections	Wafer type to AS 2129 Table D		1
Actuation	Spring Close		1
Lockable	Not Applicable		1
Supplier Name			2
Manufacturer			2
Model			2
Valve Standard			2
Dimensional Standard			2
Body Material	316 Stainless Steel CF8M		3
Disc Material	316 Stainless Steel CF8M		3
Seat Material	EPDM		3
Spring/Pin Material	316 Stainless Steel		3
Gasket/Seal Material	PTFE		3
Design Pressure (kPa)	700kPa(g)		3
Design Temperature (°C)	-10 to 200°C		3
Surface Protection	Manufacturer's Standard		2

NOTES		
1	Data supplied by EPC The Supplier shall comply with this data.	
2	Data shall be provided by Supplier.	
3	Data proposed by EPC The Supplier shall confirm data or state alternative.	

VCD1S4	Swing Check Valve – Stainless steel, flanged to AS2129 Table D, for hazardous reagent applications.
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Item	EPC Data	Supplier Data	Notes
Type	Swing Check Valve, Bolted Bonnet		1
Service	Hazardous Reagents		1
Size Range (mm)	DN15 to DN32		1
End Connections	Flanged, FF to AS 2129 Table D	Flanged, RF to AS 2129 Table D	1
Actuation	Swing Close (gravity)		1
Lockable	Not Applicable		1
Supplier Name		Emerson	2
Manufacturer		Arita Unimech	2
Model		SSC-A1-SUS	2
Valve Standard		ASME B16.34	2
Dimensional Standard		ASME B16.10	2
Body Material	316 Stainless Steel CF8M	ASTM A182 F316	3
Disc Material	316 Stainless Steel CF8M	ASTM A182 F316	3
Seat Material	316 Stainless Steel CF8M	ASTM A182 F316	3
Spring/Pin Material	316 Stainless Steel	SS316	3
Gasket/Seal Material	PTFE	SS + Graphite	3
Design Pressure (kPa)	700kPa(g)	700 kPa(g)	3
Design Temperature (°C)	-10 to 200°C	-0 to 200°C	3
Surface Protection	Manufacturer's Standard	Manufacturer's Standard	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.

VCD1S5	Dual Disc Check Valve – Stainless steel, wafer to AS2129 Table D for general service and non-hazardous reagent applications.
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Item	EPC Data	Supplier Data	Notes
Type	Dual Disc Check Valve		1
Service	General Service and Non-Hazardous Reagent Applications		1
Size Range (mm)	DN750		1
End Connections	Wafer type to AS 2129 Table D		1
Actuation	Spring Close		1
Lockable	Not Applicable		1
Supplier Name		Emerson	2
Manufacturer		APV - GSL	2
Model		Supercheck ASGBD15CVF	2
Valve Standard		Manufacturers Standard	2
Dimensional Standard		Manufacturers Standard	2
Body Material	316 Stainless Steel CF8M	A351 CF8M	3
Disc Material	316 Stainless Steel CF8M	A351 CF8M	3
Seat Material	FKM	Viton	3
Spring/Pin Material	316 Stainless Steel	SS316	3
Gasket/Seal Material	PTFE	N/A	3
Design Pressure (kPa)	700kPa(g)	700 kPa(g)	3
Design Temperature (°C)	0 to 100°C	0 to 100°C	3
Surface Protection	Manufacturer's Standard	Manufacturer's Standard	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.

VCD1T1	Check Valve – PFA lined Stainless Steel, Wafer Type for Hydrochloric acid service.
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Item	EPC Data	Supplier Data	Notes
Type	Swing Check Valve, Bolted Bonnet	Wafer Poppet Check Valve	1
Service	Hydrochloric Acid		1
Size Range (mm)	DN50 to DN150	DN50 to DN100	1
End Connections	Flanged, FF to AS 2129 Table D	Wafer	1
Actuation	Swing Close (gravity)	Spring Close	1
Lockable	Not Applicable		1
Supplier Name		Emerson	2
Manufacturer		KDV	2
Model		Series 24 Check Valve	2
Valve Standard		Manufacturer's Standard	2
Dimensional Standard		Manufacturer's Standard	2
Body Material	Polypropylene	304 SS	3
Disc Material	Polypropylene	PTFE	3
Seat Material	Polypropylene	PFA Lined 304SS	3
Spring/Pin Material	Not Applicable	Hastelloy C276	3
Gasket/Seal Material		PFA Lined 304SS	2
Design Pressure (kPa)	700kPa(g)	1800kPa(g)	3
Design Temperature (°C)	-10 to 50°C	-29 to 50°C	3
Surface Protection	Nil	Nil	2

NOTES		
1	Data supplied by EPC	The Supplier shall comply with this data.
	Data shall be provided by Supplier.	
2	Data proposed by EPC	The Supplier shall confirm data or state alternative.

VCE1C1	Dual Disc Check Valve – Carbon steel, wafer to AS2129 Table E for general service applications.
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Item	EPC Data	Supplier Data	Notes
Type	Dual Disc Check Valve		1
Service	General Service and Non-Hazardous Reagent Applications		1
Size Range (mm)	DN50 to DN600		1
End Connections	Wafer type to suit AS 2129 Table E		1
Actuation	Spring Close		1
Lockable	Not Applicable		1
Supplier Name		Emerson	2
Manufacturer		GSL - Supercheck	2
Model		ASGBE15SVF-94	2
Valve Standard		API 594	2
Dimensional Standard		API 594 & ASME B16.10	2
Body Material	Carbon Steel	ASTM A216 WCB	3
Disc Material	316 Stainless Steel CF8M	ASTM A351 CF8M	3
Seat Material	Viton	Viton	3
Spring/Pin Material	316 Stainless Steel	Pin: SS316 Spring: Inconel X750	3
Gasket/Seal Material	PTFE	N/A	3
Design Pressure (kPa)	1,400kPa(g)	1,400kPa(g)	3
Design Temperature (°C)	-10 to 200°C	-10 to 150°C	3
Surface Protection	Manufacturer's Standard	Manufacturer's Standard	2

NOTES		
1	Data supplied by EPC The Supplier shall comply with this data.	
2	Data shall be provided by Supplier.	
3	Data proposed by EPC The Supplier shall confirm data or state alternative.	

VCE1S1	Dual Disc Check Valve – Stainless steel, wafer to AS2129 Table E for general service and hazardous reagent applications.
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Item	EPC Data	Supplier Data	Notes
Type	Dual Disc Check Valve		1
Service	General and Hazardous Reagent Applications		1
Size Range (mm)	DN50 to DN600		1
End Connections	Wafer type to AS 2129 Table E		1
Actuation	Spring Close		1
Lockable	Not Applicable		1
Supplier Name		Emerson	2
Manufacturer		Arita - Unimech	2
Model		SCV-W-SUS	2
Valve Standard		DIN3202	2
Dimensional Standard		DIN3202	2
Body Material	316 Stainless Steel CF8M	316 Stainless Steel CF8M	3
Disc Material	316 Stainless Steel CF8M	316 Stainless Steel CF8M	3
Seat Material	FKM	FKM	3
Spring/Pin Material	316 Stainless Steel	316 Stainless Steel	3
Gasket/Seal Material	PTFE	Shim - PTFE	3
Design Pressure (kPa)	1,400kPa(g)	1,400kPa(g)	3
Design Temperature (°C)	-10 to 200°C	-10 to 150°C	3
Surface Protection	Manufacturer's Standard	Manufacturer's Standard	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.

VCE1S3	Swing Check Valve – Stainless steel, flanged to AS2129 Table E, Fire-safe/Anti-static for Oxygen Service.
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Item	EPC Data	Supplier Data	Notes
Type	Swing Check Valve, Bolted Bonnet, fire safe to API 607 and fitted with anti-static device		1
Service	Oxygen Service		1
Size Range (mm)	DN50 to DN600		1
End Connections	Flanged, FF to AS 2129 Table E		1
Actuation	Swing Close (gravity)		1
Lockable	Not Applicable		1
Supplier Name			2
Manufacturer			2
Model			2
Valve Standard			2
Dimensional Standard			2
Body Material	316 Stainless Steel CF8M		3
Disc Material	316 Stainless Steel CF8M		3
Seat Material	EPDM		3
Spring/Pin Material	316 Stainless Steel		3
Gasket/Seal Material	PTFE		3
Design Pressure (kPa)	1,400kPa(g)		3
Design Temperature (°C)	-10 to 120°C		3
Surface Protection	Nil		2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.
4	Valves to be cleaned and passivated in accordance to ASTM G93-03 and supplied in vacuum packed containers with dessicant and marked as cleaned for oxygen service.
5	Inspection and record keeping of valves to be in accordance with EIGA "Oxygen Pipeline and Piping Systems"

VCE1S4	Inline Single Disc Check Valve – Stainless steel, wafer to AS2129 Table E for Hydrogen Peroxide
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Item	EPC Data	Supplier Data	Notes
Type	Inline Single Disc Check Valve		1
Service	Hydrogen Peroxide		1
Size Range (mm)	DN25 to DN100		1
End Connections	Wafer to AS 2129 Table E		1
Actuation	Spring Close		1
Lockable	Not Applicable		1
Supplier Name		Emerson	2
Manufacturer		HMA - Gestra	2
Model		RKE 86A	2
Valve Standard		Manufacturers Standard	2
Dimensional Standard		Short overall length to EN 558	2
Body Material	316 Stainless Steel CF8M	1.4408 (ASTM A351 CF8M)	3
Disc Material	316 Stainless Steel CF8M	1.4571 Stainless Steel	3
Seat Material	Viton	FPM	3
Spring/Guide Material	316 Stainless Steel	1.4571 Stainless Steel	3
Gasket/Seal Material	Viton	1.4408 (ASTM A351 CF8M)	3
Design Pressure (kPa)	1,400kPa(g)	1,400kPa(g)	3
Design Temperature (°C)	-10 to 200°C	-10 to 200°C	3
Surface Protection	Manufacturer's Standard	N/A	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.
4	Approved for end of service use (vacuum breakers) as per EU Pressure Equipment Directive (PED)

VCE1T1	Check Valve – Wafer Type for Hydrochloric acid service.
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Item	EPC Data	Supplier Data	Notes
Type	Check Valve, Bolted Bonnet	Wafer Poppet Check Valve	1
Service	Hydrochloric Acid		1
Size Range (mm)	DN50 to DN150	DN50 to DN100	1
End Connections	Flanged, FF to AS 2129 Table E	Wafer	1
		Spring Close	1
Lockable	Not Applicable		1
Supplier Name		Emerson	2
Actuation	Swing Close (gravity)		
Manufacturer		KDV	2
Model		Series 24 Check Valve	2
Valve Standard		Manufacturer's Standard	2
Dimensional Standard		Manufacturer's Standard	2
Body Material	Polypropylene	304 SS	3
Disc Material	Polypropylene	PTFE	3
Seat Material	Polypropylene	PFA Lined 304SS	3
Spring/Pin Material	Not Applicable	Hastelloy C276	3
Gasket/Seal Material		PFA Lined 304SS	2
Design Pressure (kPa)	1,400kPa(g)	1800kPa(g)	3
Design Temperature (°C)	-10 to 50°C	-29 to 50°C	3
Surface Protection	Nil	Nil	2

NOTES		
1	Data supplied by EPC The Supplier shall comply with this data.	
	Data shall be provided by Supplier.	
2	Data proposed by EPC The Supplier shall confirm data or state alternative.	

VDD1L1	Diaphragm Valve - Straight through type, cast iron, rubber lined, flanged to AS 2129 Table D for slurry service applications.
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Item	EPC Data	Supplier Data	Notes
Type	Diaphragm Valve, Straight Through Type, Lined		1
Service	Slurry		1
Size Range (mm)	DN50 to DN350		1
End Connections	Flanged, FF to AS 2129 Table D	Flanged, FF to AS 2129 Table D (Table E up to DN80)	1
Actuation	Hand Wheel		1
Lockable	Yes (provision for locking)		1
Supplier Name		Emerson	2
Manufacturer		KDV	2
Model		Series 37 Straight Through (ST), V7KCSC10BCN	2
Valve Standard		EN13397:2002 (replaced BS5156:1985)	2
Dimensional Standard		Face to Face: EN558-1 Series 7	2
Body & Bonnet Material	Cast Iron	Cast Iron	3
Diaphragm Material	Natural Rubber	Natural Rubber	3
Lining Material	Soft Natural Rubber	Soft Natural Rubber	3
Spindle Material	316 Stainless Steel	316 Stainless Steel	3
Design Pressure	700kPa(g)	700kPa(g)	3
Design Temperature	-10 to 80°C	-10 to 80°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.

VDD1L2	Diaphragm Valve - Weir type, cast iron, butyl rubber diaphragm and lining, flanged to AS 2129 Table D for Acid/Reagent service applications.
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Item	EPC Data	Supplier Data	Notes
Type	Diaphragm Valve, Weir Type, lined		1
Service	Acid/Reagent Applications		1
Size Range (mm)	DN50 to DN250		1
End Connections	Flanged, FF to AS 2129 Table D	Flanged, FF to AS 2129 Table D	1
Actuation	Hand wheel		1
Lockable	Yes (provision for locking)		1
Supplier Name		Emerson	2
Manufacturer		KDV	2
Model		Series 35 Weir Type (WT), V7KCBC30BDN	2
Valve Standard		EN13397:2002 (replaced BS5156:1985)	2
Dimensional Standard		Face to Face: EN558-1 Series 7	2
Body & Bonnet Material	Cast Iron	Cast Iron	3
Diaphragm Material	Butyl Rubber	Butyl Rubber	3
Lining Material	Butyl Rubber	Butyl Rubber	3
Spindle Material	316 Stainless Steel	316 Stainless Steel	3
Design Pressure	700kPa(g)	700kPa(g)	3
Design Temperature	0 to 100°C	0 to 100°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.

VDD1S1	Diaphragm Valve - Weir type, stainless steel, unlined, rubber diaphragm, flanged to AS 2129 Table D for general service and non-hazardous reagent applications.		
Item	EPC Data	Supplier Data	Notes
Type	Diaphragm Valve, Weir Type, Unlined		1
Service	General Service and Non-Hazardous Reagent Applications		1
Size Range (mm)	DN25 to DN250		1
End Connections	Flanged, FF to AS 2129 Table D	Flanged, FF to AS 2129 Table D	1
Actuation	Hand Wheel		1
Lockable	Yes (provision for locking)		1
Supplier Name		Emerson	2
Manufacturer		KDV	2
Model		Series 35 Weir Type (WT), V7KIOB40BMVK	2
Valve Standard		EN13397:2002 (replaced BS5156:1985)	2
Dimensional Standard		Face to Face: EN558-1 Series 7	2
Body & Bonnet Material	Stainless Steel CF8M	Stainless Steel CF8M	3
Diaphragm Material	Nitrile Rubber	Nitrile Rubber	3
Lining Material	None	None	3
Spindle Material	316 Stainless Steel	316 Stainless Steel	3
Design Pressure	700 kPa(g)	700 kPa(g)	3
Design Temperature	0 to 100°C	0 to 100°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2
NOTES			
1	Data supplied by EPC The Supplier shall comply with this data.		
2	Data shall be provided by Supplier.		
3	Data proposed by EPC The Supplier shall confirm data or state alternative.		

VDD1S2	Diaphragm Valve - Weir type, stainless steel, unlined, butyl rubber diaphragm, flanged to AS 2129 Table D for pregnant eluate and electrowinning service.		
Item	EPC Data	Supplier Data	Notes
Type	Diaphragm Valve, Weir Type, Unlined		1
Service	Pregnant Eluate / Electrowinning		1
Size Range (mm)	DN25 to DN250		1
End Connections	Flanged, FF to AS 2129 Table D	Flanged, FF to AS 2129 Table D	1
Actuation	Hand wheel		1
Lockable	Yes (provision for locking)		1
Supplier Name		Emerson	2
Manufacturer		KDV	2
Model		Series 35 Weir Type (WT), V7KIOB30BMVK	2
Valve Standard		EN13397:2002 (replaced BS5156:1985)	2
Dimensional Standard		Face to Face: EN558-1 Series 7	2
Body & Bonnet Material	316 Stainless Steel CF8M	Stainless Steel CF8M	3
Diaphragm Material	Butyl Rubber	Butyl Rubber	3
Lining Material	None	None	3
Spindle Material	316 Stainless Steel	316 Stainless Steel	3
Design Pressure	700 kPa(g)	700 kPa(g)	3
Design Temperature	0 to 120°C	0 to 120°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2
NOTES			
1	Data supplied by EPC The Supplier shall comply with this data.		
2	Data shall be provided by Supplier.		
3	Data proposed by EPC The Supplier shall confirm data or state alternative.		

VDE1L2	Diaphragm Valve - Weir type, cast iron, butyl rubber diaphragm and lining, flanged to AS 2129 Table E for Acid/Reagent service applications.
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Item	EPC Data	Supplier Data Manual Valve	Supplier Data Actuated Valve	Notes
Type	Diaphragm Valve, Weir Type, lined			1
Service	Acid/Reagent Applications			1
Size Range (mm)	DN25 to DN150	DN25 to DN150	DN25 to DN150	1
End Connections	Flanged, FF AS 2129 Table E	Flanged, FF AS 2129 Table E	Flanged, FF AS 2129 Table E	1
Actuation	Hand wheel	Hand wheel	Pneumatic Actuator	1
Lockable	Yes (provision for locking)	Yes	N/A	1
Supplier Name		Emerson	Emerson	2
Manufacturer		KDV	KDV	2
Model		WT Series 35	Series 0HRT (DN25-32) Series 2AHRT (DN40-150)	2
Valve Standard		EN 13397-2001 (BS 5156:1985)	EN 12266-2 2002 (BS 6755 Part 1)	2
Dimensional Standard		EN 558-1 Series 7 (BS 5156)	EN 558-1 Series 7 (BS 5156)	2
Body & Bonnet Material	Cast Iron	Cast Iron	Cast Iron	3
Diaphragm Material	Butyl Rubber	Butyl Rubber	Butyl Rubber	3
	Butyl Rubber	Butyl Rubber	Butyl Rubber	3
	316 Stainless Steel	316 Stainless Steel	316 Stainless Steel	3
Design Pressure	1,000kPag	1,000kPag	1,000kPag	3
Lining Material	0 to 100°C	0 to 100°C	0 to 100°C	3
Design Temperature	0 to 100°C	0 to 100°C	0 to 100°C	3
Spindle Material	Manufacturers Standard	Manufacturers Standard	Manufacturers Standard	2
Surface Protection	Manufacturers Standard	Manufacturers Standard	Manufacturers Standard	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
	Data shall be provided by Supplier.
2	Data proposed by EPC The Supplier shall confirm data or state alternative.

VDE1S1	Diaphragm Valve - Weir type, stainless steel, unlined, rubber diaphragm, flanged to AS 2129 Table E for general service.		
Item	EPC Data	Supplier Data	Notes
Type	Diaphragm Valve, Weir Type, Unlined		1
Service	General		1
Size Range (mm)	DN25 to DN250		1
End Connections	Flanged, FF to AS 2129 Table E	Flanged, FF to AS 2129 Table E	1
Actuation	Hand wheel		1
Lockable	Yes (provision for locking)		1
Supplier Name		Emerson	2
Manufacturer		KDV	2
Model		Series 35 Weir Type (WT), V7EIOB40BMVK	2
Valve Standard		EN13397:2002 (replaced BS5156:1985)	2
Dimensional Standard		Face to Face: EN558-1 Series 7	2
Body & Bonnet Material	316 Stainless Steel CF8M	Stainless Steel CF8M	3
Diaphragm Material	Nitrile Rubber	Nitrile Rubber	3
Lining Material	None	None	3
Spindle Material	316 Stainless Steel	316 Stainless Steel	3
Design Pressure	1,400 kPa(g)	1,400 kPa(g)	3
Design Temperature	0 to 100°C	0 to 100°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2
NOTES			
1	Data supplied by EPC The Supplier shall comply with this data.		
2	Data shall be provided by Supplier.		
3	Data proposed by EPC The Supplier shall confirm data or state alternative.		

VDE1S2	Diaphragm Valve - Weir type, stainless steel, unlined, FKM rubber diaphragm, flanged to AS 2129 Table E for High Temperature Eluate.
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Item	EPC Data	Supplier Data	Notes
Type	Diaphragm Valve, Weir Type, Unlined		1
Service			1
Size Range (mm)	DN25 to DN250		1
End Connections	High Temperature Process Eluate flanged, RPT to AS 2129 Table E (3.0% w/v Cyanide & 3.0% w/v Caustic)		1
Actuation	Hand wheel		1
Lockable	Yes (provision for locking)		1
Supplier Name		Emerson	2
Manufacturer		KDV	2
Model		Series 35 Weir Type	2
Valve Standard		EN13397:2002 (replaced BS5156:1985)	2
Dimensional Standard		Face to Face: EN558-1 Series 7	2
Body & Bonnet Material	316 Stainless Steel CF8M	Stainless Steel CF8M	3
Diaphragm Material	FKM Rubber	FKM Rubber	3
Lining Material	None	None	3
Spindle Material	316 Stainless Steel	316 Stainless Steel	3
Design Pressure	1,400 kPa(g)	1,400 kPa(g)	3
Design Temperature	0 to 150°C	0 to 150°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
	Data shall be provided by Supplier.
2	Data proposed by EPC The Supplier shall confirm data or state alternative.

VDTD1L1	Dart Valve - yDart type, mild steel, rubber lined, flanged to AS 2129 Table D for slurry service applications.
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Item	EPC Data	Supplier Data	Notes
Type	yDart Valve, Full Bore, Lined		1
Service	Slurry flow control		1
Size Range (mm)	DN100 to DN600		1
End Connections	Flanged, FF to AS 2129 Table D		1
Actuation	Pneumatic c/w positioner ; Fail Closed on loss of air and power.		1
Lockable	N/A		1
Supplier Name		Emerson	2
Manufacturer		Dual Valves Australasia Pty Ltd	2
Model		yDart YDP200B	2
Valve Standard		Manufacturer's standard	2
Dimensional Standard		Manufacturer's standard	2
Body Material	Mild Steel - natural rubber lined	Mild Steel polyurethane Lined	3
Plug Material	Mild Steel - polyurethane coated plug	Mild Steel - polyurethane coated plug	3
Shaft Material	316 stainless steel integral with disc	Shaft material SAF2507	3
Seat Material	Polyurethane	Polyurethane	3
Design Pressure	700kPa(g)	700kPa(g)	3
Design Temperature	0 to 80°C	75°C maximum temperature	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2

NOTES		
1	Data supplied by EPC The Supplier shall comply with this data.	
2	Data shall be provided by Supplier.	
3	Data proposed by EPC The Supplier shall confirm data or state alternative.	

VF03S1	Float Valve - Stainless Steel, Viton Seal, BSP threaded for scheme water applications.		
Item	EPC Data	Supplier Data	Notes
Type	Float valve		1
Service	Scheme Water		1
Size Range (mm)	DN15 to DN50		1
End Connections	Threaded, BSPT Male		1
Actuation	Mechanical Float and Lever		1
Lockable	N/A		1
Supplier Name		Emerson	2
Manufacturer		Philmac	2
Model		500 series	2
Valve Standard		Manufacturer's Standard	2
Dimensional Standard	Manufacturer's Standard	Manufacturer's Standard	3
Body Material	316 Stainless Steel	316 Stainless Steel	3
Float Material	Plastic rated for 80°C	Plastic rated for 80°C	3
Seal Material	Viton A	Viton A	3
Seat Material	316 Stainless Steel	316 Stainless Steel	3
Plunger Material	316 Stainless Steel	316 Stainless Steel	3
Design Pressure	700 kPa(g)	700 kPa(g)	3
Design Temperature	-10 to 80°C	-10 to 80°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	3
NOTES			
1	Data supplied by EPC The Supplier shall comply with this data.		
2	Data shall be provided by Supplier.		
3	Data proposed by EPC The Supplier shall confirm data or state alternative.		

VGD1C1	Gate Valve - Bonnet, carbon steel, flanged to AS2129 Table D for Reagents services.
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Item	EPC Data	Supplier Data	Notes
Type	Gate Valve, Bonnet		1
Service	Reagents		1
Size Range (mm)	DN150 to DN300		1
End Connections	Flanged, FF to AS 2129 Table D	ASME B16.5 CLASS 150 RF	1
Actuation	Hand wheel		1
Lockable	Yes (provision for locking)		1
Supplier Name		Emerson	2
Manufacturer		Australian Pipeline Valve	2
Model		Gate Valve Model 250~300AP47XUF	2
Valve Standard		API 600 & ASME B16.34	2
Dimensional Standard	ASME B16.10	ASME B16.10	3
Body Material	Carbon steel	ASTM A216 WCB	3
Wedge Material	Carbon steel	ASTM A216 WCB +13CR	3
Shaft / Stem Material	Carbon steel	ASTM A182 F6A	3
Gland Packing	PTFE	FLEXIBLE GRAPHITE	3
Design Pressure	700kPa(g)	2000kPa(g)	3
Design Temperature	0 to 80°C	-29 to 538°C	3
Surface Protection	Coated to Manufacturers Standard	Manufacturers Standard	3

NOTES		
1	Data supplied by EPC The Supplier shall comply with this data.	
2	Data shall be provided by Supplier.	
3	Data proposed by EPC The Supplier shall confirm data or state alternative.	

VG1S1	Gate Valve - Bonnet, stainless steel, flanged to AS2129 Table D for general services and low hazard reagents.
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Item	EPC Data	Supplier Data	Notes
Type	Gate Valve, Bonnet		1
Service	Gland water		1
Size Range (mm)	DN150 to DN300		1
End Connections	Flanged, FF to AS 2129 Table D		1
Actuation	Hand wheel		1
Lockable	No		1
Supplier Name			2
Manufacturer			2
Model			2
Valve Standard			2
Dimensional Standard	ASME B16.10		3
Body Material	316 Stainless steel		3
Wedge Material	316 Stainless steel		3
Shaft / Stem Material	316 Stainless steel		3
Gland Packing	PTFE		3
Design Pressure	700kPa(g)		3
Design Temperature	0 to 150°C		3
Surface Protection	Coated to Manufacturers Standard		3

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.

VG1C1	Gate Valve - Bonnet, carbon steel, flanged to AS2129 Table D for Reagents/Air services.
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Item	EPC Data	Supplier Data	Notes
Type	Gate Valve, Bonnet		1
Service	Reagents & Air		1
Size Range (mm)	DN150 to DN350		1
End Connections	Flanged, FF to AS 2129 Table E		1
Actuation	Hand wheel		1
Lockable	Yes (provision for locking)		1
Supplier Name		Emerson	2
Manufacturer		Arita	2
Model		GG0101T1	2
Valve Standard		ASME B16.34	2
Dimensional Standard	ASME B16.10	ASME B16.10	3
Body Material	Carbon steel	ASTM A216 WCB	3
Wedge Material	Carbon steel	ASTM A216 WCB +13CR	3
Shaft / Stem Material	Carbon steel	ASTM A276 420	3
Gland Packing	PTFE	SS304 + Graphite	3
Design Pressure	1,400kPa(g)	1,900kPa(g)	3
Design Temperature	0 to 80°C	0 to 100°C	3
Surface Protection	Coated to Manufacturers Standard	Coated to Manufacturers Standard	3

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.

VG1F1	Gate Valve - OS&Y, Ductile Iron, flanged to AS2129 Table E for isolation duty of fire water service.
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Item	EPC Data	Supplier Data	Notes
Type	Gate Valve, OS& Y, Fire Rated		1
Service	Fire Water		1
Size Range (mm)	DN50 to DN200		1
End Connections	Flanged, FF to AS 2129 Table E		1
Actuation	Hand wheel		1
Lockable	No		1
Supplier Name		Emerson	2
Manufacturer		Challenger	2
Model		RSGV3299	2
Valve Standard		AWWA C515 & FM/UL approved	2
Dimensional Standard		AWWA C515	2
Body Material	Ductile Iron	Ductile Iron ASTM A536 65-45-12	3
Wedge Material	EPDM coated ductile iron	EPDM coated ductile iron	3
Shaft / Stem Material	304 Stainless Steel	304 Stainless Steel	3
Gland Packing	Graphite	Graphite	3
Design Pressure	1,400kPa(g)	1,400kPa(g)	3
Design Temperature	0 to 80°C	0 to 52°C	3
Surface Protection	Red Epoxy Coated to Manufacturers Standard	Red Epoxy Coated to Manufacturers Standard	3

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.

VGLE151	Globe Valve - Stainless Steel, PTFE Seat, flanged to AS 2129 Table E for oxygen applications.
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Item	EPC Data	Manual Valve Supplier Data	Actuated Valve Supplier Data	Notes
Type	Globe Valve, Body & Bonnet Style			1
Service	Oxygen			1
Size Range (mm)	DN15 to DN100			1
End Connections	Flanged, FF to AS 2129 Table E		ANSI Class 150 Raised Face	1
Actuation	Handwheel			1
Lockable	Yes			1
Supplier Name			Emerson	2
Manufacturer			Fisher	2
Model			24000SVF Size 1	2
Valve Standard			Manufacturers Standard	2
Dimensional Standard	MSS SP-80		EN 588-2 (ISA 575.03)	3
Body & Bonnet Material	316 Stainless Steel		A351 CF3M	3
Wedge Material	316 Stainless Steel		316 SS	3
Stem Material	316 Stainless Steel		316 SS	3
Seats Material	PTFE		PTFE	3
Stem Packing Material	PTFE		PTFE	3
Design Pressure	1,400 kPa(g)		1,400 kPa(g)	3
Design Temperature	-10 to 200°C		-46 to 232°C	3
Surface Protection	Manufacturers Standard		Manufacturers Standard	3

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.
4	Valves to be cleaned and passivated in accordance to ASTM A380/A380M and ASTM A967/A967M and supplied in vacuum packed containers with dessicant and marked as cleaned for oxygen service.

VKGD1C1	Knife Gate Valve - Delta, Bidirectional shear, Ni resist, Metal Seat, lugged to AS2129 Table D for severe slurry service applications.
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Item	EPC Data	Supplier Data	Notes
Type	Delta Kinfe Gate Valve, Bidirectional Shear, Lug Type, Narrow Body		1
Service	Abbrasive Mill Slurry		1
Size Range (mm)	DN50 to DN750		1
End Connections	Lugged body to suit AS 2129 Table D, Metric body lugs tapping		1
Actuation	≤DN250 – Hand wheel ≥DN300 – Gearbox with hand wheel		1
Lockable	Yes	Mechanical lockout, open/closed with standard pin	3
Supplier Name		Emerson	2
Manufacturer		Clarkson	2
Model		KS1	2
Valve Standard		MSS SP-135	2
Dimensional Standard		MSS SP-135 SHORT	2
Body Material	Ni Resist	DN50 to 500: A439 D2 Ni Resist DN600 to 750: WCB	3
Gate Material	316 Hard Chrome Coated Stainless Steel	Duplex 2205 Stainless Steel	3
Wear Ring Material	Ni Resist	A439 D2 Ni Resist	3
Stem Material	316 Stainless Steel	316 Stainless Steel	3
Seal Material	Buna N	Natural Rubber	3
Packing Material	N/A	N/A	3
Deflection Cone Material	Not required	N/A	3
Design Pressure	700 kPa(g)	285 psi MAWP (1,965 kPa(g))	3
Design Temperature	0 to 80°C	0 to 80°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard TES-B-48	2
Shroud / Gate Guard	Yes - Actuated Only	Included	3

NOTES		
1	Data supplied by EPC	The Supplier shall comply with this data.
2	Data shall be provided by Supplier.	
3	Data proposed by EPC	The Supplier shall confirm data or state alternative.

VKGD1L1	Knife Gate Valve - Bidirectional, Ductile Iron, Natural Rubber replaceable sleeve, narrow body, lugged to AS2129 Table D for slurry service applications.
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Item	EPC Data	Supplier Data	Notes
Type	Kinfe Gate Valve, Bidirectional, Lug Type, Narrow Body, Push-Through Gate		1
Service	Slurry		1
Size Range (mm)	DN50 to DN600		1
End Connections	Lugged body to suit AS 2129 Table D, Metric body lugs tapping		1
Actuation	≤DN250 – Hand wheel ≥DN300 – Gearbox with hand wheel		1
Lockable	Yes	Mechanical lockout, open/closed with standard pin	3
Supplier Name		Emerson	2
Manufacturer		Clarkson	2
Model		KGD	2
Valve Standard		MSS SP-81	2
Dimensional Standard		MSS SP-81	2
Body Material	Ductile Iron with replaceable elastomer sleeve	Ductile Iron with replaceable elastomer sleeve	3
Gate Material	316 Stainless Steel	316 Stainless Steel	3
Stem Material	316 Stainless Steel	316 Stainless Steel	3
Sleeve Material	Natural Rubber with reinforcing ring	Natural Rubber with reinforcing ring	3
Packing Material	Packingless top seal	Packingless top seal	3
Deflection Cone Material	Not required	N/A	3
Design Pressure	700 kPa(g)	DN50 to 400: 150 psi MAWP (1,034 kPa(g)) DN450 to 600: 90 psi MAWP (620 kPa(g))	3
Design Temperature	0 to 80°C	0 to 80°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard TES-B-48	2
Shroud / Gate Guard	Yes - Actuated Only	Included	3

NOTES		
1	Data supplied by EPC The Supplier shall comply with this data.	
2	Data shall be provided by Supplier.	
3	Data proposed by EPC The Supplier shall confirm data or state alternative.	

VKGD1L2	Knife Gate Valve - Bidirectional, ductile iron, natural rubber replaceable sleeve, wide body, flanged to AS2129 Table D for slurry service.
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Item	EPC Data	Supplier Data	Notes
Type	Kinfe Gate Valve, Bidirectional, Flanged Type, Wide Body, Push-Through Gate		1
Service	Slurry		1
Size Range (mm)	DN80 to DN1200		1, 4, 5
End Connections	Flanged body to suit AS 2129 Table D FF, Metric body lugs tapping		1
Actuation	Gearbox with hand wheel	DN80 to 750: Pneumatic DN800 to 1200: Hydraulic	1
Lockable	Yes	Mechanical lockout, open/closed with standard pin	1
Supplier Name		Emerson	2
Manufacturer		Clarkson	2
Model		KGA+	2
Valve Standard		Manufacturers Standard	2
Dimensional Standard		Wide face to face (manufacturing standard)	2
Body Material	Ductile Iron with replaceable elastomer sleeve	WCB with replaceable elastomer sleeve	3
Gate Material	316 Stainless Steel	DN80 to 800: Duplex 2205 Stainless Steel DN900 to 1200: 316 Stainless Steel ⁴	3
Stem Material	316 Stainless Steel	Plated Rod for pneumatic/hydraulic actuation	3
Sleeve Material	Natural Rubber with reinforcing ring	Natural Rubber with reinforcing ring	3
Packing Material	Packingless top seal	Packingless top seal	3
Deflection Cone Material	Not required	Not required	3
Design Pressure	700kPa(g)	DN80 to 800: 700 kPa(g) DN900 to 1050: 520 kPa(g) ⁴ DN1200: 340 kPa(g) ⁴	3
Design Temperature	0 to 80°C	0 to 80°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard TES-B-48	2
Shroud / Gate Guard	Yes - Actuated Only	Included	3

NOTES	
1	Data supplied by EPC The Supplier shall comply with this data.
2	Data shall be provided by Supplier.
3	Data proposed by EPC The Supplier shall confirm data or state alternative.
4	DN900 and greater Knifegate only suitable to be installed at pump suction (i.e. low pressure). If required to be suitable for 700 kPa(g), gate to be Duplex 2205 Stainless Steel. Refer to vendor catalogue for pressure ranges.
5	Size range to be typically DN650 and up. Smaller bore valving to be as per VKGD1L1 spec, with exception of Thickener U/F Slurry service

VKGD1L3	Knife Gate Valve - Bidirectional, ductile iron body, 2205 duplex gate, butyl rubber replaceable sleeve, wide body, flanged to AS2129 Table D for acidic slurry service applications.
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Item	EPC Data	Supplier Data	Notes
Type	Kinfe Gate Valve, Bidirectional, Flanged Type, Wide Body, Push-Through Gate		1
Service	Slurry		1
Size Range (mm)	DN80 to DN800		1
End Connections	Flanged body to suit AS 2129 Table D FF, Metric body lugs tapping		1
Actuation	≤DN250 – Hand wheel ≥DN300 – Gearbox with hand wheel		1
Lockable	Yes	Mechanical lockout, open/closed with standard pin	1
Supplier Name		Emerson	2
Manufacturer		Clarkson	2
Model		KGD	2
Valve Standard		MSS SP-81	2
Dimensional Standard		MSS SP-81	2
Body Material	Ductile Iron with replaceable elastomer sleeve	Ductile Iron with replaceable elastomer sleeve	3
Gate Material	2205 Duplex Stainless Steel	2205 Duplex Stainless Steel	3
Stem Material	2205 Duplex Stainless Steel	2205 Duplex Stainless Steel	3
Sleeve Material	Butyl Rubber with reinforcing ring	Butyl Rubber with reinforcing ring	3
Packing Material	Packingless top seal	Packingless top seal	3
Deflection Cone Material	Not required	N/A	3
Design Pressure	700kPa(g)	DN50 to 400: 150 psi MAWP (1,034 kPa(g)) DN450 to 600: 90 psi MAWP (620 kPa(g))	3
Design Temperature	0 to 80°C	0 to 80°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard TES-B-48	2
Shroud / Gate Guard	Yes - Actuated Only	Included	3

NOTES		
1	Data supplied by EPC	The Supplier shall comply with this data.
2	Data shall be provided by Supplier.	
3	Data proposed by EPC	The Supplier shall confirm data or state alternative.

VKGE1C1	Knife Gate Valve - Delta, Bidirectional shear, Ni resist, Metal Seat, lugged to AS2129 Table E for severe slurry service applications.
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Item	EPC Data	Supplier Data	Notes
Type	Delta Kinfe Gate Valve, Bidirectional Shear, Lug Type, Narrow Body		1
Service	Abbrasive Mill Slurry		1
Size Range (mm)	DN50 to DN750		1
End Connections	Lugged body to suit AS 2129 Table E, Metric body lugs tapping		1
Actuation	≤DN250 – Hand wheel ≥DN300 – Gearbox with hand wheel		1
Lockable	Yes	Mechanical lockout, open/closed with standard pin	3
Supplier Name		Emerson	2
Manufacturer		Clarkson	2
Model		KS1	2
Valve Standard		MSS SP-135	2
Dimensional Standard		MSS SP-135 SHORT	2
Body Material	Ni Resist	DN50 to 500: A439 D2 Ni Resist DN600 to 750: WCB	3
Gate Material	316 Hard Chrome Coated Stainless Steel	Duplex 2205 Stainless Steel	3
Wear Ring Material	Ni Resist	A439 D2 Ni Resist	3
Stem Material	316 Stainless Steel	316 Stainless Steel	3
Seal Material	Buna N	Natural Rubber	3
Packing Material	N/A	N/A	3
Deflection Cone Material	Not required	Not required	3
Design Pressure	700 kPa(g)	285 psi MAWP (1,965 kPa(g))	3
Design Temperature	0 to 80°C	0 to 80°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard TES-B-48	2
Shroud / Gate Guard	Yes - Actuated Only	Included	3

NOTES		
1	Data supplied by EPC The Supplier shall comply with this data.	
2	Data shall be provided by Supplier.	
3	Data proposed by EPC The Supplier shall confirm data or state alternative.	

VKGE1L1	Knife Gate Valve - Bidirectional, ductile iron, natural rubber replaceable sleeve, wide body, flanged to AS2129 Table E for high pressure slurry service applications.
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Item	EPC Data	Supplier Data	Notes
Type	Kinfe Gate Valve, Bidirectional, High Pressure, Flanged Type, Push-Through Gate		1
Service	Slurry		1
Size Range (mm)	DN50 to DN600		1
End Connections	Flanged body to suit AS 2129 Table E FF, Metric body lugs tapping		1
Actuation	≤DN250 – Hand wheel ≥DN300 – Gearbox with hand wheel		1
Lockable	Yes	Mechanical lockout, open/closed with standard pin	1
Supplier Name		Emerson	2
Manufacturer		Clarkson	2
Model		KGA+	2
Valve Standard		Manufacturers Standard	2
Dimensional Standard		Wide face to face (manufacturing standard)	2
Body Material	Ductile Iron with replaceable elastomer sleeve	Ductile Iron with replaceable elastomer sleeve	3
Gate Material	316 Stainless Steel	Duplex 2205	3
Stem Material	316 Stainless Steel	316 Stainless Steel	3
Sleeve Material	Natural Rubber with reinforcing ring	Natural Rubber with reinforcing ring	3
Packing Material	Packingless top seal	Packingless top seal	3
Deflection Cone Material	Not required	Not required	3
Design Pressure	1,400 kPa(g)	1,380 kPa(g)	3
Design Temperature	0 to 80°C	0 to 80°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard TES-B-48	2
Shroud / Gate Guard	Yes - Actuated Only	Included	3

NOTES		
1	Data supplied by EPC	The Supplier shall comply with this data.
2	Data shall be provided by Supplier.	
3	Data proposed by EPC	The Supplier shall confirm data or state alternative.

VPID1L1	Pinch Valve – Closed body type, cast iron body, replaceable natural rubber sleeve, flanged to AS2129 Table D for isolation and flow control duty for slurry service applications.
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Item	EPC Data	Supplier Data	Notes
Type	Pinch Valve - Closed Body Type		1
Service	Slurry		1
Size Range (mm)	DN15 to DN600		1
End Connections	Flanged, FF to AS2129 Table D		1
Actuation	≤DN250 – Hand wheel ≥DN300 – Gearbox with hand wheel		1
Lockable	Yes		1
Supplier Name		Emerson	2
Manufacturer		Jindex	2
Model		CMR06OT-MYY58-TD40XX	2
Valve Standard		Manufacturer's standard	2
Dimensional Standard		Manufacturer's standard	2
Body/Bonnet Material	Cast Iron to ASTM A 126 Class B	Cast Iron	3
Sleeve	Replaceable type, Linatex (Natural Rubber)	High wear reinforced Natural Rubber	3
Stem	Forged Stainless Steel to ASTM A 182 Type F304. PTFE Coated	SS316	3
Design Pressure	700kPa(g)	600 kPag	3
Design Temperature	65°C	65°C	3
Surface Protection	Manufacturers Standard	Manufacturers Standard	2

NOTES		
1	Data supplied by EPC The Supplier shall comply with this data.	
2	Data shall be provided by Supplier.	
3	Data proposed by EPC The Supplier shall confirm data or state alternative.	

VTWD1C1	Three way temperature control valve, carbon steel, flanged to AS2129 Table D for cooling water services.		
Item	EPC Data	Supplier Data	Notes
Type	Three way temperature control valve	Three way ball valve	1
Service	Cooling Water		1
Size Range (mm)	DN25 to DN150		1
End Connections	Flanged, FF to AS 2129 Table D		1
Actuation	Hand wheel		1
Lockable	No		1
Supplier Name		Emerson	2
Manufacturer		Emerson	2
Model		KTM MB11 T-Port with Flow Plan G (Pattern 2) Port C - inlet from chiller Port B - outlet to equipment Port A - bypass around equipment	2
Valve Standard		Manufacturers Standard	2
Dimensional Standard	ASME B16.10 Short Pattern	Manufacturers Standard	3
Body Material	Carbon steel	A216 Carbon Steel	3
Ball Material	316SS	A351 CF8M Stainless Steel	3
Shaft / Stem Material	316SS	A276 316 Stainless Steel	3
Gland Packing	PTFE	Reinforced PTFE	3
Design Pressure	700kPa(g)	700kPa(g)	3
Design Temperature	0 to 120°C	0 to 120°C	3
Surface Protection	Coated to Manufacturers Standard	Coated to Manufacturers Standard	3
NOTES			
1	Data supplied by EPC The Supplier shall comply with this data.		
2	Data shall be provided by Supplier.		
3	Data proposed by EPC The Supplier shall confirm data or state alternative.		